Subset Sum in Almost-Linear Time (Draft)

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Abstract

We give a deterministic algorithm for finding a subset of the input integers $1 \leq a_1, \ldots, a_n \leq \Delta$ that sums to the target t in $\mathcal{O}(n + \Delta^{1+o(1)})$ time. This improves upon the previous $\widetilde{\mathcal{O}}(n + \Delta^{1.5})$-time algorithm by Chen, Lian, Mao, and Zhang (SODA 2024), and achieves almost tight dependence on Δ assuming SETH.

We use an inverse theorem for subset sums due to Conlon, Fox, and Pham (2023), and develop a quotient variant of economic cap covering from the polytope approximation literature.

This is a working draft for time-stamping purpose. The mathematical content is complete. It is pending a technical overview section, local expositional enhancements, and acknowledgements and AI disclosure.

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1 Introduction

1.1 Background

We revisit the Subset Sum problem, a fundamental NP-hard computational task with connections to combinatorics, discrete optimization, and cryptography.

Problem 1. Subset Sum (decision version)
Input: Integers $1 \le a_1, a_2, \ldots, a_n \le \Delta$ and a target t (without loss of generality, $\Delta \le t \le n\Delta$)[a].
Task: Decide whether there exists $(x_1, \ldots, x_n) \in \{0,1\}^n$ such that $\sum_{i=1}^n a_i x_i = t$.

[a] To justify this, remove the useless integers $a_i > t$, reset $\Delta := \max_i a_i$, and return NO if $t > n\Delta$.

The Subset Sum problem (as well as the more difficult 0-1 Knapsack[1]) can be solved in pseudopolynomial time $\mathcal{O}(nt) = \mathcal{O}(n^2\Delta)$ by a standard dynamic program [Bel57] (see the textbook [KPP04] for an extensive treatment of related problems and algorithms). We study the fine-grained complexity of Subset Sum in the pseudopolynomial-time regime:

How fast can we solve Subset Sum *with n input integers bounded by Δ?*

Reference	**Time in n, Δ**	**Time in n, t, Δ**
Bellman [Bel57]	$\mathcal{O}(n^2\Delta)$	$\mathcal{O}(nt)$
Pisinger [Pis99]	$\mathcal{O}(n\Delta)$	–
Koiliaris and Xu [KX17]	–	$\widetilde{\mathcal{O}}(n + t\sqrt{n})$
Bringmann [Bri17]	–	$\widetilde{\mathcal{O}}(n + t)$
Chen, Lian, Mao, and Zhang [CLMZ24c]	$\widetilde{\mathcal{O}}(n + \sqrt{n}\Delta)$	$\widetilde{\mathcal{O}}(n + \sqrt{\Delta t})$
Galil and Margalit [GM91] + [Bri17]$^\triangle$	$\widetilde{\mathcal{O}}(n + \Delta^{3/2})$	–
Eisenbrand and Weismantel [EW20] + [Bri17]	$\widetilde{\mathcal{O}}(n + \Delta^{2})$	–
Polak, Rohwedder, and Węgrzycki [PRW21]	$\widetilde{\mathcal{O}}(n + \Delta^{5/3})$	–
Chen, Lian, Mao, and Zhang [CLMZ24b]; [Jin23]	$\widetilde{\mathcal{O}}(n + \Delta^{3/2})$	–
Chen, Lian, Mao, and Zhang [CLMZ24c]$^\triangle$	$\widetilde{\mathcal{O}}(n + \Delta^{5/4})$	–
This work	$\mathcal{O}(n + \Delta^{1+o(1)})$	–

Table 1: Pseudopolynomial-time algorithms for Subset Sum (possibly randomized). Entries marked with $^\triangle$ require distinct $a_1, \ldots, a_n$. Throughout this paper, $\widetilde{\mathcal{O}}(T)$ denotes $\mathcal{O}(T \cdot (\log T)^{\mathcal{O}(1)})$.

Work on this question dates back more than three decades. Early approaches were based on additive number theory and combinatorics. In particular, the existence of long arithmetic progressions in subset sums [AF88, Fre93, Sár94] was used to design faster algorithms in certain regimes [Fre88, CFG89, Fre90, GM91, Cha99a]. For example, Chaimovich, Freiman, and Galil's algorithm runs in $\widetilde{\mathcal{O}}(n + \Delta)$ time when $\Delta^{3/2}\sqrt{\log \Delta} \ll t \le \frac{1}{2}\sum_{i=1}^n a_i$ and all a_i are distinct [CFG89]. See Table 2 in Section 1.3 and the survey [Cha99b] for further results of this kind.

Later, Pisinger [Pis99] solved Subset Sum in $\mathcal{O}(n\Delta)$ time via a refined dynamic program, achieving the first general improvement over the baseline $\mathcal{O}(n^2\Delta)$ time complexity.

[1] In 0-1 Knapsack, given the knapsack capacity $t \in \mathbb{Z}_{\ge 0}$ and n items with weights $w_1, \ldots, w_n \in \mathbb{Z} \cap [0, \Delta]$ and profits $p_1, \ldots, p_n \in \mathbb{Z}_{\ge 0}$, the task is to find $(x_1, \ldots, x_n) \in \{0,1\}^n$ that maximizes $\sum_{i=1}^n p_i x_i$ subject to $\sum_{i=1}^n w_i x_i \le t$.

Using fast Fourier transform (FFT) with divide and conquer, Koiliaris and Xu [KX17, KX19] solved Subset Sum in $\widetilde{\mathcal{O}}(n + t\sqrt{n})$ deterministic time. This was improved in a landmark result of Bringmann [Bri17] to an $\widetilde{\mathcal{O}}(n + t)$-time randomized algorithm (derandomized by Chan [Cha26]).

Recent advances in Subset Sum algorithms came from a blend of proximity bounds (e.g., [EW20]), FFT, and additive combinatorial structure (e.g., [BW21, GM91]). The previous state of the art was an $\widetilde{\mathcal{O}}(n + \Delta^{3/2})$-time algorithm by Chen, Lian, Mao, and Zhang [CLMZ24b] (and independently by the unpublished note [Jin23]). Under the assumption that $a_1, \ldots, a_n$ are distinct, the same authors further improved the time complexity to $\widetilde{\mathcal{O}}(n + \Delta^{5/4})$ [CLMZ24c, CLMZ26]. They also achieved $\widetilde{\mathcal{O}}(n + \sqrt{\Delta t})$ time without the distinctness assumption. See Table 1 for a summary of known results.

Despite the exciting progress, the answer to the following question remains elusive:

Main open problem: Can Subset Sum *be solved in* $\widetilde{\mathcal{O}}(n + \Delta)$ *time?*

This question has been repeatedly asked in the literature on fine-grained algorithms for combinatorial optimization, e.g., [ABJ+19, BW21, PRW21, ABHS22a, BC22, Cas24, Bri24, Jin24, CLMZ24a, FW25, CLMZ24c, CLMZ25, Che25], and was described as "the most important open question in this line of research" in a survey talk given by Bringmann in 2021[2].

Under the Strong Exponential Time Hypothesis (SETH) [IPZ01], Subset Sum cannot be solved in $\widetilde{\mathcal{O}}(n + \Delta^{0.99})$ time. In fact, even a $2^{o(n)} \cdot t^{1-\delta}$-time algorithm (for any positive constant δ) would refute SETH, as shown by Abboud, Bringmann, Hermelin, and Shabtay [ABHS22b].

1.2 Our result

We *almost* (up to a subpolynomial factor) answer the main open problem and achieve SETH-optimality. With little extra effort, our algorithm can solve the following slightly more general problem (sometimes called Bounded Subset Sum), and can find a witness when one exists.

Problem 2. Single-Constraint Bounded Integer Feasibility (search version)
Input: Integers $a_i \in [-\Delta, \Delta]$ and $u_i \in [1, U]$ for $1 \le i \le n$, and an integer b ($|b| \le nU\Delta$).
Task: Find a solution to $\sum_{i=1}^{n} a_i x_i = b, x_i \in \{0, 1, \ldots, u_i\}$, or report that none exists.

Theorem 1.1. *There is a deterministic algorithm that solves Problem 2 (which generalizes* Subset Sum*) in* $\mathcal{O}(n + \Delta^{1+o(1)})$ *time (in the word-RAM model with* $\Theta(\log(nU\Delta))$*-bit words).*

Our algorithm runs in time $\mathcal{O}(n + C_d \Delta^{1+\mathcal{O}(\frac{1}{d})})$, where $d \ge 1$ is a dimension parameter as discussed below, and C_d is a (fast-growing) computable function of d. Letting d grow sufficiently slowly with Δ yields Theorem 1.1.

Our approach for Subset Sum lies at the intersection of additive combinatorics and computational geometry. We use the inverse theorem of Conlon, Fox, and Pham [CFP23] to model (almost all of) the input integers $a_1, \ldots, a_n$ as elements of a generalized arithmetic progression (GAP) $f(B \cap \mathbb{Z}^d)$ with $f\colon \mathbb{R}^d \to \mathbb{R}$ linear and a box $B \subseteq \mathbb{R}^d$, such that, writing $a_i = f(\boldsymbol{x}_i)$, $\boldsymbol{x}_i \in B \cap \mathbb{Z}^d$, every integer point deep inside the zonotope $Z = \{\sum_i \varepsilon_i \boldsymbol{x}_i : \varepsilon_i \in [0, 1]\}$ must be expressible as a subset sum of $(\boldsymbol{x}_i)_i$.

Then, the remaining possible subset sums correspond to points in a thin shell near the boundary of the zonotope Z. To compute them, we follow the polytope approximation literature (e.g., [AdFM17, AdFM23]) to efficiently cover the shell by a collection of caps. Due to convexity, we can

[2] Link: https://youtu.be/lTw0flxXuKE?t=2473

compute the subset sums in each cap in near-linear time by adapting Bringmann's algorithm [Bri17]. The total time is thus near-linear in the shell volume.

To control the shell volume, we remove the redundancy caused by the non-injectivity of f: define the lattice $\Lambda = \ker(f) \cap \mathbb{Z}^d$, and measure shell volume in the quotient space $\mathbb{R}^d/\Lambda$. We then develop and apply a quotient variant of cap covering. We also have to control the thickness of the shell when Z is not fat. This is done by repeatedly applying [CFP23] to partition the input integers, so that groups with thin zonotopes also have thin shells. See Section 2 for a more detailed overview.

1.3 Related work and open questions

It remains an open question to improve our algorithm to $\mathcal{O}((n+\Delta)\operatorname{poly}\log\Delta)$ time. To tackle it, one may start with the perhaps easier Partition problem, which is the special case of Subset Sum with $t = \frac{1}{2}\sum_{i=1}^{n} a_i$. Analogous questions for several related problems have already been resolved: Unbounded Subset Sum[3] can be solved in $\widetilde{\mathcal{O}}(n+\Delta)$ time [JR19]. Modular Subset Sum[4] can be solved in $\widetilde{\mathcal{O}}(n+m)$ time [ABJ+19, ABB+21, CI21, BN21, Pot21]. Chen, Lian, Mao, and Zhang [CLMZ24a, CLMZ25] gave a $(1-\varepsilon)$-approximation algorithm for Partition in $\widetilde{\mathcal{O}}(n+1/\varepsilon)$ time.

Our techniques might lead to further progress on related problems in combinatorial optimization. In particular, several previous papers identified the lack of an $\widetilde{\mathcal{O}}(n+\Delta)$-time Subset Sum algorithm as a barrier for resolving their open problems: Bringmann and Wellnitz [BW21, Section 6] on dense Subset Sum instances in the super-linear-time regime, Fischer and Wennmann [FW25, Page 4] on minimizing tardy processing time on a single machine, and Chen, Hu, Mao, and Zhang [CHMZ26, Section 1.1] on removing the nondegeneracy assumption in their algorithm for solving multidimensional dense Subset Sum instances.

For 0-1 Knapsack, Bringmann [Bri24] and Jin [Jin24] gave $\widetilde{\mathcal{O}}(n+\Delta^2)$-time exact algorithms, and Chen, Lian, Mao, and Zhang [CLMZ24d] and Mao [Mao24] gave $\widetilde{\mathcal{O}}(n+1/\varepsilon^2)$-time approximation algorithms. See the survey of Lin Chen [Che25].

Bringmann's $\widetilde{\mathcal{O}}(n+t)$-time algorithm [Bri17, Cha26] actually computes *all* attainable subset sums in $[0,t]$. This all-target feature might seem impossible in $\mathcal{O}(n+\Delta^{1+o(1)})$ time, since the output size may well exceed Δ. Interestingly, the output set is actually expressible as the union of $\mathcal{O}(\Delta\log(n\Delta))$ arithmetic progressions of the same common difference (Lemma D.1). Can we efficiently compute such a compact representation? Such an algorithm would have positive implications for Węgrzycki's Bounded Support Subset Sum question [Węg25, Problem 2]; see Appendix D for details.

Subset Sum has been studied in the exponential-time regime and in the average-case setting as well; see e.g., [HS74, Ned26, LO85, JW26] and the references therein.

Reference	Time	Assumptions (in addition to *)
Chaimovich, Freiman, and Galil [CFG89]	$\widetilde{\mathcal{O}}(n+\Delta^2/n^2)$	$t \gg \Delta^{3/2}\sqrt{\log\Delta}$
Freiman [Fre90]	$\mathcal{O}_\varepsilon(n)$	$t \gg \Delta^{5/3}(\log\Delta)^{4/3}$, $S \geq \varepsilon n\Delta$
Chaimovich [Cha99a]	$\mathcal{O}(n^{7/4})$	$n \gg \sqrt{\Delta\log\Delta}$, $\lvert t - S/2\rvert \ll n^2$
Galil and Margalit [GM91]	$\widetilde{\mathcal{O}}(n+\Delta^2/n^2)$	$nt \gg \sqrt{\Delta}\,S\log\Delta$
Bringmann and Wellnitz [BW21]	$\widetilde{\mathcal{O}}(n)$	$n^2 t \gg \Delta S\log(2n)$

Table 2: A non-exhaustive list of previous algorithms for Subset Sum (decision version) under the following assumptions*: the input integers $a_1,\dots,a_n$ are distinct, and $t \leq S/2$, where $S \coloneqq \sum_{i=1}^{n} a_i$. Here $f \gg g$ means $f \geq Cg$ for an absolute positive constant C.

[3] Unbounded Subset Sum asks whether there exists $(x_1,\dots,x_n) \in \mathbb{Z}_{\geq 0}^n$ such that $\sum_{i=1}^{n} a_i x_i = t$.

[4] Modular Subset Sum asks whether there exists $(x_1,\dots,x_n) \in \{0,1\}^n$ such that $\sum_{i=1}^{n} a_i x_i \equiv t \pmod m$.

2 Technical Overview

Under construction.

3 Preliminaries

3.1 Basic notation

Let $[n] = \{1, 2, \ldots, n\}$. Let $\log x$ denote $\log_2 x$. We assume $\Delta \geq 2$ to simplify some mathematical expressions.

Use $\ll_d, \mathcal{O}_d(\cdot)$, etc., to hide factors depending on a parameter d. Such factors are also denoted by C_d or c_d, etc. They are always computable functions of d.

Let $\widetilde{\mathcal{O}}(T)$ and $\widetilde{\mathcal{O}}_d(T)$ denote $\mathcal{O}(T \cdot (\log T)^{\mathcal{O}(1)})$ and $\mathcal{O}_d(T \cdot (\log T)^{\mathcal{O}_d(1)})$, respectively.

A (finite) *multiset* A is represented by a list of pairs of element x together with its *multiplicity* $\mu_A(x)$, over all $x \in \operatorname{supp} A \coloneqq \{x : \mu_A(x) > 0\}$. It has *size* $|A| = \sum_x \mu_A(x)$. If $\mu_A(x) \leq \mu_B(x)$ for all x, then write $A \subseteq B$, and define their *multiset difference* $B \setminus A$ by the multiplicity function $\mu_{B \setminus A} \coloneqq \mu_B - \mu_A$. The *multiset union (sum)* $A \sqcup B$ is defined by $\mu_{A \sqcup B} \coloneqq \mu_A + \mu_B$. For a multiset A, the notation $\{f(x) : x \in A\}$ (also written as $f(A)$) builds a multiset that preserves the multiplicities.

The *set of subset sums* of a (multi)set A is

$$\Sigma(A) = \Big\{ \sum_{a \in B} a : B \subseteq A \Big\}.$$

The *sumset* (or *Minkowski sum*) of two sets A, B is $A + B = \{a + b : a \in A, b \in B\}$. Define $A - B$ analogously. For $k \in \mathbb{Z}_{>0}$, the *k-fold sumset* of A is $kA \coloneqq \underbrace{A + A + \cdots + A}_{k \text{ times}}$.

We denote the set dilation by $k \cdot A = \{ka : a \in A\}$ to distinguish it from the k-fold sumset kA, but *with the following exceptions* to avoid notational clutter: For a lattice L or a convex set K, we may write rL, rK for their r-factor dilations.

For $r = p/q \in \mathbb{Q}$ in lowest terms, define $\mathsf{maxbitlen}(r) = \max\{\log(2 + |p|), \log(2 + |q|)\}$. Then, r can be encoded as a fraction using $\mathcal{O}(\mathsf{maxbitlen}(r))$ bits. We extend this notation to other objects in Section 6.

3.2 Sparse convolution

Generalizing the fast Fourier transform (FFT), the following algorithm multiplies sparse polynomials in near-linear output-sensitive time.

Lemma 3.1 (Nonnegative sparse convolution [BFN22, Theorem 1.1])**.** *Given polynomials $f(x), g(x)$ (represented as lists of non-zero terms) with degrees at most N and all coefficients in $\mathbb{Z} \cap [0, U]$, their product $h(x) = f(x)g(x)$ can be computed in $\mathcal{O}(t \operatorname{poly}\log(NU))$ time deterministically, where t denotes the number of non-zero terms in $h(x)$.*

In particular, the sumset of $A, B \subseteq [N]$ can be computed in $\mathcal{O}(|A + B| \operatorname{poly}\log N)$ time.

3.3 Geometry and lattices

Write $\operatorname{vol}_d(\cdot)$ for d-dimensional volume. For $A \subseteq \mathbb{R}^d$, write $\operatorname{conv}(A)$ and $\operatorname{span}(A)$ for the convex hull and linear span of A. A *(convex) polytope* is the convex hull of finitely many points. A *vertex* of

a polytope K is a point $\boldsymbol{v} \in K$ such that $\boldsymbol{v} \notin \operatorname{conv}(K \setminus \{\boldsymbol{v}\})$. A *convex body* in $\mathbb{R}^d$ is a compact convex set in $\mathbb{R}^d$ with nonempty interior.

Let $\langle \cdot, \cdot \rangle$ denote the Euclidean inner product. For a linear function $f(\boldsymbol{x}) = \langle \boldsymbol{x}, \boldsymbol{u} \rangle$ and a bounded set S, write $\operatorname{width}_f(S) = \sup f(S) - \inf f(S)$.

Denote the *Minkowski difference* (or *erosion*) by $A \ominus B = \{\boldsymbol{x} : \boldsymbol{x} + B \subseteq A\}$. If A is convex, then $A \ominus B$ is convex.

Denote the line segment joining $\boldsymbol{x}, \boldsymbol{y}$ by $[\boldsymbol{x}, \boldsymbol{y}] = \{t\boldsymbol{x} + (1-t)\boldsymbol{y} : 0 \le t \le 1\}$. The *zonotope* generated by a multiset $X = (\boldsymbol{x}_i)_{i=1}^{|X|}$ is

$$\mathcal{Z}(X) = \sum_{i=1}^{|X|} [0, \boldsymbol{x}_i] = \Big\{ \sum_{i=1}^{|X|} \varepsilon_i \boldsymbol{x}_i : 0 \le \varepsilon_i \le 1 \Big\}.$$

Similarly, define the *centered zonotope* by $\mathcal{Z}_0(X) = \sum_{i=1}^{|X|} [-\boldsymbol{x}_i/2, \boldsymbol{x}_i/2]$.

A matrix $B = (\boldsymbol{b}_1, \dots, \boldsymbol{b}_{d'}) \in \mathbb{R}^{d \times d'}$ with linearly independent columns generates the *lattice* $\mathcal{L}(B) = B\mathbb{Z}^{d'} = \{\sum_{i=1}^{d'} n_i \boldsymbol{b}_i : n_i \in \mathbb{Z}\}$ of *rank* d'. Its *determinant* is $\det \mathcal{L}(B) = \sqrt{\det(B^\top B)}$, which equals $|\det B|$ when $d' = d$. More generally, for arbitrary $X \subseteq \mathbb{R}^d$, denote

$$\mathcal{L}(X) = \Big\{ \sum_{i=1}^{m} n_i \boldsymbol{x}_i : m \ge 0,\ \boldsymbol{x}_i \in X,\ n_i \in \mathbb{Z} \Big\}.$$

If $X \subseteq \mathbb{Z}^d$, then $\mathcal{L}(X)$ is a lattice. In this paper, we only deal with rational lattices $L \subseteq \mathbb{Q}^d$, and our algorithm represents a rational lattice by its Hermite normal form (HNF) (Section 6.3) by default.

For a lattice $\Lambda \subseteq \mathbb{R}^d$, write π_Λ for the quotient map $\mathbb{R}^d \to \mathbb{R}^d/\Lambda$. We also write vol_d for the Haar measure on $\mathbb{R}^d/\Lambda$, normalized so that $\operatorname{vol}_d(\pi_\Lambda(A))$ equals the Euclidean volume $\operatorname{vol}_d(A)$ for every measurable $A \subseteq \mathbb{R}^d$ on which π_Λ is injective.

Lemma 3.2 (cf. [TV06, Eq. (3.10)])**.** *Let $\Lambda, \Gamma \subseteq \mathbb{R}^d$ be lattices, $K \subseteq \mathbb{R}^d$ be an origin-symmetric compact convex set, and $s \ge 1$. Then,*

$$|\pi_\Lambda(sK \cap \Gamma)| \le (2s+1)^d |\pi_\Lambda(K \cap \Gamma)|.$$

In particular, $|sK \cap \Gamma| \le (2s+1)^d |K \cap \Gamma|$.

Proof. Let $q = \lceil 2s \rceil$. Partition $sK \cap \Gamma$ into its at most q^d residue classes modulo $q\Gamma$. Choose a representative $\boldsymbol{a}$ in each nonempty class. For every $\boldsymbol{x}$ in the class of $\boldsymbol{a}$, since $\boldsymbol{x} - \boldsymbol{a} \in 2sK \subseteq qK$, we have $(\boldsymbol{x} - \boldsymbol{a})/q \in K \cap \Gamma$. Since $\pi_\Lambda((\boldsymbol{x} - \boldsymbol{a})/q)$ determines $\pi_\Lambda(\boldsymbol{x})$, summing over all q^d residue classes gives $|\pi_\Lambda(sK \cap \Gamma)| \le q^d |\pi_\Lambda(K \cap \Gamma)|$. The claim then follows from $q \le 2s + 1$. □

3.4 Progressions and s-generating sets

We use a slight variant of convex progressions (the standard definition takes $\Gamma = \mathbb{Z}^d$):

Definition 3.3 (Progression $f(K \cap \Gamma)$)**.** Let Γ be a rank-d lattice, $K \subseteq \operatorname{span}(\Gamma)$ be an origin-symmetric compact convex set such that $\dim \operatorname{span}(K \cap \Gamma) = d$, and $f\colon \operatorname{span}(\Gamma) \to \mathbb{R}^k$ be a linear map. The set $f(K \cap \Gamma)$ is called a *progression* of *rank* d.

By abuse of notation, a progression $f(K \cap \Gamma)$ is understood to include its representation (K, Γ, f), since the same set may have different representations.

For $s \ge 1$, we say the progression $f(K \cap \Gamma)$ is *s-proper* if f is injective on $sK \cap \Gamma$ (we abbreviate 1-proper to *proper*).

The progression $f(K \cap \Gamma)$ with f being the identity map is simply referred to as $K \cap \Gamma$.

An origin-symmetric generalized arithmetic progression (GAP) is a progression $f(K \cap \mathbb{Z}^d)$ where K is a box $\prod_{i=1}^{d}[-m_i, m_i]$ with $m_i \in \mathbb{Z}_{>0}$. More details on GAPs are given in Section 5.1 where they are needed.

Definition 3.4 (s-generating)**.** Let $f(K \cap \Gamma)$ be a proper progression (Definition 3.3). For $s \geq 1$, if a multiset X of vectors in $K \cap \Gamma$ satisfies $\Sigma(X) \supseteq (\boldsymbol{\tau} + sK) \cap \Gamma$ for some $\boldsymbol{\tau} \in \Gamma$, then we say X is *s-generating* for $K \cap \Gamma$, and we also say the multiset $f(X)$ is *s-generating* for $f(K \cap \Gamma)$.

Lemma 3.5. *Let $K \cap \Gamma$ be a progression of rank $d \geq 1$, and let X be a multiset supported on $K \cap \Gamma$. If a submultiset $X' \subseteq X$ is d-generating for $K \cap \Gamma$, then $(\mathcal{Z}(X) \ominus |X'|K) \cap \Gamma \subseteq \Sigma(X)$.*

Proof. Let $\boldsymbol{y} \in (\mathcal{Z}(X) \ominus |X'|K) \cap \Gamma$. We will prove $\boldsymbol{y} \in \Sigma(X)$ by finding $\boldsymbol{z}' \in \Sigma(X \setminus X')$ such that $\boldsymbol{y} - \boldsymbol{z}' \in \Sigma(X')$.

By Definition 3.4, there is $\boldsymbol{\tau} \in \Gamma$ with $(\boldsymbol{\tau} + dK) \cap \Gamma \subseteq \Sigma(X')$. Since $\mathbf{0} \in dK \cap \Gamma$, we have $\boldsymbol{\tau} \in \Sigma(X')$. Since $\boldsymbol{\tau} \in \Sigma(X')$, $\mathcal{Z}(X') - \boldsymbol{\tau}$ is a Minkowski sum of $|X'|$ segments of the form $[0, \boldsymbol{x}]$ or $[-\boldsymbol{x}, 0]$ where $\boldsymbol{x} \in X'$. Each segment is contained in K, so $\mathcal{Z}(X') - \boldsymbol{\tau} \subseteq |X'|K$. By definition of $\boldsymbol{y}$, this implies $\boldsymbol{y} + (\mathcal{Z}(X') - \boldsymbol{\tau}) \subseteq \mathcal{Z}(X)$, so

$$\boldsymbol{z} \coloneqq \boldsymbol{y} - \boldsymbol{\tau} \in \mathcal{Z}(X) \ominus \mathcal{Z}(X') = \mathcal{Z}(X \setminus X'),$$

where we used the identity $(K_1 + K_2) \ominus K_2 = K_1$ for compact convex sets K_1, K_2 [Sch14, Lemma 3.1.11].

Write $X \setminus X' = (\boldsymbol{x}_i)_{i=1}^{r}$, so $\{\boldsymbol{\alpha} \in [0,1]^r : \boldsymbol{z} = \sum_i \alpha_i \boldsymbol{x}_i\}$ is a non-empty polyhedron. Choose $\boldsymbol{\alpha}$ to be one of its vertices, which can have at most d fractional coordinates. Rounding $\boldsymbol{\alpha}$ down to $\boldsymbol{\alpha}' \in \{0,1\}^r$ gives $\boldsymbol{z}' = \sum_i \alpha_i' \boldsymbol{x}_i \in \Sigma(X \setminus X')$ with $\boldsymbol{z} - \boldsymbol{z}' \in dK$. Thus, $\boldsymbol{y} - \boldsymbol{z}' \in (\boldsymbol{\tau} + dK) \cap \Gamma \subseteq \Sigma(X')$, and hence $\boldsymbol{y} \in \Sigma(X)$. □

3.5 AP decompositions

Definition 3.6 (AP decomposition)**.** For finite $A \subseteq \mathbb{Z}$ and $r \in \mathbb{Z}_{>0}$, the *AP decomposition of A with common difference r* is the unique decomposition of A into disjoint arithmetic progressions of common difference r, each maximal in A.

The *size* of the AP decomposition is the number of arithmetic progressions contained, which equals $|A \setminus (A + r)|$ (since their starting points are precisely the elements of $A \setminus (A + r)$).

Algorithmically, the AP decomposition is represented by the set $A \bmod r \subseteq \{0, \ldots, r-1\}$ as a sorted list, together with a list $\mathcal{D}_i$ for every $i \in A \bmod r$, which consists of the pairs (a, ℓ) representing the arithmetic progressions $\{a, a+r, \ldots, a+(\ell-1)r\}$ with $a \equiv i \pmod r$, in increasing order of a.

For example, the AP decomposition of $A = \{4, 5, 6, 7, 9, 10\}$ with common difference $r = 2$ is $\{4, 6\} \sqcup \{10\} \sqcup \{5, 7, 9\}$. It has size $|A \setminus (A + r)| = |\{4, 5, 10\}| = 3$.

Using binary search, it is straightforward to implement the following operations on AP decompositions. We omit the proof here.

Lemma 3.7.

- *Initialization: Given integer arithmetic progressions $A_1, \ldots, A_k$ of the same common difference r, we can construct the AP decomposition of $\bigcup_{i=1}^{k} A_i$ of common difference r in $\mathcal{O}(k \log k)$ time.*

Given a size-k AP decomposition of an integer set A, we can perform:

- *Membership: given an integer a, decide whether $a \in A$ in $\mathcal{O}(\log k)$ time.*
- *Restriction: given an interval $[\ell, u]$, compute the AP decomposition of $A \cap [\ell, u]$ in $\mathcal{O}(k)$ time.*

4 Main Algorithm

In this section, we formally state all the algorithmic components, and use them to solve Problem 2 in $\mathcal{O}(n + \Delta^{1+o(1)})$ time, thereby proving Theorem 1.1.

To solve Problem 2, we first use a standard proximity argument to reduce Problem 2 to the Subset Sum problem on an input multiset of size $\mathrm{poly}(\Delta)$. The step reduces the multiplicities of multisets to $\mathcal{O}(\log \Delta)$ bits, in order to simplify the time complexity for manipulating multisets. This proximity argument can be found in e.g., [PRW21, Lemma 2.1], and was also used in [CLMZ24b, Observation 6.1] for their $\widetilde{\mathcal{O}}(n + \Delta^{1.5})$-time algorithm. We omit the proof here.

Lemma 4.1. *In $\mathcal{O}(n+\Delta)$ time, an instance of Problem 2 can be reduced to a Subset Sum instance with an input multiset A supported on $\{1, 2, \dots, \Delta\}$ with $|A| \le \mathcal{O}(\Delta^2)$.*

We henceforth work with the Subset Sum instance produced by Lemma 4.1.

4.1 Partitioning into groups

We will iteratively apply the Conlon–Fox–Pham theorem [CFP23] to partition the input integer multiset A into a logarithmic number of groups $A_1, \dots, A_k$ and a small exceptional part A^{ex}. Each group A_i is embedded into a low-dimensional progression $f(H_i \cap \Gamma_i)$, with $A_i = f(X_i)$ and $\mathrm{supp}(X_i) \subseteq H_i \cap \Gamma_i$, so that the vector multiset X_i satisfies a relative fatness condition and contains a small generating submultiset X_i'. These conditions (formally stated in Eq. (4.1) below) are useful for our single-group algorithm. We also ensure certain extra compatibility conditions between the groups, which are useful for merging the answers from the groups.

We now formally state the conditions required for each group. Let Γ be a lattice, $H \subseteq \mathrm{span}\,\Gamma$ be an origin-symmetric polytope, $f\colon \mathrm{span}\,\Gamma \to \mathbb{R}$ be a linear map, and $X' \subseteq X$ be two nonempty multisets with $\mathrm{supp}(X) \subseteq H \cap \Gamma$. For $\lambda \ge 1$, define the following conditions (recalling that $\mathcal{Z}_0(\cdot)$ denotes the centered zonotope):

$$f(\mathrm{supp}\, X) \subseteq [1, \Delta] \cap \mathbb{Z}, \text{ and } f(H \cap \Gamma) \text{ is a proper progression (see Definition 3.3)}, \tag{4.1a}$$

$$H \subseteq \frac{\lambda}{|X|} \mathcal{Z}_0(X), \tag{4.1b}$$

$$X' \text{ is } (\mathrm{rank}\,\Gamma)\text{-generating for } H \cap \Gamma \text{ (see Definition 3.4)}. \tag{4.1c}$$

We now state the main partitioning lemma. Write Π_U for the orthogonal projection onto the subspace U.

Lemma 4.2. *Let $\boldsymbol{e}_1, \dots, \boldsymbol{e}_d$ be the standard basis of $\mathbb{R}^d$, and for $1 \le d' \le d$ identify $\mathbb{R}^{d'}$ with the subspace $\mathrm{span}\{\boldsymbol{e}_1, \dots, \boldsymbol{e}_{d'}\}$ of $\mathbb{R}^d$.*

Given a nonempty multiset A of $\Delta^{\mathcal{O}(1)}$ integers in $[1, \Delta]$, we can construct in $\widetilde{\mathcal{O}}_d(\Delta^{1+2/d})$ time a partition into submultisets,

$$A = A^{\mathrm{ex}} \sqcup \bigsqcup_{i=1}^{k} A_i, \text{ where } |A^{\mathrm{ex}}| = \mathcal{O}_d(\Delta^{1/d} \log^2 \Delta),\ k = \mathcal{O}_d(\log \Delta),$$

integers $1 \le d_1 \le d_2 \le \cdots \le d_k \le d$, and nonempty vector multisets X_i', X_i with $X_i' \subseteq X_i$, such that

$$\mathrm{supp}(X_i) \subseteq \left(\mathbb{Z} \cap [-\Delta^{\mathcal{O}(1)}, \Delta^{\mathcal{O}(1)}]\right)^{d_i} \text{ and } f(X_i) = A_i,$$

where $f\colon \mathbb{R}^d \to \mathbb{R}$ *is given by* $f(\boldsymbol{x}) = x_1$*, and, writing*

$$\Gamma_i \coloneqq \mathcal{L}(X_i) \quad \textit{and} \quad H_i \coloneqq \operatorname{conv}\Big(\pm \bigcup_{j=i}^{k} \Pi_{\mathbb{R}^{d_i}}(\operatorname{supp} X_j)\Big),$$

for $i \in [k]$*, the following properties hold:*

(i) *The conditions in* (4.1) *hold for* $(\Gamma, H, f, X', X) = (\Gamma_i, H_i, f|_{\operatorname{span}(\Gamma_i)}, X_i', X_i)$ *with* $\lambda = \mathcal{O}_d(\log \Delta)$*. Moreover, the algorithm also returns* $\boldsymbol{\tau}_i \in \Gamma_i$ *witnessing* (4.1c)*, i.e.,* $\boldsymbol{\tau}_i + (\Gamma_i \cap (\operatorname{rank} \Gamma_i) H_i) \subseteq \Sigma(X_i')$*.*

(ii) $|X_i'| = \mathcal{O}_d(\Delta^{1/d})$*.*

(iii) *For each* $i \in [k]$*,* $\bigcup_{j=i}^{k} \Pi_{\mathbb{R}^{d_i}}(\operatorname{supp} X_j) \subseteq \Gamma_i$*.*

(iv) *For* $1 \le i \le k$*,* $\sum_{j=1}^{i} |A_j| = \mathcal{O}_d(|A_i| \log \Delta)$*.*

We remark that the computed vectors $\boldsymbol{\tau}_i$ are needed only for the witness recovery steps.

We prove Lemma 4.2 in Section 7 by iteratively applying the algorithmic Conlon–Fox–Pham theorem (Section 5).

4.2 Solving one group and merging groups

We now state the algorithm for computing the subset sums of one group and representing them as an AP decomposition (Definition 3.6).

Lemma 4.3. *Let* $X' \subseteq X$ *be two multisets of vectors in* $\mathbb{Z}^d$*. Suppose* (4.1) *holds for* X', X*,* $\Gamma = \mathcal{L}(X), H = \operatorname{conv}(\pm \operatorname{supp} X)$*, a linear map* $f\colon \mathbb{R}^d \to \mathbb{R}$ *with integer coefficients, and an unknown* $\lambda \ge 1$*. Then, we have the following algorithms:*

- $\mathsf{OneGroup}(X, X', f)$ *computes an AP decomposition of* $f(\Sigma(X))$ *of size* $\mathcal{O}_d(\lambda |X'| \Delta)$*.*
- $\mathsf{OneGroupWitness}(X, X', f, \boldsymbol{\tau}, t)$*, where* $\boldsymbol{\tau} + (\Gamma \cap (\operatorname{rank} \Gamma) H) \subseteq \Sigma(X')$ *and* $t \in f(\Sigma(X))$*, computes a submultiset* $W \subseteq X$ *such that* $\sum_{\boldsymbol{w} \in W} f(\boldsymbol{w}) = t$*.*

Both algorithms run in $\widetilde{\mathcal{O}}_d(\lambda |X'| \Delta)$ *time, provided the vector coordinates and multiplicities in* X *and coefficients of* f *have* $\mathcal{O}(\log \Delta)$ *bits.*

We prove Lemma 4.3 in Section 11.

We now state the algorithm for merging the subset sums of one group $f(X)$ with an accumulated partial result that consists of subset sums of $f(Y)$ restricted to an interval J. The algorithm returns the combined subset sums $f(\Sigma(X \sqcup Y))$ restricted to an interval I, defined so that any pair of $a \in f(\Sigma(X))$ and $b \in f(\Sigma(Y))$ with $a + b \in I$ must satisfy $b \in J$, and thus b is included in the input. The algorithm is efficient if the length of J is not much larger than $|X| \cdot \Delta$.

Lemma 4.4. *Let* $X' \subseteq X$ *be two multisets of vectors in* $\mathbb{Z}^d$*, and let* Y *be a multiset of vectors in* $\mathcal{L}(X)$*. Suppose* (4.1) *holds for* $X', X, \Gamma = \mathcal{L}(X)$*,* $H = \operatorname{conv}(\pm \operatorname{supp}(X \sqcup Y))$*, a linear map* $f\colon \mathbb{R}^d \to \mathbb{R}$ *with integer coefficients, and an unknown* $\lambda \ge 1$*. Assume* $S_X = f(\Sigma(X))$ *is implicitly given by access to its AP decomposition.*

Let $J \subseteq f(\mathcal{Z}(Y))$ *be a nonempty closed interval (with integer endpoints), and* $S_Y = f(\Sigma(Y)) \cap J$ *be implicitly given by access to its AP decomposition. Let* I *be a closed interval (with integer endpoints) satisfying*

$$I \subseteq f(\mathcal{Z}(X \sqcup Y)) \textit{ and } (I - f(\mathcal{Z}(X))) \cap f(\mathcal{Z}(Y)) \subseteq J. \tag{4.2}$$

Then we have the following algorithms:

- $\mathsf{Merge}(X, X', Y, f, S_X, J, S_Y, I)$ *computes an AP decomposition of*

$$(S_X + S_Y) \cap I = f(\Sigma(X \sqcup Y)) \cap I,$$

  where the equality follows from (4.2). The decomposition has size $\mathcal{O}_d\Big(\lambda|X'|\left(\Delta + \frac{|J|}{|X|}\right)\Big)$*, where* $|J|$ *denotes interval length.*
- $\mathsf{MergeWitness}(X, X', Y, f, S_X, J, S_Y, I, \boldsymbol{\tau}, t)$*, where* $\boldsymbol{\tau}$ *satisfies* $\boldsymbol{\tau} + (\Gamma \cap (\operatorname{rank}\Gamma)H) \subseteq \Sigma(X')$ *and* $t \in (S_X + S_Y) \cap I$*, computes* $a \in S_X$ *and* $b \in S_Y$ *with* $a + b = t$*.*

Both algorithms run in $\widetilde{\mathcal{O}}_d\Big(\lambda|X'|\left(\Delta + \frac{|J|}{|X|}\right)\Big)$ *time, provided the vector coordinates and multiplicities in* X, Y*, and coefficients of* f *have* $\mathcal{O}(\log \Delta)$ *bits.*

We prove Lemma 4.4 in Section 12.

4.3 Main algorithm

Algorithm 1: Main algorithm for Subset Sum with target t (decision version)

Function $\mathsf{SubsetSum}_d(A, t)$**:**
  Partition $A = A^{\mathrm{ex}} \sqcup \bigsqcup_{i=1}^{k} A_i$ by Lemma 4.2, and compute $\Sigma(A^{\mathrm{ex}})$ by standard DP
  if $k = 0$ **then return** $\mathbf{1}[t \in \Sigma(A^{\mathrm{ex}})]$
  for $i \in [k]$ **do** compute the interval I_i from Eq. (4.3)
  for $i \in [k]$ **do** find an AP decomposition of $\Sigma(A_i)$ by $\mathsf{OneGroup}(X_i, X_i', f)$ (Lemma 4.3)
  $S_k \leftarrow \Sigma(A_k) \cap I_k$ (Lemma 3.7)
  for $i = k-1, \ldots, 1$ **do**
    $Y_i \leftarrow \bigsqcup_{j=i+1}^{k} \Pi_{\mathbb{R}^{d_i}} X_j$
    $S_i \leftarrow \mathsf{Merge}(X_i, X_i', Y_i, f, \Sigma(A_i), I_{i+1}, S_{i+1}, I_i)$ (Lemmas 4.4 and 3.7)
  return $\mathbf{1}[(t - \Sigma(A^{\mathrm{ex}})) \cap S_1 \neq \varnothing]$ (Lemma 3.7)

We now describe the main algorithm (for a fixed $d \geq 1$), with pseudocode given in Algorithm 1. Use Lemma 4.2 to partition the input multiset A (with $\operatorname{supp}(A) \subseteq \mathbb{Z} \cap [1, \Delta]$) into groups $A_1, A_2, \ldots, A_k$ and a small exceptional part A^{ex}. Compute the subset sums of each group using $\mathsf{OneGroup}$ (Lemma 4.3), and then merge their results by using Merge (Lemma 4.4) successively in decreasing order $A_k, A_{k-1}, \ldots, A_1$. Finally, merge the result with the subset sums of A^{ex} computed by brute-force dynamic programming. After each merging step, restrict the obtained partial subset sums to an interval that excludes those useless values which can never contribute to the target subset sum t.

We now define the intervals. Given the partition in Lemma 4.2, for $i \in [k]$, write

$$W_i = \sum_{a \in A_i} a, \quad W_{\geq i} = \sum_{j=i}^{k} W_j, \quad \text{and } W^{\mathrm{ex}} = \sum_{a \in A^{\mathrm{ex}}} a.$$

We assume $0 \leq t \leq \sum_{a \in A} a$ (otherwise, t is not a subset sum of A). Since A^{ex} and $A_1, \ldots, A_{i-1}$ contribute total sum at most $W^{\mathrm{ex}} + \sum_{j<i} W_j$, the relevant interval for the subset sums of the suffix $A_i \sqcup \cdots \sqcup A_k$ is

$$I_i = [0, W_{\geq i}] \cap \Big[t - W^{\mathrm{ex}} - \sum_{j<i} W_j,\ t\Big], \tag{4.3}$$

which is non-empty due to $0 \le t \le \sum_{a\in A} a$, and satisfies

$$I_{i+1} \supseteq (I_i - [0, W_i]) \cap [0, W_{\ge i+1}] \text{ for all } i \in [k-1]. \tag{4.4}$$

Lemma 4.5. *Given integer $d \ge 1$, a multiset A of integers in $[1, \Delta]$, and an integer t, $\mathsf{SubsetSum}_d(A, t)$ (Algorithm 1) correctly decides whether there is a submultiset $B \subseteq A$ with $\sum_{a\in B} a = t$ in $\widetilde{\mathcal{O}}_d(\Delta^{1+2/d})$ time. In the YES case, one can recover a witness B in the same time complexity.*

Proof.

Correctness. In Algorithm 1, if the partition step returns $k = 0$, then the answer is clearly correct. We now assume $k \ge 1$. For each $i \in [k]$, the partition step (Lemma 4.2) guarantees that the conditions (4.1) hold for $(X_i, X_i', H_i, \mathcal{L}(X_i))$ where $H_i = \operatorname{conv}(\pm \bigcup_{j=i}^{k} \Pi_{\mathbb{R}^{d_i}}(\operatorname{supp} X_j))$ and $\lambda = \mathcal{O}_d(\log \Delta)$. Clearly, the conditions (4.1) remain valid with H_i replaced by its subset $\operatorname{conv}(\pm \operatorname{supp} X_i)$. Thus, Lemma 4.3 applies, so $\mathsf{OneGroup}(X_i, X_i', f)$ correctly computes an AP decomposition of $f(\Sigma(X_i)) = \Sigma(A_i)$.

We now prove by backward induction that the computed sets S_i (represented as AP decompositions) satisfy

$$S_i = \Sigma(A_i \sqcup \cdots \sqcup A_k) \cap I_i \text{ for all } i \in [k].$$

The base case $i = k$ immediately follows from the definition $S_k = \Sigma(A_k) \cap I_k$. Now consider $1 \le i < k$, and assume the displayed inductive hypothesis holds for $i+1$. Since $Y_i = \bigsqcup_{j=i+1}^{k} \Pi_{\mathbb{R}^{d_i}} X_j$ by definition and $f(\boldsymbol{x}) = x_1$ by Lemma 4.2, we have $f(Y_i) = \bigsqcup_{j=i+1}^{k} f(X_j) = A_{i+1} \sqcup \cdots \sqcup A_k$, and the induction hypothesis gives $S_{i+1} = f(\Sigma(Y_i)) \cap I_{i+1}$. By Eqs. (4.3) and (4.4), the requirement (4.2) is satisfied by $(I, J, X, Y) = (I_i, I_{i+1}, X_i, Y_i)$, so Lemma 4.4 applies, and thus Merge correctly computes $f(\Sigma(X_i \sqcup Y_i)) \cap I_i = \Sigma(A_i \sqcup \cdots \sqcup A_k) \cap I_i$, completing the induction step.

Setting $i = 1$ in the previous displayed property gives

$$S_1 = \Sigma(A_1 \sqcup \cdots \sqcup A_k) \cap I_1 = \Sigma(A \setminus A^{\mathrm{ex}}) \cap \Big[t - \sum_{a\in A^{\mathrm{ex}}} a,\ t\Big].$$

Thus, $t \in \Sigma(A)$ if and only if there exists $t' \in \Sigma(A^{\mathrm{ex}})$ such that $t - t' \in S_1$. This proves the correctness of Algorithm 1.

Running time. Write $s = \Delta^{1/d}$. By Lemma 4.2, the partitioning step takes $\widetilde{\mathcal{O}}_d(\Delta^{1+2/d})$ time, and produces $k = \mathcal{O}_d(\log \Delta)$ groups and an exceptional part of size $|A^{\mathrm{ex}}| = \widetilde{\mathcal{O}}_d(s)$. Hence, $W^{\mathrm{ex}} = \sum_{a\in A^{\mathrm{ex}}} a \le \widetilde{\mathcal{O}}_d(s\Delta)$, and standard dynamic programming computes $\Sigma(A^{\mathrm{ex}})$ in $\mathcal{O}(|A^{\mathrm{ex}}| \cdot W^{\mathrm{ex}}) = \widetilde{\mathcal{O}}_d(s^2\Delta)$ time.

Recall $|X_i'| = \mathcal{O}_d(s)$, $\lambda = \mathcal{O}_d(\log \Delta)$, and $k = \mathcal{O}_d(\log \Delta)$ from Lemma 4.2. By Lemma 4.3, each call to $\mathsf{OneGroup}$ takes $\widetilde{\mathcal{O}}_d(\lambda |X_i'| \Delta) = \widetilde{\mathcal{O}}_d(s\Delta)$ time, with total time $\widetilde{\mathcal{O}}_d(s\Delta)$ over the k calls.

Now we consider the Merge call at iteration i. By (4.3),

$$|I_{i+1}| \le W^{\mathrm{ex}} + \sum_{j=1}^{i} W_j \le W^{\mathrm{ex}} + \Delta \sum_{j=1}^{i} |A_j|.$$

Thus, by Lemma 4.4, the Merge call takes time

$$\widetilde{\mathcal{O}}_d\left(\lambda |X_i'| \left(\Delta + \frac{|I_{i+1}|}{|A_i|}\right)\right) \le \widetilde{\mathcal{O}}_d\left(\lambda |X_i'| \Delta + \lambda \frac{|X_i'|}{|A_i|} W^{\mathrm{ex}} + \lambda |X_i'| \Delta \frac{\sum_{j=1}^{i} |A_j|}{|A_i|}\right)$$

$$= \widetilde{\mathcal{O}}_d(s\Delta),$$

where the last term is bounded using Lemma 4.2 (iv). The total time for Merge calls over all $k-1$ iterations is therefore $\widetilde{\mathcal{O}}_d(s\Delta)$.

The final step can be tested by looping over all $t' \in \Sigma(A^{\mathrm{ex}})$ and checking whether $t - t' \in S_1$ using Lemma 3.7, in $\mathcal{O}((W^{\mathrm{ex}}+1)\log\Delta) = \widetilde{\mathcal{O}}_d(s\Delta)$ time.

Thus, the total time of Algorithm 1 is $\widetilde{\mathcal{O}}_d(\Delta^{1+2/d})$.

Witness Recovery. The standard dynamic programming with backtracing can find a witness $B' \subseteq A^{\mathrm{ex}}$ with $\sum_{a\in B'} a = t'$. Then, we can find a witness $B'' \subseteq A_1 \sqcup \cdots \sqcup A_k$ with $\sum_{a\in B''} a = t - t'$ by backtracing using the procedures OneGroupWitness and MergeWitness provided by Lemmas 4.3 and 4.4 with the vectors $\boldsymbol{\tau}_i$ computed by Lemma 4.2 (i). The total time remains $\widetilde{\mathcal{O}}_d(\Delta^{1+2/d})$. □

Proof of Theorem 1.1. After applying the reduction of Lemma 4.1 in $\mathcal{O}(n+\Delta)$ time, run the algorithm of Lemma 4.5 (which is uniform in the parameter d) on the Subset Sum instance for $d = 1, \ldots, \lfloor \log\Delta \rfloor$ in parallel at equal speed, and return the answer of the first completed run. Since the run with parameter d takes time $C_d\Delta^{1+2/d}(\log\Delta)^{C_d}$ for some C_d, their overall time complexity is $\mathcal{O}(\log\Delta)\cdot\min_{d\le\lfloor\log\Delta\rfloor}\{C_d\Delta^{1+2/d}(\log\Delta)^{C_d}\}$, which is clearly in $\mathcal{O}(\Delta^{1+o(1)})$. Hence, the total time is $\mathcal{O}(n+\Delta^{1+o(1)})$ as claimed. □

5 Algorithmic Conlon–Fox–Pham Theorem

Conlon, Fox, and Pham proved a structural theorem that almost characterizes the set of subset sums using a generalized arithmetic progression (GAP) [CFP23, Theorem 1.5]. In this section, we prove the following algorithmic version of the Conlon–Fox–Pham theorem which efficiently finds such a GAP.

Theorem 5.1. *For any $\beta > 1$, there are constants $C > 1, d \ge 1$ such that the following holds.*

Let A be a multiset of integers in $[-\Delta,\Delta]\setminus\{0\}$, and $s \in [\Delta^{1/\beta}, C^{-1}\min\{\Delta, |A|/\log\Delta\}]\cap\mathbb{Z}$. We can compute multisets $A' \subseteq \hat{A} \subseteq A$ and an s-proper origin-symmetric GAP $P \supseteq \mathrm{supp}(\hat{A})$ of rank at most d, such that $|A| - |\hat{A}| \le Cs\log\Delta$ and $|A'| \le Cs$, and A' is s-generating for P.

The algorithm runs in deterministic time $\widetilde{\mathcal{O}}_\beta(s\,|s(\mathrm{supp}(A)\cup\{0\})|)$, provided $|A| \le \Delta^{\mathcal{O}(1)}$.

Since $s(\mathrm{supp}(A)\cup\{0\}) \subseteq [-s\Delta, s\Delta]$, the time complexity can also be bounded by $\widetilde{\mathcal{O}}_\beta(s^2\Delta)$.

Our actual application requires a vector generalization of Theorem 5.1, which we state in Section 5.4. The same generalization also appeared in [PZ25].

To prove Theorem 5.1, we streamline the original proof in [CFP23], avoiding some technical complications that are not algorithmically friendly, and making its central idea more transparent. In particular, our adapted proof bypasses certain non-constructive definitions (e.g., [CFP23, Definitions 2.25, 2.28, 2.33]), and completely avoids the probabilistic steps in [CFP23, Section 2.5]. We achieve this by converting some of their indirect arguments into potential decrement steps in our algorithm. As in [CFP23], we use Freiman's theorem for low-doubling sets [Fre93] (for us, its algorithmic version [RW24]), but we do not need its strengthening due to Bilu [Bil99] which was used in [CFP23].

5.1 Preliminaries

A rank-d *generalized arithmetic progression (GAP) in* $\mathbb{R}^k$ is a set of the form

$$Q = \boldsymbol{x}_Q + f_Q(B_Q \cap \mathbb{Z}^d) \subseteq \mathbb{R}^k,$$

where $\boldsymbol{x}_Q \in \mathbb{R}^k$, $f_Q \colon \mathbb{R}^d \to \mathbb{R}^k$ is linear, and $B_Q = \prod_{i=1}^d [a_i, b_i]$ is a box with integer endpoints $a_i < b_i$. A GAP Q is understood to include its representation $(\boldsymbol{x}_Q, f_Q, B_Q)$, since the same set may have different representations. It has *widths* $b_i - a_i + 1 \geq 2$ and *volume* $\mathrm{vol}(Q) = \prod_{i=1}^d (b_i - a_i + 1) \geq |Q|$.

We say Q is *origin-symmetric* if $\boldsymbol{x}_Q = \mathbf{0}$ and $a_i = -b_i$ for all i. We say Q is *proper* if f_Q is injective on $B_Q \cap \mathbb{Z}^d$. For $r \in \mathbb{Z}_{>0}$, its r-fold sumset is the GAP $rQ = r\boldsymbol{x}_Q + f_Q((rB_Q) \cap \mathbb{Z}^d)$. Define the GAP $rQ - r'Q$ analogously.

We say Q is an r-proper GAP if the GAP rQ is proper.

If Q is proper and $\boldsymbol{x}_Q = \mathbf{0}$, for $\boldsymbol{q} \in Q$, let $f_Q^{-1}(\boldsymbol{q})$ denote the unique $\boldsymbol{u} \in B_Q \cap \mathbb{Z}^d$ such that $f_Q(\boldsymbol{u}) = \boldsymbol{q}$.

We need the following simple estimates:

Lemma 5.2. *Let Q be a rank-d GAP, and $p \in \mathbb{Z}_{>0}, r \in \mathbb{Z}_{\geq 0}$. Then, $|pQ - rQ| \leq (p+r)^d |Q|$, $\mathrm{vol}(pQ - rQ) \leq (p+r)^d \mathrm{vol}(Q)$, and $\mathrm{vol}(pQ) \geq (p/2)^d \mathrm{vol}(Q)$.*

Proof. The first inequality follows from the observation that $pQ - rQ$ can be covered by the union of $(p+r)^d$ translates of Q. The second inequality follows similarly.

Suppose Q has widths w_i for $i \in [d]$. Then, pQ has widths $p(w_i - 1) + 1 \geq (p/2)w_i$ due to $w_i \geq 2$, giving the third inequality. □

We work with GAPs in $\mathbb{Z}$ until Section 5.4.

Randolph and Węgrzycki [RW24, Theorem 5] proved an algorithmic version of Freiman's theorem. We need the following strengthened version.

Lemma 5.3 (Algorithmic Freiman's theorem)**.** *Let $K \geq 1$, and let $A \subseteq \mathbb{Z} \cap [-\Delta, \Delta]$ satisfy $|A + A| \leq K|A|$ and $0 \in A$. There is a deterministic algorithm that computes in $\mathcal{O}_K(|A|) \operatorname{poly}\log \Delta$ time an origin-symmetric 2-proper GAP P of rank $d = \mathcal{O}_K(1)$ and volume $\mathcal{O}_K(|A|)$ which contains $A + A$, such that $\mathcal{L}(f_P^{-1}(A)) = \mathbb{Z}^d$, and the representation of P has $\mathcal{O}_K(\log \Delta)$ bits.*

Compared to the original randomized algorithm by [RW24], Lemma 5.3 is deterministic and additionally achieves 2-properness and the $\mathcal{L}(f_P^{-1}(A)) = \mathbb{Z}^d$ condition (assuming $0 \in A$). In Appendix A we describe how to strengthen [RW24] to obtain Lemma 5.3.

The following lemma of Conlon, Fox, and Pham [CFP23] shows that the sumset of many dense subsets of an integer box contains a large box, provided each subset generates the integer lattice. It generalizes a result of Lev [Lev10] in the one-dimensional case.

Lemma 5.4 ([CFP23, Lemma 2.15] rephrased)**.** *For any $c \in (0, 1]$ and $d \in \mathbb{Z}_{>0}$, there are constants $\gamma > 0$ and $C \geq 1$ such that the following holds.*

Let $Q = \mathbb{Z}^d \cap \prod_{i=1}^d [-w_i, w_i]$ with $\min_i w_i \geq C$. For $\ell \geq C$, let subsets $A_1, \ldots, A_\ell \subseteq Q$ satisfy $|A_j| \geq c|Q|$, $\mathbf{0} \in A_j$, and $\mathcal{L}(A_j) = \mathbb{Z}^d$ for all $j \in [\ell]$. Then $A_1 + \cdots + A_\ell$ contains a translate of $\lfloor \gamma \ell \rfloor Q$.

5.2 Structure of s-fold sumsets

The following lemma is extracted and simplified from [CFP23, Lemmas 2.22 and 2.26]. It embeds a given Δ-bounded integer set into a rank-d GAP P, so that its s-fold sumset, where $s \geq \Delta^{\Omega_d(1)}$, has size comparable to $\mathrm{vol}(sP)$.

Lemma 5.5. *For any $\beta \geq 1$, there is a constant $d_\beta \geq 1$ such that the following holds.*

For any set $A \subseteq \mathbb{Z} \cap [-\Delta, \Delta]$ with $0 \in A$ and any integer $s \geq \Delta^{1/\beta}$, there exists an origin-symmetric $2s$-proper GAP $P \supseteq A$ of rank $d \leq d_\beta$ such that $|sA| \gg_\beta s^d |P|$ and $\mathcal{L}(f_P^{-1}(A)) = \mathbb{Z}^d$.

Moreover, the GAP P can be found by a deterministic algorithm in $\mathcal{O}_\beta(|sA|) \cdot \operatorname{poly}\log(s\Delta)$ time.

The lower bound on $|sA|$ is nearly tight, as $|sA| \le |sP| \le s^d|P|$ by Lemma 5.2.

To prove Lemma 5.5, we use the following lemma:

Lemma 5.6. *Given positive integers K, q, and a set $A \subseteq \mathbb{Z} \cap [-\Delta, \Delta]$ with $0 \in A$ such that $|2qA| \le K|qA|$, we can deterministically compute an origin-symmetric $2q$-proper GAP $P \supseteq A$ of rank $d = \mathcal{O}_K(1)$, such that $q^d|P| \ll_K |qA|$ and $\mathcal{L}(f_P^{-1}(A)) = \mathbb{Z}^d$, in time $\mathcal{O}_K(|qA|)\operatorname{poly}\log(q\Delta)$.*

Proof. Assume $|A| \ge 2$; otherwise, $A = \{0\}$, and we return $P = \{0\}$. We can compute qA by repeated squaring using the sparse convolution algorithm (Lemma 3.1), in total $\mathcal{O}(|qA|)\operatorname{poly}\log(q\Delta)$ time since each of the $\mathcal{O}(\log q)$ intermediate sumsets has size at most $|qA|$. Then, apply the algorithmic Freiman's theorem (Lemma 5.3) to qA to obtain an origin-symmetric 2-proper GAP $Q = f_Q(B_Q \cap \mathbb{Z}^d)$ containing $2qA$, where $B_Q = \prod_{i=1}^d [-w_i, w_i]$, of rank $d = \mathcal{O}_K(1)$ and size $|Q| \ll_K |qA|$, with $\mathcal{L}(f_Q^{-1}(qA)) = \mathbb{Z}^d$, in $\mathcal{O}_K(|qA|)\operatorname{poly}\log(q\Delta)$ time.

We first claim that, for any $g_1, \dots, g_m \in A$ where $1 \le m \le 2q$, it holds that

$$f_Q^{-1}(g_1 + \cdots + g_m) = \sum_{j=1}^{m} f_Q^{-1}(g_j). \tag{5.1}$$

To show this claim, denote $x_j = \sum_{i=1}^{j} g_i$ for $0 \le j \le m$. Since $0 \in A$, all x_j belong to $2qA \subseteq Q$. The equality $x_j + 0 = x_{j-1} + g_j$ and the 2-properness of Q imply $f_Q^{-1}(x_j) = f_Q^{-1}(x_{j-1}) + f_Q^{-1}(g_j)$ for $j \in [m]$. Then, Eq. (5.1) follows by induction on j.

Eq. (5.1) implies $f_Q^{-1}(qA) \subseteq q f_Q^{-1}(A)$, so $\mathbb{Z}^d = \mathcal{L}(f_Q^{-1}(qA)) \subseteq \mathcal{L}(f_Q^{-1}(A))$, and thus $\mathcal{L}(f_Q^{-1}(A)) = \mathbb{Z}^d$. For every $g \in A$, Eq. (5.1) also implies $2q f_Q^{-1}(g) = f_Q^{-1}(2qg) \in B_Q \cap \mathbb{Z}^d$, and hence $f_Q^{-1}(g) \in (\frac{1}{2q} B_Q) \cap \mathbb{Z}^d$. Combining these two properties gives $\mathbb{Z}^d = \mathcal{L}(f_Q^{-1}(A)) \subseteq \mathcal{L}((\frac{1}{2q} B_Q) \cap \mathbb{Z}^d)$, which implies the half-widths w_i of the box B_Q must satisfy $w_i \ge 2q$ for all $i \in [d]$. Thus, the GAP

$$P = f_Q\left(B_P \cap \mathbb{Z}^d\right), \; B_P = \prod_{i=1}^{d} \left[-\left\lfloor \frac{w_i}{2q} \right\rfloor, \left\lfloor \frac{w_i}{2q} \right\rfloor \right],$$

contains A, with $f_P^{-1}(g) = f_Q^{-1}(g)$ for every $g \in A$. Thus $\mathcal{L}(f_P^{-1}(A)) = \mathbb{Z}^d$ as claimed. Since $2qB_P \subseteq B_Q$ and Q is proper, P is $2q$-proper. Therefore, Lemma 5.2 implies $q^d|P| \le \operatorname{vol}(2qP) = |2qP| \le |Q| \ll_K |qA|$ as claimed. □

Proof of Lemma 5.5. Assume $|A| \ge 2$; otherwise, $A = \{0\}$, and we may return $P = \{0\}$. We assume that s is sufficiently large in terms of β; otherwise, $s = \mathcal{O}_\beta(1)$, and $\Delta \le s^\beta = \mathcal{O}_\beta(1)$, so we may return the one-dimensional arithmetic progression P consisting of all multiples of $\gcd(A)$ in $[-\Delta, \Delta]$, which has size $\mathcal{O}_\beta(1)$ and satisfies $|sA| \ge s + 1 \gg_\beta s|P|$ and $\mathcal{L}(f_P^{-1}(A)) = \mathbb{Z}$ as claimed.

Choose h divisible by 4 so that $2^h \le s < 2^{h+4}$. Then h can be assumed to be sufficiently large in terms of β. Since $|2^h A| \le 2 \cdot 2^h \Delta + 1 \le 3s^{\beta+1}$, by telescoping and averaging, there exists an integer $j \in [h/4, h/2 - 1]$ such that, writing $q = 2^j \ge 2^{h/4}$,

$$\frac{|2qA|}{|qA|} \le \left(\frac{|2^{h/2}A|}{|2^{h/4}A|} \right)^{1/(h/2 - h/4)} \le (3s^{\beta+1})^{4/h} = (3s^{\beta+1})^{\mathcal{O}(1/\log s)} = 2^{\mathcal{O}(\beta)}.$$

Thus $|2qA| \le K_\beta |qA|$ for $K_\beta = \mathcal{O}_\beta(1)$. By Lemma 5.6, there exists an origin-symmetric $2q$-proper GAP $Q = f_Q(B_Q \cap \mathbb{Z}^{d'}) \supseteq A$ of rank $d' = \mathcal{O}_\beta(1)$ such that $q^{d'}|Q| \ll_\beta |qA|$ and $\mathcal{L}(f_Q^{-1}(A)) = \mathbb{Z}^{d'}$.

Let $T = q f_Q^{-1}(A)$ and $\ell = 2^h/q \ge 2^{h/2}$. Then $\mathbf{0} \in T \subseteq qB_Q \cap \mathbb{Z}^{d'}$, $\mathcal{L}(T) = \mathbb{Z}^{d'}$, and the density of T in $qB_Q \cap \mathbb{Z}^{d'}$ can be bounded as

$$|qB_Q \cap \mathbb{Z}^{d'}| = \operatorname{vol}(qQ) \underbrace{\le}_{\text{Lemma 5.2}} q^{d'}|Q| \ll_\beta |qA| = |T|.$$

Since both $q \geq 2^{h/4}$ and $\ell \geq 2^{h/2}$ are sufficiently large in terms of β, Lemma 5.4 applies to the box $qB_Q \cap \mathbb{Z}^{d'}$ and ℓ copies of T, showing that $\ell T = 2^h f_Q^{-1}(A)$ contains a translate of the box $rqB_Q \cap \mathbb{Z}^{d'}$ for some $r = \lfloor \gamma \ell \rfloor \gg_\beta \ell$. Applying f_Q shows that $2^h A$ contains a translate of rqQ. Since $r \gg_\beta \ell$, $d' = \mathcal{O}_\beta(1)$, and $2^h \leq s < 2^{h+4} = 16\ell q$, this implies

$$|2sA| \leq |32\ell qQ| \underbrace{\ll_\beta}_{\text{Lemma 5.2}} |rqQ| \leq |2^h A| \leq |sA|.$$

Therefore, Lemma 5.6 applies to A and s, and returns the required GAP P in $\mathcal{O}_\beta(|sA|)\operatorname{poly}\log(s\Delta)$ time. □

We record the following corollary:

Corollary 5.7. *Let $\beta \geq 1$, $A \subseteq \mathbb{Z} \cap [-\Delta, \Delta]$, and integer $s \geq \Delta^{1/\beta}$. Then, for $m \in \mathbb{Z}_{>0}$, $|msA - msA| \ll_{\beta,m} |sA|$.*

Proof. We may assume $0 \in A$ by translation invariance. By Lemma 5.5, there exists an origin-symmetric GAP $P \supseteq A$ of rank $d = \mathcal{O}_\beta(1)$ satisfying $s^d|P| \ll_\beta |sA|$. Thus Lemma 5.2 gives

$$|msA - msA| \leq |2msP| \leq (2ms)^d |P| \ll_{\beta,m} |sA|.$$ □

5.3 Growing subset sums

In this section, we prove Theorem 5.1 by adapting [CFP23, Sections 2.6 and 3]. We will choose C sufficiently large in terms of β. The hypothesis in Theorem 5.1 implies $\Delta \geq Cs$, so Δ is also sufficiently large in terms of β; we use this assumption later in the proof.

Let d_β be the constant from Lemma 5.5. Let $G_\beta \geq 1$ be the implicit constant in Corollary 5.7 with $m = 2$ and 2β in place of β. Choose $\gamma_\beta \in (0, 1/G_\beta)$. Let $\ell_\beta \in \mathbb{Z}_{>0}$, whose value will be determined later in the proof of Theorem 5.1. Recall $s \geq \Delta^{1/\beta}$.

Given the input multiset A, our algorithm is given in $\mathsf{Main}(A)$ of Algorithm 2. Starting from $\hat{A} = A$, each **while** iteration constructs a submultiset A' of the current remainder $\hat{A}$. If the condition at Line 6 is violated, then we remove A' from $\hat{A}$ and continue the loop. Otherwise, we return $\hat{A}$, A', and a GAP computed from $\hat{A}$. The submultiset A' chosen at each iteration is $A' = G^{(1)} \sqcup \cdots \sqcup G^{(\ell_\beta)}$, where each $G^{(i)}$ is chosen by Greedy and has size s.

Viewing the sumset size $|2s(\operatorname{supp}(\hat{A}) \cup \{0\})|$ as a potential function, each failure of the check at Line 6 causes a multiplicative potential drop, allowing us to bound the number of **while** iterations as follows:

Lemma 5.8. *There is a constant $u_\beta \geq 2$ depending on β, such that, if*

$$|A| \geq 2s\ell_\beta u_\beta \log \Delta, \tag{5.2}$$

then $\mathsf{Main}(A)$ (Algorithm 2) is well-defined and terminates within $u_\beta \log \Delta$ iterations of the **while** *loop, and the returned submultisets $A' \subsetneq \hat{A} \subseteq A$ satisfy $|A'| = s\ell_\beta$ and $|A| - |\hat{A}| \leq s\ell_\beta u_\beta \log \Delta$.*

Proof. The sumset size $|2s(\operatorname{supp}(\hat{A}) \cup \{0\})|$ lies in $[1, 4s\Delta + 1]$ and is non-increasing across the **while** iterations. Since each failure at Line 6 decreases the size by at least a factor of $(1 + \gamma_\beta)$, the total number of **while** iterations is at most $1 + \log(4s\Delta + 1)/\log(1 + \gamma_\beta) \leq u_\beta \log \Delta$ for some u_β depending only on β. Each **while** iteration makes ℓ_β calls to Greedy, and a failed iteration removes a submultiset A' of size $s\ell_\beta$ from $\hat{A}$. Therefore, the total number of removed elements is $|A| - |\hat{A}| \leq s\ell_\beta u_\beta \log \Delta$, and there always remain enough elements in $\hat{A}$ assuming (5.2). □

Algorithm 2: Algorithm for Theorem 5.1

1 **Function** Main(A):
2 $\hat{A} \leftarrow A$
3 **while** True **do**
4 **for** $i \leftarrow 1, 2, \ldots, \ell_\beta$ **do** $G^{(i)} \leftarrow \mathsf{Greedy}(\hat{A} \setminus (G^{(1)} \sqcup \cdots \sqcup G^{(i-1)}))$
5 $A' \leftarrow G^{(1)} \sqcup \cdots \sqcup G^{(\ell_\beta)}$
6 **if** $|2s(\mathrm{supp}(\hat{A}) \cup \{0\})| < (1+\gamma_\beta)|2s(\mathrm{supp}(\hat{A} \setminus A') \cup \{0\})|$ **then break**
7 **else** $\hat{A} \leftarrow \hat{A} \setminus A'$
8 Compute the rank-d GAP P using Lemma 5.5 for $\mathrm{supp}(\hat{A}) \cup \{0\}$ and s
9 **return** $\hat{A}, A', P$
10 **Function** Greedy(X):
  Input: a submultiset $X \subseteq A$ with $|X| \geq s$
  Output: a submultiset $G \subseteq X$ with $|G| = s$
11 $G_0 \leftarrow \varnothing$
12 **for** $j \leftarrow 1, 2, \ldots, s$ **do**
13 Pick $g_j \in X \setminus G_{j-1}$ maximizing $|\Sigma(G_{j-1} \sqcup \{g_j\})|$
14 $G_j \leftarrow G_{j-1} \sqcup \{g_j\}$
15 **return** G_s

For the rest of the analysis, assume (5.2) holds, and consider the last **while** iteration in Main(A). Let $G^{(1)}, \ldots, G^{(\ell_\beta)}$ be the submultisets chosen at Line 4 in this iteration, and let the return values be $\hat{A}$, $A' = G^{(1)} \sqcup \cdots \sqcup G^{(\ell_\beta)}$, and $P = f_P(B_P \cap \mathbb{Z}^d)$, where $B_P = \prod_{j=1}^d [-w_j, w_j]$ with $w_j \in \mathbb{Z}_{>0}$. Define sets

$$Y_0 = \mathrm{supp}(\hat{A}) \cup \{0\} \text{ and } X_0 = \mathrm{supp}(\hat{A} \setminus A') \cup \{0\}.$$

Since $A' \subsetneq \hat{A}$ and $0 \notin \hat{A}$, we have $|X_0| \geq 2$. Since the **if** condition at Line 6 passes and $\gamma_\beta \in (0,1)$, we have

$$|2sY_0| < 2|2sX_0|. \tag{5.3}$$

We now show that the multisets $G^{(i)}$ returned by Greedy have many distinct subset sums:

Lemma 5.9. *For every $i \in [\ell_\beta]$, $|\Sigma(G^{(i)})| \gg_\beta |sY_0|$.*

The proof of Lemma 5.9 uses the following lemma from [CFP23]:

Lemma 5.10 ([CFP23, Lemma 2.41])**.** *Let S, A be finite non-empty sets of integers, and let $k \in \mathbb{Z}_{>0}$ be such that $|kA| \geq 2|S|$. Then there exists $a \in A$ such that $|(S+a) \setminus S| \geq \frac{|S|}{2k}$.*

Proof of Lemma 5.9. Define

$$t_0 = 1 \text{ and } t_h = |2^h X_0|/2 \text{ for } h \geq 1.$$

Since $|X_0| \geq 2$, $t_h \geq (2^h+1)/2$.

Fix $i \in [\ell_\beta]$ and consider the call to Greedy(X) with $X = \hat{A} \setminus (G^{(1)} \sqcup \cdots \sqcup G^{(i-1)})$, which returns $G^{(i)} = \{g_1, \ldots, g_s\}$. Write the prefix $G_j = \{g_1, \ldots, g_j\}$ for $0 \leq j \leq s$. Then,

$$X_0 \setminus \{0\} = \mathrm{supp}(\hat{A} \setminus A') \subseteq \mathrm{supp}(X \setminus G^{(i)}) \subseteq \mathrm{supp}(X \setminus G_{j-1})$$

for every $j \in [s]$. Write $\Sigma_j = \Sigma(G_j)$. For $h \geq 1$, if $|\Sigma_{j-1}| \leq t_h$, then Lemma 5.10 applied to $(S, A, k) = (\Sigma_{j-1}, X_0, 2^h)$ shows that there exists $g \in X_0 \subseteq \mathrm{supp}(X \setminus G_{j-1}) \cup \{0\}$ such that $|(\Sigma_{j-1} + g) \setminus \Sigma_{j-1}| \geq 2^{-h-1}|\Sigma_{j-1}|$, and thus $g \neq 0$. The greedy rule at Line 13 then implies

$$|\Sigma_j| \geq (1 + 2^{-h-1})|\Sigma_{j-1}|.$$

Thus, by telescoping,

$$|\{j \in [s] : |\Sigma_j| \in [t_{h-1}, t_h]\}| \leq 1 + 2^{h+1} \log \frac{t_h}{t_{h-1}}. \tag{5.4}$$

Let h^* be the largest integer such that $t_{h^*} \leq |\Sigma_s|$. Then $h^* \leq \log |\Sigma_s| + 1 = \mathcal{O}(\log \Delta)$. For $h > \frac{1}{2\beta} \log \Delta$, Corollary 5.7 gives $t_h / t_{h-1} = \mathcal{O}_\beta(1)$. For the remaining h, the bound in Eq. (5.4) is $\mathcal{O}(\Delta^{1/(2\beta)} \log \Delta)$. Thus, summing Eq. (5.4) over all $h \in [1, h^* + 1]$ gives

$$s \leq \sum_{h=1}^{h^*+1} \left(1 + 2^{h+1} \log \frac{t_h}{t_{h-1}}\right) \leq \mathcal{O}(\Delta^{1/(2\beta)} \log^2 \Delta) + \mathcal{O}_\beta(2^{h^*}),$$

where the first inequality is due to $t_0 \leq |\Sigma_1| \leq \cdots \leq |\Sigma_s| < t_{h^*+1}$. Since $s \geq \Delta^{1/\beta}$, the inequality above implies $2^{h^*} \gg_\beta s$. In particular, $2^{h^*} \geq \Delta^{1/(2\beta)}$ (for Δ sufficiently large in terms of β), so Corollary 5.7 implies $|2^{h^*} X_0| \gg_\beta |2sX_0|$. Thus,

$$|sY_0| \leq |2sY_0| \underbrace{\leq}_{\text{Eq. (5.3)}} 2|2sX_0| \ll_\beta |2^{h^*} X_0| = 2t_{h^*} \leq 2|\Sigma_s| = 2|\Sigma(G^{(i)})|,$$

which finishes the proof of Lemma 5.9. □

Now we analyze the rank-d GAP P computed at Line 8.

Lemma 5.11. $\mathcal{L}(f_P^{-1}(X_0)) = \mathbb{Z}^d$.

Proof. By Lemma 5.5, P is $2s$-proper and $\mathcal{L}(f_P^{-1}(Y_0)) = \mathbb{Z}^d$. Suppose for contradiction that $\mathcal{L}(f_P^{-1}(X_0)) \neq \mathbb{Z}^d$. Then there exists $y \in Y_0$ such that $f_P^{-1}(y) \notin \mathcal{L}(f_P^{-1}(X_0))$, which implies

$$2sf_P^{-1}(X_0) \cap (f_P^{-1}(y) + sf_P^{-1}(X_0)) = \varnothing.$$

Since $2sP$ is proper, this implies $2sX_0 \cap (y + sX_0) = \varnothing$. Since $0 \in X_0 \subseteq Y_0$, we have $2sX_0 \subseteq 2sY_0$ and $y + sX_0 \subseteq 2sY_0$, so $|2sY_0| \geq |2sX_0| + |sX_0|$. By Corollary 5.7 and the choice of G_β, $|2sX_0| \leq G_\beta |sX_0|$. Hence, $|2sY_0| \geq (1 + 1/G_\beta)|2sX_0|$, which contradicts the **if** condition at Line 6, since $\gamma_\beta < 1/G_\beta$. □

Lemma 5.12. *For every $i \in [\ell_\beta]$, $\mathcal{L}(f_P^{-1}(G^{(i)})) = \mathbb{Z}^d$.*

Proof. Fix i and suppose for contradiction that $L = \mathcal{L}(f_P^{-1}(G^{(i)})) \neq \mathbb{Z}^d$. Then, by Lemma 5.11, there exists $a \in \mathrm{supp}(\hat{A} \setminus A')$ such that $f_P^{-1}(a) \notin L$. For every step j in the call of Greedy returning $G^{(i)}$, a is available at Line 13. We claim $\Sigma(G_{j-1}) \cap (a + \Sigma(G_{j-1})) = \varnothing$, and thus the greedy choice at Line 13 must yield $|\Sigma(G_j)| = 2|\Sigma(G_{j-1})|$. Indeed, if $a + \sum_{x \in X'} x = \sum_{y \in Y'} y$ for some submultisets $X', Y' \subseteq G_{j-1} \subseteq G^{(i)}$, the s-properness of P would give

$$f_P^{-1}(a) = \sum_{y \in Y'} f_P^{-1}(y) - \sum_{x \in X'} f_P^{-1}(x) \in L,$$

a contradiction.

Since $|\Sigma(G_j)| = 2|\Sigma(G_{j-1})|$ for every $j \in [s]$,

$$2^s = |\Sigma(G_s)| \leq 2s\Delta + 1 \leq 3\Delta^2,$$

which contradicts $s \geq \Delta^{1/\beta}$ for sufficiently large Δ. Thus, $\mathcal{L}(f_P^{-1}(G^{(i)})) = \mathbb{Z}^d$ as claimed. □

Proof of Theorem 5.1. We first analyze the time complexity of $\mathsf{Main}(A)$ (Algorithm 2). Consider step j of $\mathsf{Greedy}(X)$, and write $S = \Sigma(G_{j-1})$ and $U = \operatorname{supp}(X) \cup \{0\}$. For all $g \in \mathbb{Z}$, we have $|\Sigma(G_{j-1} \sqcup \{g\})| = 2|S| - r_g$, where $r_g = |S \cap (S+g)|$. Using sparse convolution (Lemma 3.1), we compute all nonzero r_g in $\mathcal{O}(|S-S| \operatorname{poly}\log\Delta)$ time. Then, we find the desired maximizer g_j by scanning $\operatorname{supp}(X \setminus G_{j-1})$, and compute $\Sigma(G_j) = S \cup (S+g_j)$. Since $S \subseteq sU$,

$$|S-S| + |\operatorname{supp}(X)| \le 2|sU - sU| \underbrace{\ll_\beta}_{\text{Corollary 5.7}} |sU|.$$

Hence all s steps in $\mathsf{Greedy}(X)$ take $\mathcal{O}_\beta(s|sU| \operatorname{poly}\log\Delta)$ total time. The two $2s$-fold sumsets at Line 6 can be computed by repeated squaring with sparse convolution (Lemma 3.1) in $|2s(\operatorname{supp}(A)\cup\{0\})| \operatorname{poly}\log\Delta \ll_\beta |s(\operatorname{supp}(A)\cup\{0\})| \operatorname{poly}\log\Delta$ time by Corollary 5.7. Each **while** iteration makes $\ell_\beta = \mathcal{O}_\beta(1)$ greedy calls, and Lemma 5.8 bounds the number of iterations by $\mathcal{O}_\beta(\log\Delta)$. Finally, Lemma 5.5 at Line 8 takes $\mathcal{O}_\beta(|s(\operatorname{supp}(\hat{A})\cup\{0\})| \operatorname{poly}\log\Delta)$ time. Therefore, the total time complexity is $\mathcal{O}_\beta(s|s(\operatorname{supp}(A)\cup\{0\})| \operatorname{poly}\log\Delta)$, as claimed.

Now we prove the correctness of Algorithm 2. By Lemma 5.8, $A' \subseteq \hat{A} \subseteq A$, $|A'| = s\ell_\beta$, and $|A| - |\hat{A}| \le s\ell_\beta u_\beta \log\Delta$. By Lemma 5.5, $P = f_P(B_P \cap \mathbb{Z}^d) \supseteq \operatorname{supp}(\hat{A}) \cup \{0\}$ is an origin-symmetric $2s$-proper GAP of rank $d \le d_\beta$. Next, it remains to prove that $A' = G^{(1)} \sqcup \cdots \sqcup G^{(\ell_\beta)}$ is s-generating for P.

For every $i \in [\ell_\beta]$, we have $\mathbf{0} \in \Sigma(f_P^{-1}(G^{(i)})) \subseteq sB_P \cap \mathbb{Z}^d$ and

$$|\Sigma(f_P^{-1}(G^{(i)}))| = |\Sigma(G^{(i)})| \underbrace{\gg_\beta}_{\text{Lemma 5.9}} |s(\operatorname{supp}(\hat{A})\cup\{0\})| \underbrace{\gg_\beta}_{\text{Lemma 5.5}} s^d|P| \ge |sB_P \cap \mathbb{Z}^d|.$$

By Lemma 5.12, $\mathcal{L}(\Sigma(f_P^{-1}(G^{(i)}))) = \mathbb{Z}^d$. The box $sB_P \cap \mathbb{Z}^d$ has half-widths at least $s \ge \Delta^{1/\beta}$. Thus, after choosing ℓ_β sufficiently large in terms of β, Lemma 5.4 applies to the dense sets $\Sigma(f_P^{-1}(G^{(i)}))$ ($i \in [\ell_\beta]$) in the box $sB_P \cap \mathbb{Z}^d$, and gives $\boldsymbol{\tau} \in \mathbb{Z}^d$ such that

$$(\boldsymbol{\tau} + sB_P) \cap \mathbb{Z}^d \subseteq \sum_{i=1}^{\ell_\beta} \Sigma(f_P^{-1}(G^{(i)})) = \Sigma(f_P^{-1}(A')).$$

Therefore, by Definition 3.4, A' is s-generating for P, as desired. The proof is finished after choosing the constant C in Theorem 5.1 to be larger than $2\ell_\beta u_\beta$ and sufficiently large in terms of β as mentioned earlier, and setting the rank upper bound in Theorem 5.1 to d_β. □

5.4 Generalization to vectors

Following Pham and Zakharov [PZ25], we generalize the algorithmic Conlon–Fox–Pham theorem (Theorem 5.1) from integers to vectors. We also bound the widths of the obtained GAP and its s-fold sumset size for later applications in Section 7.

Theorem 5.13. *For any $\beta > 1$ and $k \in \mathbb{Z}_{>0}$, there are constants $C > 1, d \ge 1$ such that the following holds.*

Let A be a multiset of nonzero vectors in $(\mathbb{Z}\cap[-\Delta,\Delta])^k$, and $s \in [\Delta^{1/\beta}, C^{-1}\min\{\Delta, |A|/\log\Delta\}]\cap\mathbb{Z}$. We can compute multisets $A' \subsetneq \hat{A} \subseteq A$ and an s-proper origin-symmetric GAP $P \supseteq \operatorname{supp}(\hat{A})$ of rank $m \le d$, such that $|A| - |\hat{A}| \le Cs\log\Delta$ and $|A'| \le Cs$, and A' is s-generating for P.

Moreover, writing $P = f_P(B_P \cap \mathbb{Z}^m)$ with $B_P = \prod_{j=1}^m [-w_j, w_j]$ and $w_j \in \mathbb{Z}_{>0}$, we have $w_j \le C\Delta$ and $|sP| = \operatorname{vol}(sP) \le C|s(\operatorname{supp}(\hat{A})\cup\{\mathbf{0}\})|$.

The algorithm runs in deterministic time $\widetilde{\mathcal{O}}_{\beta,k}(s\,|s(\operatorname{supp}(A)\cup\{\mathbf{0}\})|)$ provided $|A| \le \Delta^{\mathcal{O}(1)}$.

An integer encoding lets us apply Theorem 5.1; s-generation then allows us to lift the GAP generators back to vectors without carries. The proof is given in Appendix B.

6 Basic Geometric Algorithms

In this section, we collect some standard algorithms for optimization and lattices.

Rational arithmetic. We need to account for the time complexity of rational arithmetic operations. Recall that for a rational number $r = p/q$ in lowest terms, $\mathsf{maxbitlen}(r) := \max\{\log(2+|p|), \log(2+|q|)\}$.

For a rational matrix M, define $\mathsf{maxbitlen}(M) = \max_{i,j} \mathsf{maxbitlen}(M_{i,j})$; similarly define $\mathsf{maxbitlen}(\boldsymbol{v})$ for a rational vector $\boldsymbol{v}$. For a rational linear map f, let $\mathsf{maxbitlen}(f) = \mathsf{maxbitlen}(M_f)$ where M_f is its matrix in the standard bases. For a rational polytope K, let $\mathsf{maxbitlen}(K) = \max_{\boldsymbol{v}} \mathsf{maxbitlen}(\boldsymbol{v})$ where $\boldsymbol{v}$ ranges over its vertices.

For a rational lattice L, let $\mathsf{maxbitlen}(L) = \mathsf{maxbitlen}(H_L)$ where the matrix H_L is the HNF representation (see Section 6.3) of L.

For a multiset A, define $\mathsf{maxbitlen}(A) = \max\{\log(2+|A|), \max_{a\in\operatorname{supp}(A)} \mathsf{maxbitlen}(a)\}$.

We use the shorthand $\mathsf{maxbitlen}(x_1,\dots,x_k) = \max_{1\le i\le k} \mathsf{maxbitlen}(x_i)$. In these definitions, a maximum over an empty family is taken to be 1.

For a rational polytope $K \subseteq \mathbb{R}^d$, a vector $\boldsymbol{u} \in \mathbb{Q}^d$, and a lattice $L \subseteq \mathbb{Q}^d$,

$$\boldsymbol{x} \in K \cap (\boldsymbol{u}+L) \quad\Longrightarrow\quad \mathsf{maxbitlen}(\boldsymbol{x}) = \mathcal{O}_d(\mathsf{maxbitlen}(K,\boldsymbol{u},L)). \tag{6.1}$$

Indeed, K bounds the coordinate magnitudes, and a common denominator of $\boldsymbol{u}$ and H_L clears the coordinates of every point in $\boldsymbol{u}+L$.

6.1 Linear optimization oracles

Definition 6.1. For a rational polytope $K \subseteq \mathbb{R}^d$, a *linear optimization oracle* receives a query $\boldsymbol{c} \in \mathbb{Q}^d$ and returns a *vertex* $\boldsymbol{x}$ of K that maximizes $\langle \boldsymbol{c}, \boldsymbol{x}\rangle$ (or reports $K = \varnothing$). Let $T_{\text{query}}(K)$ be such that K has a linear optimization oracle that answers a query $\boldsymbol{c}$ in $T_{\text{query}}(K)\cdot(\mathsf{maxbitlen}(\boldsymbol{c}))^{\mathcal{O}_d(1)}$ time.

Note that we are using the convention that the returned maximizer must be a vertex.

Lemma 6.2. *Let X be a multiset of vectors in $\mathbb{Z}^d$, and $b_0 = \mathsf{maxbitlen}(X)$. Then $\mathsf{maxbitlen}(\mathcal{Z}(X)) = \mathcal{O}_d(b_0)$ and $T_{\text{query}}(\mathcal{Z}(X)) = (|\operatorname{supp} X|+1)\cdot b_0^{\mathcal{O}_d(1)}$.*

Proof. Return the sum of $\mu_X(\boldsymbol{v})\boldsymbol{v}$ over $\boldsymbol{v} \in \operatorname{supp} X$ for which $(\langle \boldsymbol{c}, \boldsymbol{v}\rangle, -v_1, \dots, -v_d)$ is lexicographically positive. This maximizes the objective and then minimizes the output coordinates successively, giving a vertex of $\mathcal{Z}(X)$ in one scan of $\operatorname{supp} X$. Every vertex of $\mathcal{Z}(X)$ is a subset sum of X, giving $\mathsf{maxbitlen}(\mathcal{Z}(X)) = \mathcal{O}_d(b_0)$. □

A *strong separation oracle* for a rational polytope $K \subseteq \mathbb{R}^d$, given $\boldsymbol{x} \in \mathbb{Q}^d$, certifies $\boldsymbol{x} \in K$ or returns $\boldsymbol{u} \in \mathbb{Q}^d \setminus \{0\}$ and $a \in \mathbb{Q}$ satisfying

$$K \subseteq \{\boldsymbol{y} \in \mathbb{R}^d : \langle \boldsymbol{u}, \boldsymbol{y}\rangle \le a\}, \quad \langle \boldsymbol{u}, \boldsymbol{x}\rangle > a.$$

For rational polytopes, linear optimization and strong separation are equivalent in oracle-polynomial time [GLS88, Theorem 6.4.9]. We use the following standard consequences; optimizing vertices can be obtained by lexicographic tie-breaking [GLS88, Remark 6.5.2].

For a Cartesian product $K = \prod_i K_i$ of rational polytopes, the vertices of K are precisely the tuples of vertices of the K_i, so $\mathsf{maxbitlen}(K) \le \max_i \mathsf{maxbitlen}(K_i)$. Optimizing over each K_i separately gives a linear optimization oracle for K with $T_{\text{query}}(K) \le \sum_i T_{\text{query}}(K_i)$.

Lemma 6.3. *Let $K_0 \subseteq \mathbb{R}^d$ be a rational polytope given by a linear optimization oracle. Let $A \in \mathbb{Q}^{m\times d}$, $\boldsymbol{a} \in \mathbb{Q}^m$, and $f\colon \mathbb{R}^d \to \mathbb{R}^{d'}$ be a rational linear map ($d' \le d$). Define $K = f(\{\boldsymbol{x} \in K_0 : A\boldsymbol{x} \le \boldsymbol{a}\})$. Let $b_0 = \mathsf{maxbitlen}(K_0, A, \boldsymbol{a}, f)$. Then $\mathsf{maxbitlen}(K) = \mathcal{O}_d(b_0)$ and $T_{\text{query}}(K) \le (T_{\text{query}}(K_0) + m)b_0^{\mathcal{O}_d(1)}$.*

In particular, suppose a polytope K is given by $K = \{\boldsymbol{x} \in \mathbb{R}^d : A\boldsymbol{x} \le \boldsymbol{a}\}$, where $A \in \mathbb{Q}^{m\times d}$, $\boldsymbol{a} \in \mathbb{Q}^m$, and $b_0 = \mathsf{maxbitlen}(A, \boldsymbol{a})$. Then $\mathsf{maxbitlen}(K) = \mathcal{O}_d(b_0)$ and $T_{\text{query}}(K) = mb_0^{\mathcal{O}_d(1)}$.

Proof. Put $P = \{\boldsymbol{x} \in K_0 : A\boldsymbol{x} \le \boldsymbol{a}\}$. By [GLS88, Lemma 6.2.4], K_0 admits an inequality description with $\mathsf{maxbitlen}(\alpha) = \mathcal{O}_d(b_0)$ for every coefficient α. Together with $A\boldsymbol{x} \le \boldsymbol{a}$, this description defines P, giving $\mathsf{maxbitlen}(P) = \mathcal{O}_d(b_0)$. Every vertex of K is the image of a vertex of P, giving the claimed bound on $\mathsf{maxbitlen}(K)$.

Let $\boldsymbol{c} \in \mathbb{Q}^{d'}$ and set $b = \mathsf{maxbitlen}(\boldsymbol{c})$. By [GLS88, Theorem 6.4.9], the optimization oracle for K_0 gives a strong separation oracle. To separate $\boldsymbol{x}$ from P, check the explicit inequalities and whether $\boldsymbol{x} \in K_0$. A violated inequality or a separator for K_0 gives a separator for P. Use the same theorem to optimize $\boldsymbol{c}^{\mathsf{T}} f(\boldsymbol{x})$ over P. To obtain a vertex of K, break ties by successively maximizing the coordinates of $f(\boldsymbol{x})$, fixing each previous optimum as an equality [GLS88, Remark 6.5.2]. These reductions use $(b_0+b)^{\mathcal{O}_d(1)}$ oracle calls, with $\mathsf{maxbitlen}$ of each query bounded by $(b_0+b)^{\mathcal{O}_d(1)}$. Each separation query checks $m + \mathcal{O}_d(1)$ inequalities and invokes the oracle for K_0. The total time is therefore $(b_0+b)^{\mathcal{O}_d(1)}(T_{\text{query}}(K_0) + m)$, which gives the stated bound on $T_{\text{query}}(K)$.

For the explicit-inequality case, Cramer's rule gives $\mathsf{maxbitlen}(K) = \mathcal{O}_d(b_0)$. Hence K lies in a rational box K_0 with $\mathsf{maxbitlen}(K_0) = \mathcal{O}_d(b_0)$ and $T_{\text{query}}(K_0) = b_0^{\mathcal{O}_d(1)}$. Apply the general case with the identity map to obtain the claimed query bound. □

6.2 Parallelotope approximations

For an explicitly listed finite set $E \subseteq \mathbb{Q}^d$, optimize over $\operatorname{conv} E$ by scanning E and breaking ties lexicographically in the coordinates. The selected point is a vertex of $\operatorname{conv} E$.

We choose input vectors in whose basis all points have bounded coordinates by determinant exchanges, as in [AK08].

Lemma 6.4. *Let $K \subseteq \mathbb{R}^d$ be a nonempty rational polytope given by a linear optimization oracle, $d' = \dim \operatorname{span} K$, and $b_0 = \mathsf{maxbitlen}(K)$. Then, one can find d' vertices $\boldsymbol{p}_1, \dots, \boldsymbol{p}_{d'}$ of K such that*

$$K \subseteq \left\{ \sum_{i=1}^{d'} a_i \boldsymbol{p}_i : |a_i| \le 2 \right\},$$

in $b_0^{\mathcal{O}_d(1)} T_{\text{query}}(K)$ time.

Proof. First find vertices $\boldsymbol{p}_1, \dots, \boldsymbol{p}_{d'}$ of K forming a basis of $\operatorname{span} K$. Starting with no points, let V be the span of the points chosen so far. Compute a basis of $V^\perp$ and maximize both signs of each corresponding linear functional over K. A positive maximum gives a vertex of $K \setminus V$; add it to the list and repeat. If all maxima are zero, $K \subseteq V$. At most d points are added, so this takes $\mathcal{O}_d(1)$ calls.

Let φ_i be the coefficient functionals of the basis $(\boldsymbol{p}_i)$. Maximize both signs of each φ_i over K. Whenever a returned point $\boldsymbol{x}$ satisfies $|\varphi_i(\boldsymbol{x})| > 2$, replace $\boldsymbol{p}_i$ by $\boldsymbol{x}$, update the coefficient functionals, and repeat. When all these maxima are at most 2, the claimed containment holds. Each replacement multiplies the absolute determinant of the basis, in any fixed coordinates on $\operatorname{span} K$, by $|\varphi_i(\boldsymbol{x})| > 2$.

Compute the coefficient functionals as the rows of $M^{-1}P$, where $P\colon \mathbb{R}^d \to \mathbb{R}^{d'}$ is a fixed coordinate projection injective on $\operatorname{span} K$ and $M = (P\boldsymbol{p}_1, \dots, P\boldsymbol{p}_{d'})$. Since $\mathsf{maxbitlen}(\boldsymbol{p}_i) \le b_0$ for

every selected point, $2^{-\mathcal{O}_d(b_0)} \le |\det M| \le 2^{\mathcal{O}_d(b_0)}$ throughout. Thus there are $\mathcal{O}_d(b_0)$ replacements, each requiring $\mathcal{O}_d(1)$ calls. All query directions $\boldsymbol{c}$ satisfy $\mathsf{maxbitlen}(\boldsymbol{c}) = \mathcal{O}_d(b_0)$, and the additional time is $b_0^{\mathcal{O}_d(1)}$. □

Lemma 6.5. *Let $K \subseteq \mathbb{R}^d$ be a convex body. There are an origin-symmetric parallelotope B and a point $\boldsymbol{c}$ such that $\boldsymbol{c} + B \subseteq K \subseteq \boldsymbol{c} + \beta_d B$ for some $\beta_d \ge 1$ depending only on d.*

If K is a rational polytope given by a linear optimization oracle, and $b_0 = \mathsf{maxbitlen}(K)$, then one can compute such rational $B, \boldsymbol{c}$ with $\mathsf{maxbitlen}(B, \boldsymbol{c}) = \mathcal{O}_d(b_0)$ in $b_0^{\mathcal{O}_d(1)} T_{\text{query}}(K)$ time.

Proof. Choose an inscribed simplex with vertices $\boldsymbol{v}_1, \dots, \boldsymbol{v}_{d+1}$ such that every $\boldsymbol{x} \in K$ has barycentric coordinates $\boldsymbol{x} = \sum_i a_i \boldsymbol{v}_i$, $\sum_i a_i = 1$, with $|a_i| \le 2$. A maximum-volume simplex has this property: replacing $\boldsymbol{v}_i$ by $\boldsymbol{x}$ multiplies its volume by $|a_i|$, giving even $|a_i| \le 1$. For a rational polytope, apply Lemma 6.4 to $K \times \{1\}$ to compute such a simplex.

Set $\boldsymbol{c} = (\sum_i \boldsymbol{v}_i)/(d+1)$, $\boldsymbol{q}_i = \boldsymbol{v}_i - \boldsymbol{v}_{d+1}$, and $B = \frac{1}{d(d+1)} \sum_{i=1}^d [-\boldsymbol{q}_i, \boldsymbol{q}_i]$. For $|t_i| \le 1/(d(d+1))$, the barycentric coordinates of $\boldsymbol{c} + \sum_i t_i \boldsymbol{q}_i$ are $1/(d+1) + t_i$ for $i \le d$ and $1/(d+1) - \sum_i t_i$ for the last coordinate. They are nonnegative, so $\boldsymbol{c} + B \subseteq K$. Conversely, $\boldsymbol{x} - \boldsymbol{c} = \sum_{i=1}^d (a_i - 1/(d+1)) \boldsymbol{q}_i$ has coefficients of absolute value at most 3, giving $K \subseteq \boldsymbol{c} + 3d(d+1)B$. The oracle for K also optimizes over $K \times \{1\}$, so Lemma 6.4 gives the stated running time. Rational linear algebra gives $\mathsf{maxbitlen}(\boldsymbol{c}, B) = \mathcal{O}_d(b_0)$. □

6.3 Lattice algorithms

A matrix $H \in \mathbb{Z}^{d \times d'}$ with linearly independent columns is in *(column) Hermite normal form* (HNF) if, writing $r_j = \max\{i : H_{i,j} \ne 0\}$, the following properties hold:

- $r_1 < \cdots < r_{d'}$,
- $H_{r_j,j} > 0$ for all $1 \le j \le d'$,
- $0 \le H_{r_j,k} < H_{r_j,j}$ for all $1 \le j < k \le d'$.

It is well-known that every integer lattice $L \subseteq \mathbb{Z}^d$ with $\operatorname{rank} L = d'$ has a unique HNF representation $H \in \mathbb{Z}^{d \times d'}$ such that $L = H\mathbb{Z}^{d'}$.

For a rational lattice $L \subseteq \mathbb{Q}^d$, writing q for the least positive integer with $qL \subseteq \mathbb{Z}^d$, and writing H for the unique HNF representation of qL, the unique HNF representation of L is $H_L = H/q$.

Lemma 6.6 ([SL96, Theorem 5])**.** *Given $q \in \mathbb{Z}_{>0}$ and $\boldsymbol{z}_1, \dots, \boldsymbol{z}_m \in \mathbb{Z}^d$ with $b_0 = \mathsf{maxbitlen}(q, \boldsymbol{z}_1, \dots, \boldsymbol{z}_m)$, one can compute the HNF representation H_L of the lattice $L = \mathcal{L}(\boldsymbol{z}_1/q, \dots, \boldsymbol{z}_m/q)$ with $\mathsf{maxbitlen}(H_L) = \mathcal{O}_d(b_0)$ in $(m+1)\operatorname{poly}(d, b_0)$ time.*

Lemma 6.7 ([SL96, Theorem 12])**.** *Given a lattice $\Gamma \subseteq \mathbb{Q}^d$ and a rational linear map $f\colon \mathbb{R}^d \to \mathbb{R}$ with $b_0 = \mathsf{maxbitlen}(\Gamma, f)$, one can compute the lattice $\Lambda = \Gamma \cap \ker f$ and $g \in \mathbb{Q}_{\ge 0}$ such that $f(\Gamma) = g\mathbb{Z}$, with $\mathsf{maxbitlen}(g, \Lambda) = \mathcal{O}_d(b_0)$ in $\operatorname{poly}(d, b_0)$ time.*

Lemma 6.8 ([LLL82])**.** *Every lattice $L \subseteq \mathbb{R}^d$ with $\operatorname{rank} L = d'$ has a basis $(\boldsymbol{b}_i)_{i=1}^{d'}$ satisfying*

$$\prod_{i=1}^{d'} \|\boldsymbol{b}_i\|_2 \le 2^{(d')^2} \det L.$$

If L is rational and $b_0 = \mathsf{maxbitlen}(L)$, one can compute such a basis with $\mathsf{maxbitlen}(\boldsymbol{b}_1, \dots, \boldsymbol{b}_{d'}) = \mathcal{O}_d(b_0)$ in $\operatorname{poly}(d, b_0)$ time.

The oracle version of Lenstra's algorithm [Len83] solves integer linear optimization over a rational polytope of fixed dimension from a strong separation oracle [GLS88, Section 6.7, p. 196].

Lemma 6.9 (Lenstra)**.** *Given a rational polytope $K \subseteq \mathbb{R}^d$ with a linear optimization oracle, vectors $\boldsymbol{c}, \boldsymbol{\tau} \in \mathbb{Q}^d$, and a lattice $L \subseteq \mathbb{Q}^d$, writing $F = K \cap (\boldsymbol{\tau} + L)$, one can find $\boldsymbol{x} \in F$ that maximizes $\langle \boldsymbol{c}, \boldsymbol{x} \rangle$ (or report $F = \varnothing$) in $b_0^{\mathcal{O}_d(1)} T_{\text{query}}(K)$ time, where $b_0 = \mathsf{maxbitlen}(K, \boldsymbol{c}, \boldsymbol{\tau}, L)$.*

Proof. Writing H for the HNF representation of L, substitute $\boldsymbol{x} = \boldsymbol{\tau} + H\boldsymbol{z}$, $\boldsymbol{z} \in \mathbb{Z}^{\operatorname{rank} L}$. The claim follows from Lenstra's algorithm and optimization–separation equivalence. □

The following lemma enumerates the values of a linear map on lattice points in a polytope.

Lemma 6.10. *Given a rational lattice $\Gamma \subseteq \mathbb{Q}^d$, a rational linear map $f\colon \mathbb{R}^d \to \mathbb{R}$, and $A \in \mathbb{Q}^{m \times d}, \boldsymbol{a} \in \mathbb{Q}^m$ which define a polytope $K = \{\boldsymbol{x} \in \mathbb{R}^d : A\boldsymbol{x} \le \boldsymbol{a}\}$, one can enumerate $f(K \cap \Gamma)$ in $(|f(K \cap \Gamma)| + 1) m b_0^{\mathcal{O}_d(1)}$ time where $b_0 = \mathsf{maxbitlen}(\Gamma, f, A, \boldsymbol{a})$.*

Proof. Let q be the product of the denominators of the entries of $M_f H_\Gamma$. Then $qf(\Gamma) \subseteq \mathbb{Z}$ and $\mathsf{maxbitlen}(q, qf) = \mathcal{O}_d(b_0)$. Replacing f by qf and dividing the enumerated values by q, we may assume $f(\Gamma) \subseteq \mathbb{Z}$. By Lemma 6.3, $\mathsf{maxbitlen}(K) = \mathcal{O}_d(b_0)$ and $T_{\text{query}}(K) = m b_0^{\mathcal{O}_d(1)}$. Minimize f over $K \cap \Gamma$ using Lemma 6.9. After returning a value a, minimize over $K \cap \Gamma \cap \{\boldsymbol{x} : f(\boldsymbol{x}) \ge a + 1\}$, and repeat until this set is empty. Integrality of f on Γ ensures that every value is returned exactly once. Since $\mathsf{maxbitlen}(K, f) = \mathcal{O}_d(b_0)$, each threshold satisfies $\mathsf{maxbitlen}(a + 1) = \mathcal{O}_d(b_0)$. Applying Lemmas 6.3 and 6.9 to each constrained polytope, which has $m + 1$ inequalities, gives $m b_0^{\mathcal{O}_d(1)}$ time per optimization. There are $|f(K \cap \Gamma)| + 1$ optimizations. □

6.4 Reparameterizing lattices

The following lattice-box approximation follows from lattice reduction; compare [TV06, Section 3.5].

Lemma 6.11. *Let $K \subseteq \mathbb{R}^d$ be a nonempty origin-symmetric compact convex set and $\Gamma \subseteq \mathbb{R}^d$ a lattice of rank d' with $\operatorname{span}(K \cap \Gamma) = \operatorname{span} \Gamma$. There are a basis $\boldsymbol{b}_1, \dots, \boldsymbol{b}_{d'}$ of Γ and positive integers $m_1, \dots, m_{d'}$ such that*

$$K \cap \operatorname{span} \Gamma \subseteq \left\{ \sum_{i=1}^{d'} t_i \boldsymbol{b}_i : |t_i| \le m_i \right\} \subseteq \kappa_d K,$$

where $\kappa_d \ge 1$ depends only on d. If K is a rational polytope given by a linear optimization oracle and Γ is rational, one can compute $\boldsymbol{b}_i, m_i$ in $b_0^{\mathcal{O}_d(1)} T_{\text{query}}(K)$ time, where $b_0 = \mathsf{maxbitlen}(K, \Gamma)$.

Proof. If $d' = 0$, the empty basis and list of widths suffice. Assume $d' > 0$. Put $E = \operatorname{span} \Gamma$ and $K_E = K \cap E$. Choose d' points in K_E maximizing the absolute determinant in orthonormal coordinates on E, and let $T\colon E \to \mathbb{R}^{d'}$ send them to the standard basis. Maximality of the determinant, symmetry, and convexity give

$$[-1/d', 1/d']^{d'} \subseteq T K_E \subseteq [-2, 2]^{d'}.$$

By Lemma 6.8, $T\Gamma$ has a basis $\boldsymbol{\beta}_1, \dots, \boldsymbol{\beta}_{d'}$. Cramer's rule and Hadamard's inequality give, for all $\boldsymbol{t} \in \mathbb{R}^{d'}$ and $i \in [d']$,

$$|t_i| \, \|\boldsymbol{\beta}_i\|_2 \le \frac{\prod_j \|\boldsymbol{\beta}_j\|_2}{\det(T\Gamma)} \left\| \sum_j t_j \boldsymbol{\beta}_j \right\|_2 \le 2^{d^2} \left\| \sum_j t_j \boldsymbol{\beta}_j \right\|_2.$$

By the spanning hypothesis, $T\Gamma$ contains d' independent vectors of Euclidean norm at most $2\sqrt{d'}$. For each i, one of these vectors has a nonzero integer i-th basis coordinate, so the coefficient bound gives $\|\boldsymbol{\beta}_i\|_2 \le 2^{d^2}\, 2\sqrt{d'}$. With $A_d = 2d\, 2^{d^2}$, the same bound gives

$$\|\boldsymbol{\beta}_i\|_\infty \le A_d, \qquad |t_i| \le \frac{A_d}{\|\boldsymbol{\beta}_i\|_\infty} \quad \text{whenever } \sum_i t_i\boldsymbol{\beta}_i \in TK_E.$$

Set $\boldsymbol{b}_i = T^{-1}\boldsymbol{\beta}_i$ and $m_i = \lceil A_d/\|\boldsymbol{\beta}_i\|_\infty \rceil$. These widths give the first containment. Since $m_i\|\boldsymbol{\beta}_i\|_\infty \le 2A_d$, the image of the box is contained in $[-2dA_d, 2dA_d]^{d'}$, hence in $\kappa_d TK_E$ for a sufficiently large κ_d depending only on d. This proves the second containment. For rational K and Γ, obtain an oracle for K_E by Lemma 6.3, with $\mathsf{maxbitlen}(K_E) = \mathcal{O}_d(b_0)$. Choose the points defining T by Lemma 6.4; they satisfy the same displayed containments. Compute $T\Gamma$ and its basis by Lemmas 6.6 and 6.8, and then compute $\boldsymbol{b}_i, m_i$ as above. These algorithms and rational linear algebra give the stated time. The containments give $\boldsymbol{b}_i, m_i\boldsymbol{b}_i \in \kappa_d K \cap \Gamma$; (6.1) gives $\mathsf{maxbitlen}(\boldsymbol{b}_i, m_i) = \mathcal{O}_d(b_0)$. □

Quotienting by short kernel directions is the rank-reduction step for convex progressions [TV08, Lemma 4.1 and Corollary 4.2]. We choose two successive scales at which these directions span the same subspace; quotienting by that subspace makes the resulting lattice box proper.

Lemma 6.12. *Given an origin-symmetric rational polytope $K \subseteq \mathbb{R}^d$ with a linear optimization oracle, a lattice $\Gamma \subseteq \mathbb{Q}^d$ such that $\operatorname{span}(K \cap \Gamma) = \mathbb{R}^d$, and a nonzero rational linear map $f\colon \mathbb{R}^d \to \mathbb{R}$, one can compute $1 \le d' \le d$, rational linear maps $\Phi\colon \mathbb{R}^d \to \mathbb{R}^{d'}$ and $\bar{f}\colon \mathbb{R}^{d'} \to \mathbb{R}$, and a box $B = \prod_{i=1}^{d'}[-m_i, m_i]$, $m_i \in \mathbb{Z}_{>0}$, such that*

- $f = \bar{f} \circ \Phi$,
- $\Phi\Gamma = \mathbb{Z}^{d'}$,
- $\Phi K \subseteq B \subseteq \kappa_d \Phi K$, *where κ_d is as in Lemma 6.11,*
- $|B \cap \mathbb{Z}^{d'}| \le C_d |f(K \cap \Gamma)|$, *where C_d depends only on d,*
- *and, $\bar{f}$ is injective on $4B \cap \mathbb{Z}^{d'}$,*

with $\mathsf{maxbitlen}(\Phi, \bar{f}, B) = \mathcal{O}_d(b_0)$ in $b_0^{\mathcal{O}_d(1)} T_{\text{query}}(K)$ time, where $b_0 = \mathsf{maxbitlen}(K, \Gamma, f)$.

Proof. Compute $\Lambda = \Gamma \cap \ker f$ by Lemma 6.7, and put $k = \operatorname{rank}\Lambda = d - 1$. Choose an integer $a > 8\kappa_d + d$ and set

$$V_j = \operatorname{span}(\Lambda \cap a^j K) \qquad (0 \le j \le k+1).$$

These nested spaces have dimension at most k, so $V_j = V_{j+1}$ for some $j \le k$. Each span can be computed by successively optimizing both signs of a basis of the orthogonal complement of the points already selected, using Lemma 6.9 over $a^j K \cap \Lambda$. Put $R = a^j$, $V = V_j$, $r = \dim V$, and $d' = d - r$. Choose independent $\boldsymbol{\lambda}_1, \dots, \boldsymbol{\lambda}_r \in \Lambda \cap RK$ spanning V. Their centered fundamental parallelotope is contained in $(rR/2)K$.

Choose a rational quotient $\Phi_0\colon \mathbb{R}^d \to \mathbb{R}^{d'}$ with kernel V, and write $\Gamma_0 = \Phi_0\Gamma$, $K_0 = \Phi_0 K$, and $f = f_0 \circ \Phi_0$. For every $t \ge 0$ and $\boldsymbol{z} \in tK_0 \cap \Gamma_0$, take a lattice lift $\boldsymbol{x} \in \Gamma$ and a point $\boldsymbol{y} \in tK$ with $\Phi_0\boldsymbol{y} = \boldsymbol{z}$. Reduce $\boldsymbol{x} - \boldsymbol{y} \in V$ modulo $\sum_i \mathbb{Z}\boldsymbol{\lambda}_i$. This gives a lattice lift

$$\boldsymbol{x}' \in (t + rR/2)K \cap \Gamma, \qquad \Phi_0\boldsymbol{x}' = \boldsymbol{z}, \qquad f(\boldsymbol{x}') = f_0(\boldsymbol{z}). \tag{6.2}$$

If $\boldsymbol{z}, \boldsymbol{z}' \in 4\kappa_d K_0 \cap \Gamma_0$ have the same f_0-value, Eq. (6.2) gives lattice lifts in $(4\kappa_d + rR/2)K$ whose difference lies in

$$\Lambda \cap (8\kappa_d + rR)K \subseteq \Lambda \cap aRK \subseteq V.$$

Hence $\boldsymbol{z} = \boldsymbol{z}'$, proving injectivity on $4\kappa_d K_0 \cap \Gamma_0$.

Apply Lemma 6.11 to K_0 and Γ_0, viewed as subsets of $\mathbb{R}^d$. It returns a basis matrix U of Γ_0 and an integer box B with $K_0 \subseteq UB \subseteq \kappa_d K_0$. Set $\Phi = U^{-1}\Phi_0$ and $\bar{f} = f_0 \circ U$; then $\bar{f}$ is injective on $4B \cap \mathbb{Z}^{d'}$.

Put $c = 1 + rR/2 = \mathcal{O}_d(1)$. The box containment, injectivity on $K_0 \cap \Gamma_0$, Eq. (6.2), and the lattice-point and quotient cases of Lemma 3.2 give

$$\begin{aligned}|B \cap \mathbb{Z}^{d'}| &\le |\kappa_d K_0 \cap \Gamma_0| \le \mathcal{O}_d(|K_0 \cap \Gamma_0|) \\ &\le \mathcal{O}_d(|f(cK \cap \Gamma)|) \le \mathcal{O}_d(|f(K \cap \Gamma)|),\end{aligned}$$

proving the cardinality bound.

All scales are $\mathcal{O}_d(1)$, and each selected lattice point $\boldsymbol{x}$ satisfies $\mathsf{maxbitlen}(\boldsymbol{x}) = \mathcal{O}_d(b_0)$ by (6.1). Rational linear algebra and Lemma 6.6 give $\mathsf{maxbitlen}(\Phi_0, \Gamma_0, f_0) = \mathcal{O}_d(b_0)$. By Lemma 6.3, $\mathsf{maxbitlen}(K_0) = \mathcal{O}_d(b_0)$, and K_0 has an optimization oracle within the claimed bound. Lemma 6.11 and (6.1) give $\mathsf{maxbitlen}(\Phi, \bar{f}, B) = \mathcal{O}_d(b_0)$ and the stated running time. □

7 Grouping the Input Integers

In this section, we prove Lemma 4.2 by repeatedly applying Theorem 5.13.

7.1 Preliminaries

We need the following standard estimate for the number of lattice points in a section.

Lemma 7.1. *Let $K \subseteq \mathbb{R}^d$ be an origin-symmetric compact convex set, $\Gamma \subseteq \mathbb{R}^d$ a lattice with $\operatorname{span}(K \cap \Gamma) = \mathbb{R}^d$, and $U \subseteq \mathbb{R}^d$ a linear subspace of dimension $d - k$. Then, for every $s > 0$, $|sK \cap \Gamma \cap U| \le (5d)^d\, s^{-k}\, |sK \cap \Gamma|$.*

Proof. Choose $\boldsymbol{v}_1, \dots, \boldsymbol{v}_k \in K \cap \Gamma$ so that $U, \boldsymbol{v}_1, \dots, \boldsymbol{v}_k$ together span $\mathbb{R}^d$. Then, the translated sets $(\Gamma \cap sK \cap U) + \sum_{i=1}^k a_i \boldsymbol{v}_i$, for integers $0 \le a_i \le \lfloor s/d \rfloor$, are disjoint subsets of $\Gamma \cap 2sK$. Hence,

$$(s/d)^k |\Gamma \cap sK \cap U| \le |\Gamma \cap 2sK|.$$

Applying Lemma 3.2 to sK gives $|\Gamma \cap 2sK| \le 5^d |\Gamma \cap sK|$. This proves the claim since $5^d d^k \le (5d)^d$. □

Given a multiset of vectors, the following lemma peels off a submultiset satisfying relative fatness, so that the remaining vectors have a smaller convex hull volume.

Lemma 7.2. *Let Y be a nonempty multiset of nonzero vectors in $\mathbb{Z}^d$, $H_Y = \operatorname{conv}(\pm Y)$, and $b_0 = \mathsf{maxbitlen}(Y)$. In $|\operatorname{supp} Y| b_0^{\mathcal{O}_d(1)}$ time, one can partition $Y = Z \sqcup W$ so that*

- *$(c_{\mathrm{peel},d}|Z|)H_Y \subseteq \mathcal{Z}_0(Z)$, where $c_{\mathrm{peel},d} \in (0,1)$ only depends on d,*
- *and $\operatorname{vol}_{\dim H_Y}(\operatorname{conv}(\pm W)) \le \frac{1}{2} \operatorname{vol}_{\dim H_Y}(H_Y)$.*

We prove Lemma 7.2 in Section 7.3.

Recall the following conventions from Section 4.1: Let $\boldsymbol{e}_1, \dots, \boldsymbol{e}_d$ be the standard basis of $\mathbb{R}^d$, and for every $1 \le d' \le d$ identify $\mathbb{R}^{d'}$ with the subspace $\operatorname{span}\{\boldsymbol{e}_1, \dots, \boldsymbol{e}_{d'}\}$ of $\mathbb{R}^d$. Write Π_U for the orthogonal projection onto the subspace U.

7.2 Proof of Lemma 4.2

We begin with an outline of the proof. The proof is split into a main stage and a post-processing stage:

- The main stage partitions A into $\mathcal{O}_d(\log \Delta)$ groups with an extra part A^{ex}, and represents the grouped integers as vectors.
- The post-processing stage merges contiguous groups into larger groups.

The only purpose of the post-processing stage is to ensure the bound in Lemma 4.2 (iv), namely that the i-th group constitutes an $\Omega_d(1/\log \Delta)$ fraction of the total size of the first i groups. The other main properties, including relative fatness and s-generation, are established in the main stage. The fatness coefficient λ degrades from $\mathcal{O}_d(1)$ to $\mathcal{O}_d(\log \Delta)$ after post-processing.

The main stage consists of iterations $i = 1, 2, \dots$, where iteration i selects the i-th group of integers $f(E_i)$ from the remaining integers $f(Y_i)$, where E_i, Y_i are multisets of vectors in $\mathbb{Z}^{d_i}$, and f is projection to the first coordinate. Initially, $d_0 = 1$ and Y_0 includes all integers of A as 1-dimensional vectors.

In iteration i, first run the algorithmic Conlon–Fox–Pham theorem (Theorem 5.13) on the remaining vectors $Y_{i-1} \setminus E_{i-1} \subseteq \mathbb{Z}^{d_{i-1}}$ to embed them into a GAP, except for $\mathcal{O}_d(s \log \Delta)$ of them which are discarded and put into the extra part. By incorporating their coordinates from the GAP embedding, each remaining vector in $Y_{i-1} \setminus E_{i-1}$ receives a new representation as a possibly higher dimensional vector, while its image under f remains unchanged. These new vectors form the multiset Y_i, which contains a small s-generating submultiset E_i' due to the Conlon–Fox–Pham theorem. We then use Lemma 7.2 to extract a submultiset of Y_i that satisfies relative fatness. Together with E_i', they form the i-th group E_i.

The main technical details are to bound the number of iterations, and to show that the dimension of the representing vectors does not exceed d. This requires keeping track of the lattice counts of the GAPs and changes in their dimensions.

Now we proceed to the formal proof of Lemma 4.2.

If $\Delta = \mathcal{O}_d(1)$, then $|A| \le \Delta^{\mathcal{O}(1)} = \mathcal{O}_d(1)$, and we can return no groups and $A^{\mathrm{ex}} = A$ to satisfy all the requirements. We henceforth assume Δ is sufficiently large in terms of d.

Basic setup. Let $C_{\mathrm{CFP},d}$ and $r_{\mathrm{CFP},d}$ be the upper bounds for the constant C and the returned rank in Theorem 5.13 for $\beta = d + 1$ and all $1 \le k \le d$. Let $C_{\mathrm{count},d} = (5d)^d$. Choose $C_{\mathrm{sc},d} > d$ sufficiently large in terms of $C_{\mathrm{CFP},d}, r_{\mathrm{CFP},d}, C_{\mathrm{count},d}$, and d. Define

$$s = \lceil C_{\mathrm{sc},d} \Delta^{1/d} \rceil.$$

Recall $\mathrm{supp}(A) \subseteq \mathbb{Z} \cap [1, \Delta]$, and $f(\boldsymbol{x}) = x_1$. For $i \ge 0$, we will define $d_i \ge 1$ non-decreasing in i, and multisets $E_i \subseteq Y_i$ with $\mathrm{supp}(Y_i) \subseteq \mathbb{Z}^{d_i}$. The initial values are defined by

$$d_0 = 1, \ Y_0 = \{a\boldsymbol{e}_1 : a \in A\}, \ \text{and } E_0 = \varnothing.$$

For $i \ge 0$, write

$$\Gamma_i = \mathcal{L}(Y_i) \text{ and } H_i = \mathrm{conv}(\{\mathbf{0}\} \cup \pm Y_i).$$

We regard $H_i \cap \Gamma_i$ as a progression of rank $\dim H_i$ in $\mathrm{span}\, \Gamma_i$, embedded in $\mathbb{R}^{d_i}$. We maintain the following four invariants for $i \ge 0$:

$$f(H_i \cap \Gamma_i) \text{ is an } s\text{-proper progression}, \tag{7.1a}$$

$$\mathrm{supp}(Y_i) \subseteq (\mathbb{Z} \cap [-\Delta_i, \Delta_i])^{d_i} \setminus \{\mathbf{0}\}, \text{ where } \Delta_i \coloneqq C_{\mathrm{sc},d}^{d_i+1} \Delta, \tag{7.1b}$$

$$N_i \coloneqq |(sH_i) \cap \Gamma_i| \le 3s\Delta \left(\frac{C_{\mathrm{count},d}}{s}\right)^{q_i}, \text{ where } q_i \coloneqq d_i - \dim(H_i), \tag{7.1c}$$

$$1 \le d_i \le d. \tag{7.1d}$$

The invariants clearly hold at $i = 0$.

In the following, we describe iteration $i \ge 1$ of the main stage, assuming the invariants hold at $i - 1$.

Applying Conlon–Fox–Pham. Let $R \coloneqq Y_{i-1} \setminus E_{i-1}$. If $|R| < C_{\mathrm{sc},d} s \log \Delta$, then we stop the iterations and pass the produced $k_0 \coloneqq i - 1$ groups to the post-processing stage described at the end of the proof. Otherwise, we apply Theorem 5.13 to R with Δ_{i-1} in place of Δ and $d + 1$ in place of β. We first verify the requirements of Theorem 5.13: By (7.1b), $\mathrm{supp}(R) \subseteq ([-\Delta_{i-1}, \Delta_{i-1}]^{d_{i-1}} \cap \mathbb{Z}^{d_{i-1}}) \setminus \{\mathbf{0}\}$. The definition of s and Δ_{i-1} gives $s \ge \Delta_{i-1}^{1/(d+1)}$ for Δ sufficiently large in terms of d. Since $|R| \ge C_{\mathrm{sc},d} s \log \Delta$ and $\log \Delta_{i-1} = \log \Delta + \mathcal{O}_d(1)$, the choice of $C_{\mathrm{sc},d}$ gives $s \le C_{\mathrm{CFP},d}^{-1} \min\{\Delta_{i-1}, |R|/\log \Delta_{i-1}\}$ for Δ sufficiently large in terms of d. Thus, Theorem 5.13 applies.

Theorem 5.13 returns $R' \subsetneq \hat{R} \subseteq R$ and an s-proper GAP containing $\mathrm{supp}(\hat{R})$,

$$g(B \cap \mathbb{Z}^m), \text{ with } g\colon \mathbb{R}^m \to \mathbb{R}^{d_{i-1}} \text{ linear, } B = \prod_{j=1}^{m} [-w_j, w_j], \text{ and } w_j \le C_{\mathrm{CFP},d} \Delta_{i-1}, \tag{7.2}$$

such that R' is s-generating for the GAP, $m \le r_{\mathrm{CFP},d}$, $|R'| \le C_{\mathrm{CFP},d} s$, and $|R| - |\hat{R}| \le C_{\mathrm{CFP},d} s \log \Delta_{i-1}$. Let g^{-1} denote the unique inverting map from $g(B \cap \mathbb{Z}^m)$ to $B \cap \mathbb{Z}^m$.

Since R' is s-generating for the GAP, there is $\boldsymbol{z} \in \mathbb{Z}^m$ such that

$$\Sigma(g^{-1}(R')) \supseteq (\boldsymbol{z} + sB) \cap \mathbb{Z}^m. \tag{7.3}$$

In particular, we have

$$\mathcal{L}(g^{-1}(R')) = \mathbb{Z}^m \tag{7.4}$$

due to $sB \cap \mathbb{Z}^m \supseteq \{0, 1\}^m$.

Computing the translate and pre-images. We compute $g^{-1}(\boldsymbol{a})$ for every $\boldsymbol{a} \in \mathrm{supp}(\hat{R})$ by a brute force enumeration of all points in $B \cap \mathbb{Z}^m$. Then, we compute a $\boldsymbol{z}$ that satisfies Eq. (7.3) as follows. By Theorem 5.13 and $|R'| \le C_{\mathrm{CFP},d} s$,

$$\big||R'|B \cap \mathbb{Z}^m\big| \ll_d |sB \cap \mathbb{Z}^m| \ll_d |s(\mathrm{supp}(R) \cup \{\mathbf{0}\})| \le N_{i-1},$$

where the last inequality follows from $R \subseteq Y_{i-1}$, Eq. (7.1c), and the definitions of H_{i-1}, Γ_{i-1}. We use standard dynamic programming to compute $S \coloneqq \Sigma(g^{-1}(R')) \subseteq |R'|B \cap \mathbb{Z}^m$ in $\widetilde{\mathcal{O}}_d(|R'|N_{i-1}) = \widetilde{\mathcal{O}}_d(sN_{i-1})$ time. Since sB and $|R'|B$ are axis-aligned boxes, we can compute the multidimensional prefix sums of the indicator function of S, and then compute $|S \cap (\boldsymbol{z} + sB)|$ for all $\boldsymbol{z} \in |R'|B \cap \mathbb{Z}^m$, in $\widetilde{\mathcal{O}}_d(N_{i-1})$ total time. We choose $\boldsymbol{z}$ for which $|S \cap (\boldsymbol{z} + sB)| = |sB \cap \mathbb{Z}^m|$, so that (7.3) holds. The overall time complexity of these steps is $\widetilde{\mathcal{O}}_d(sN_{i-1})$.

New vector representations. Let $r_g = \dim g(\mathbb{R}^m)$. We may assume $g(\boldsymbol{e}_1), \dots, g(\boldsymbol{e}_{r_g})$ are linearly independent, by relabeling the coordinates of the GAP in Eq. (7.2) if necessary. Let $d_i \coloneqq d_{i-1} + (m - r_g) \ge d_{i-1} \ge 1$, and define $\phi\colon \mathbb{R}^m \to \mathbb{R}^{d_i}$ by

$$\phi(\boldsymbol{u}) \coloneqq (g(\boldsymbol{u}), u_{r_g+1}, \dots, u_m),$$

which is an injective linear map due to the linear independence of $g(\boldsymbol{e}_1), \ldots, g(\boldsymbol{e}_{r_g})$. Clearly,

$$\Pi_{\mathbb{R}^{d_{i-1}}}(\phi(\boldsymbol{u})) = g(\boldsymbol{u}) \text{ for all } \boldsymbol{u} \in \mathbb{R}^m, \tag{7.5}$$

and $\phi(\mathbb{Z}^m) \subseteq \mathbb{Z}^{d_i}$. Define

$$\begin{aligned} Y_i &\coloneqq \phi(g^{-1}(\hat{R})), \\ E_i' &\coloneqq \phi(g^{-1}(R')) \subseteq Y_i, \end{aligned}$$

with $\operatorname{supp}(Y_i) \subseteq \mathbb{Z}^{d_i} \setminus \{\mathbf{0}\}$. Eq. (7.5) gives

$$\Pi_{\mathbb{R}^{d_{i-1}}}(Y_i) = \hat{R} \subseteq R = Y_{i-1} \setminus E_{i-1}. \tag{7.6}$$

Coordinate bounds. Take arbitrary $\boldsymbol{y} \in Y_i = \phi(g^{-1}(\hat{R}))$. For $1 \le j \le d_{i-1}$, by Eq. (7.6), we have $|y_j| \le \max_{\boldsymbol{y}' \in Y_{i-1}} \|\boldsymbol{y}'\|_\infty \le \Delta_{i-1}$. For $d_{i-1} < j \le d_i$, $|y_j| \le \max_{\boldsymbol{u} \in g^{-1}(\hat{R})} \|\boldsymbol{u}\|_\infty \le \max_{\boldsymbol{u} \in B \cap \mathbb{Z}^m} \|\boldsymbol{u}\|_\infty$, so Eq. (7.2) and the choice of $C_{\mathrm{sc},d}$ give $|y_j| \le C_{\mathrm{CFP},d}\Delta_{i-1} \le C_{\mathrm{sc},d}\Delta_{i-1}$. In the latter case, $d_i \ge d_{i-1} + 1$, so $C_{\mathrm{sc},d}\Delta_{i-1} \le \Delta_i$ by the definition of Δ_i. Thus $\|\boldsymbol{y}\|_\infty \le \Delta_i$ in either case, proving (7.1b).

Generation. By definition of E_i' and Eq. (7.4), we have $\mathcal{L}(E_i') = \phi(\mathcal{L}(g^{-1}(R'))) = \phi(\mathbb{Z}^m)$. By definition of Y_i, we have $\operatorname{supp}(Y_i) \subseteq \phi(\mathbb{Z}^m)$. Since $E_i' \subseteq Y_i$, and $\Gamma_i = \mathcal{L}(Y_i)$, this means

$$\mathcal{L}(E_i') = \Gamma_i = \phi(\mathbb{Z}^m). \tag{7.7}$$

Taking real spans gives $\operatorname{span}(Y_i) = \operatorname{span}(\Gamma_i) = \phi(\mathbb{R}^m)$. Since $H_i = \operatorname{conv}(\{\mathbf{0}\} \cup \pm Y_i)$ and $\Gamma_i = \mathcal{L}(Y_i)$, we get

$$\operatorname{span}(H_i \cap \Gamma_i) = \operatorname{span}(H_i) = \phi(\mathbb{R}^m), \text{ and } \dim(H_i) = m, \tag{7.8}$$

where the last claim follows from the injectivity of ϕ.

By definition of Y_i, $\operatorname{supp}(Y_i) \subseteq \phi(B \cap \mathbb{Z}^m)$, so $H_i \subseteq \phi(B)$. Thus,

$$\phi(sB \cap \mathbb{Z}^m) \underbrace{=}_{\phi \text{ injective}} \phi(sB) \cap \phi(\mathbb{Z}^m) \underbrace{=}_{\text{Eq. (7.7)}} (s\phi(B)) \cap \Gamma_i \supseteq (sH_i) \cap \Gamma_i. \tag{7.9}$$

Let $\boldsymbol{\tau}_i \coloneqq \phi(\boldsymbol{z}) \in \Gamma_i$ for $\boldsymbol{z}$ defined in (7.3). Then, applying ϕ to (7.3) gives

$$\Sigma(E_i') \supseteq \phi((\boldsymbol{z} + sB) \cap \mathbb{Z}^m) = \boldsymbol{\tau}_i + \phi(sB \cap \mathbb{Z}^m) \underbrace{\supseteq}_{\text{Eq. (7.9)}} \boldsymbol{\tau}_i + (sH_i \cap \Gamma_i).$$

Hence E_i' is s-generating for $H_i \cap \Gamma_i$ with translate $\boldsymbol{\tau}_i$. We also have $|E_i'| = |R'| \le C_{\mathrm{CFP},d}s$.

Since the vertices of $\boldsymbol{z} + sB$ lie in $\mathbb{Z}^m$, taking convex hulls in (7.3) and symmetrizing gives $sB \subseteq \mathcal{Z}_0(g^{-1}(R'))$. Applying ϕ gives

$$sH_i \subseteq \phi(sB) \subseteq \mathcal{Z}_0(E_i'). \tag{7.10}$$

Properness. Applying g to Eq. (7.4) gives $g(\mathbb{Z}^m) = \mathcal{L}(R') \subseteq \Gamma_{i-1}$, so

$$\Pi_{\mathbb{R}^{d_{i-1}}}(\Gamma_i) \underbrace{=}_{\text{Eq. (7.7)}} \Pi_{\mathbb{R}^{d_{i-1}}}(\phi(\mathbb{Z}^m)) \underbrace{=}_{\text{Eq. (7.5)}} g(\mathbb{Z}^m) \subseteq \Gamma_{i-1}.$$

By (7.6) and the definition of H_i, H_{i-1}, we have $\Pi_{\mathbb{R}^{d_{i-1}}}(H_i) \subseteq H_{i-1}$. Thus, $\Pi_{\mathbb{R}^{d_{i-1}}}(sH_i \cap \Gamma_i) \subseteq sH_{i-1} \cap \Gamma_{i-1}$.

By Eq. (7.5) and the s-properness of the GAP, the map $\Pi_{\mathbb{R}^{d_{i-1}}}(\phi(\cdot))$ is injective on $sB \cap \mathbb{Z}^m$. Hence, $\Pi_{\mathbb{R}^{d_{i-1}}}$ is injective on $\phi(sB \cap \mathbb{Z}^m) \supseteq sH_i \cap \Gamma_i$. Thus, $\Pi_{\mathbb{R}^{d_{i-1}}} : (sH_i) \cap \Gamma_i \to (sH_{i-1}) \cap \Gamma_{i-1}$ is an injective map.

By (7.1a), f is injective on $(sH_{i-1}) \cap \Gamma_{i-1}$. By definition of f, $f \circ \Pi_{\mathbb{R}^{d_{i-1}}} = f$. Thus, f is injective on $(sH_i) \cap \Gamma_i$. Together with (7.8), this shows $f(H_i \cap \Gamma_i)$ is an s-proper progression, proving (7.1a) for i.

Lattice count and dimension bounds. Recall $N_i := |sH_i \cap \Gamma_i|$ from Eq. (7.1c). Since the map $\Pi_{\mathbb{R}^{d_{i-1}}}$ is injective from $(sH_i) \cap \Gamma_i$ to $(sH_{i-1}) \cap \Gamma_{i-1}$, we have $N_i \le N_{i-1}$.

Let $U := g(\mathbb{R}^m)$, which equals $\Pi_{\mathbb{R}^{d_{i-1}}}(\operatorname{span}(H_i))$ by Eqs. (7.5) and (7.8). Recall $r_g := \dim U$. Since $\Pi_{\mathbb{R}^{d_{i-1}}}(H_i) \subseteq H_{i-1}$, we have $U \subseteq \operatorname{span}(H_{i-1})$ and hence $r_g \le \dim(H_{i-1})$. By applying Lemma 7.1 in $\operatorname{span}(H_{i-1})$, whose dimension is $\dim(H_{i-1}) \le d$, we get

$$|(sH_{i-1}) \cap \Gamma_{i-1} \cap U| \le C_{\text{count},d} s^{-(\dim(H_{i-1}) - r_g)} N_{i-1}.$$

Since $\Pi_{\mathbb{R}^{d_{i-1}}}$ maps injectively from $(sH_i) \cap \Gamma_i$ to $(sH_{i-1}) \cap \Gamma_{i-1} \cap U$, this means

$$N_i \le C_{\text{count},d} s^{-(\dim(H_{i-1}) - r_g)} N_{i-1}. \tag{7.11}$$

Recall $d_i := d_{i-1} + (m - r_g) = d_{i-1} + (\dim(H_i) - r_g)$ (by Eq. (7.8)), and $q_i := d_i - \dim(H_i) = d_{i-1} - r_g, q_{i-1} := d_{i-1} - \dim(H_{i-1})$ from Eq. (7.1c). Putting them together gives

$$q_i - q_{i-1} = \dim(H_{i-1}) - r_g \ge 0. \tag{7.12}$$

If $q_i > q_{i-1}$, then (7.11) gives

$$N_i \le (C_{\text{count},d}/s)^{q_i - q_{i-1}} N_{i-1}.$$

If $q_i = q_{i-1}$, then use the initial bound $N_i \le N_{i-1}$. In either case, $N_i \le (C_{\text{count},d}/s)^{q_i - q_{i-1}} N_{i-1}$ holds, proving (7.1c) for i.

By Theorem 5.13,

$$s^m \le |sB \cap \mathbb{Z}^m| \ll_d |s(\operatorname{supp}(\hat{R}) \cup \{\mathbf{0}\})| \le |s(\operatorname{supp}(Y_i) \cup \{\mathbf{0}\})| \le N_i,$$

where the third inequality uses $\Pi_{\mathbb{R}^{d_{i-1}}}(Y_i) = \hat{R}$ by Eq. (7.6), and the last inequality uses $\operatorname{supp}(Y_i) \subseteq H_i \cap \Gamma_i$. Combining this with Eq. (7.1c) and $q_i = d_i - m$ gives

$$s^{d_i - 1} \ll_d \Delta C_{\text{count},d}^{q_i} = \mathcal{O}_d(\Delta),$$

where the last step uses $q_i = d_{i-1} - r_g \le d$ by Eq. (7.1d). If $d_i \ge d+1$, this contradicts $s^d \ge C_{\text{sc},d}^d \Delta$ (by definition of s) when $C_{\text{sc},d}$ is sufficiently large in terms of d. Thus $d_i \le d$, proving (7.1d) for i.

Extracting the group. Apply Lemma 7.2 to Y_i, obtaining $Y_i = Z_i \sqcup W_i$ such that

$$\begin{aligned}(c_{\mathrm{peel},d}|Z_i|)H_i &\subseteq \mathcal{Z}_0(Z_i),\\ \mathrm{vol}_{\dim(H_i)}(\mathrm{conv}(\pm W_i)) &\le \tfrac{1}{2}\,\mathrm{vol}_{\dim(H_i)}(H_i).\end{aligned} \tag{7.13}$$

Define the multiset $E_i \coloneqq Z_i \cup E_i' \subseteq Y_i$, where the multiplicities of the union are defined by the maximum (instead of sum) of the corresponding multiplicities in the two multisets. Since $Y_i \setminus E_i \subseteq W_i$,

$$\mathrm{vol}_{\dim(H_i)}(\mathrm{conv}(\pm(Y_i \setminus E_i))) \le \tfrac{1}{2}\,\mathrm{vol}_{\dim(H_i)}(H_i). \tag{7.14}$$

By $|E_i| \le |Z_i| + |E_i'| \le |Z_i| + C_{\mathrm{CFP},d}s$, we have

$$|E_i|H_i \subseteq |Z_i|H_i + C_{\mathrm{CFP},d}sH_i \underbrace{\subseteq}_{(7.13),\,(7.10)} c_{\mathrm{peel},d}^{-1}\mathcal{Z}_0(Z_i) + C_{\mathrm{CFP},d}\mathcal{Z}_0(E_i') \subseteq (c_{\mathrm{peel},d}^{-1} + C_{\mathrm{CFP},d})\mathcal{Z}_0(E_i). \tag{7.15}$$

Set $A_i^{\mathrm{ex}} = f(R \setminus \hat{R})$. Then $|A_i^{\mathrm{ex}}| \le C_{\mathrm{CFP},d}s\log\Delta_{i-1}$ and, by (7.6) and the definition of Y_i,

$$f(Y_{i-1} \setminus E_{i-1}) = A_i^{\mathrm{ex}} \sqcup f(E_i) \sqcup f(Y_i \setminus E_i). \tag{7.16}$$

Iteration count. Since all vertices of H_i are integer vectors with coordinates at most $\mathcal{O}_d(\Delta)$, we have

$$1/d! \le \mathrm{vol}_{\dim(H_i)}(H_i) \le \mathcal{O}_d(\Delta^d). \tag{7.17}$$

By (7.12) and (7.1d), both d_i and q_i are nondecreasing integers in $[0,d]$, so there are $\mathcal{O}(d)$ maximal intervals of consecutive iterations i with (d_i,q_i) being constant. On each such interval, $\dim(H_i) = d_i - q_i$ also stays constant. For consecutive iterations $i-1, i$ in one such interval, (7.6) gives $Y_i \subseteq Y_{i-1} \setminus E_{i-1}$, so (7.14) (applied to $i-1$) gives

$$\mathrm{vol}_{\dim(H_i)}(H_i) \le \tfrac{1}{2}\,\mathrm{vol}_{\dim(H_{i-1})}(H_{i-1}).$$

Thus, by (7.17), each such interval has length $\mathcal{O}_d(\log\Delta)$. Thus the main stage terminates after k_0 iterations for some $k_0 = \mathcal{O}_d(\log\Delta)$.

Iterating (7.16) gives $A = A^{\mathrm{ex}} \sqcup \bigsqcup_{i=1}^{k_0} f(E_i)$ where

$$A^{\mathrm{ex}} \coloneqq f(Y_{k_0} \setminus E_{k_0}) \sqcup \bigsqcup_{i=1}^{k_0} A_i^{\mathrm{ex}}.$$

The termination rule gives $|Y_{k_0} \setminus E_{k_0}| \le C_{\mathrm{sc},d}s\log\Delta$. Since $|A_i^{\mathrm{ex}}| = \mathcal{O}_d(s\log\Delta_{i-1})$, we obtain $|A^{\mathrm{ex}}| = \mathcal{O}_d(s\log^2\Delta)$ as required.

Time complexity. The application of the algorithmic Conlon–Fox–Pham theorem (Theorem 5.13) takes $\widetilde{\mathcal{O}}_d(s|s(\mathrm{supp}(R)\cup\{\mathbf{0}\})|) = \widetilde{\mathcal{O}}_d(sN_{i-1})$ time because $s(\mathrm{supp}(R)\cup\{\mathbf{0}\}) \subseteq (sH_{i-1})\cap\Gamma_{i-1}$. Earlier, we argued that computing the translate $\boldsymbol{z}$ and $g^{-1}(\boldsymbol{a})$ for all $\boldsymbol{a} \in \mathrm{supp}(\hat{R})$ also takes $\widetilde{\mathcal{O}}_d(sN_{i-1})$ time.

The application of Lemma 7.2 takes $\mathcal{O}_d(|\operatorname{supp} R|)(\log\Delta)^{\mathcal{O}_d(1)}$ time, which is dominated by the time complexity of Theorem 5.13. By (7.1c), we have $N_{i-1} \le 3s\Delta$ provided $s \ge C_{\mathrm{count},d}$ (which holds when Δ is sufficiently large in terms of d). Therefore, each iteration takes $\widetilde{\mathcal{O}}_d(s^2\Delta)$ time. The total time complexity over all $k_0 = \mathcal{O}_d(\log\Delta)$ iterations is therefore $\widetilde{\mathcal{O}}_d(s^2\Delta)$ as claimed.

Post-processing stage. Now we merge some of the contiguous groups among the k_0 obtained groups based on their sizes. Consider the prefix maxima,

$$\{i_1 < \cdots < i_k\} = \{r \in [k_0] : |E_r| = \max_{j \in [r]} |E_j|\},$$

and let $i_{k+1} = k_0 + 1$. For $u \in [k]$, define

$$X_u = \bigsqcup_{r=i_u}^{i_{u+1}-1} \Pi_{\mathbb{R}^{d_{i_u}}}(E_r), \ A_u = f(X_u),$$

and define $X'_u = E'_{i_u}$. Then, $|X'_u| \le |E'_{i_u}| \le C_{\mathrm{CFP},d} s$, proving Item (ii). Since $f \circ \Pi_{\mathbb{R}^{d_{i_u}}} = f$ and $A = A^{\mathrm{ex}} \sqcup \bigsqcup_{i=1}^{k_0} f(E_i)$, we have $A = A^{\mathrm{ex}} \sqcup \bigsqcup_{u=1}^{k} f(X_u)$ as required. The returned dimensions are $d_{i_1} \le \cdots \le d_{i_k}$, which are in $[1, d]$, and $k \le k_0 = \mathcal{O}_d(\log \Delta)$, as required.

By definition of the prefix maxima i_u and i_{u+1}, we have $|E_i| \le |E_{i_u}|$ for $1 \le i < i_{u+1}$. Therefore,

$$|E_{i_u}| \le |X_u| = |A_u| \le \sum_{v=1}^{u} |A_v| = \sum_{i=1}^{i_{u+1}-1} |E_i| \le k_0 |E_{i_u}|,$$

and in particular, $\sum_{v=1}^{u} |A_v| \le k_0 |A_u|$. Together with $k_0 = \mathcal{O}_d(\log \Delta)$, this proves Lemma 4.2 (iv).

By (7.6), $E_i \sqcup \Pi_{\mathbb{R}^{d_i}}(Y_{i+1}) \subseteq Y_i$ for $1 \le i < k_0$; starting from $E_{k_0} \subseteq Y_{k_0}$ and iteratively applying this containment for $i = k_0 - 1, \dots, i_u$, we obtain

$$\bigsqcup_{v=u}^{k} \Pi_{\mathbb{R}^{d_{i_u}}}(X_v) = \bigsqcup_{i=i_u}^{k_0} \Pi_{\mathbb{R}^{d_{i_u}}}(E_i) \subseteq Y_{i_u}.$$

Thus,

$$\tilde{H}_u \coloneqq \operatorname{conv}\left(\pm \bigcup_{v=u}^{k} \Pi_{\mathbb{R}^{d_{i_u}}}(\operatorname{supp} X_v) \right) \subseteq H_{i_u}.$$

Since $|X_u| \le k_0 |E_{i_u}|$ and $E_{i_u} \subseteq X_u$,

$$|X_u| H_{i_u} \subseteq k_0 |E_{i_u}| H_{i_u} \underbrace{\subseteq}_{(7.15)} k_0 (c_{\mathrm{peel},d}^{-1} + C_{\mathrm{CFP},d}) \mathcal{Z}_0(E_{i_u}) \subseteq k_0 (c_{\mathrm{peel},d}^{-1} + C_{\mathrm{CFP},d}) \mathcal{Z}_0(X_u).$$

Since $\operatorname{supp} X_u \subseteq \tilde{H}_u \subseteq H_{i_u}$, this proves (4.1b) with $\lambda = k_0 (c_{\mathrm{peel},d}^{-1} + C_{\mathrm{CFP},d}) = \mathcal{O}_d(\log \Delta)$.

Since $E'_{i_u} \subseteq E_{i_u} \subseteq X_u$ by definition, (7.7) gives

$$\Gamma_{i_u} = \mathcal{L}(E'_{i_u}) \subseteq \mathcal{L}(X_u) \subseteq \mathcal{L}(Y_{i_u}) = \Gamma_{i_u},$$

so all containments are equalities. Thus, for all $v \ge u$, $\Pi_{\mathbb{R}^{d_{i_u}}}(\operatorname{supp} X_v) \subseteq \operatorname{supp} Y_{i_u} \subseteq \Gamma_{i_u} = \mathcal{L}(X_u)$, proving Item (iii).

Since $\operatorname{supp} X_u \subseteq \tilde{H}_u \subseteq \operatorname{span} \Gamma_{i_u}$ and $\mathcal{L}(X_u) = \Gamma_{i_u}$, we have $\operatorname{span}(\tilde{H}_u \cap \Gamma_{i_u}) = \operatorname{span} \Gamma_{i_u}$. Together with $\tilde{H}_u \subseteq H_{i_u}$, (7.1a) implies that $f(\tilde{H}_u \cap \mathcal{L}(X_u))$ is an s-proper progression, proving (4.1a). Moreover, $X'_u = E'_{i_u}$ is s-generating for $H_{i_u} \cap \Gamma_{i_u}$ with translate $\boldsymbol{\tau}_{i_u}$, and $\operatorname{rank} \mathcal{L}(X_u) \le d \le s$, so X'_u is $(\operatorname{rank} \mathcal{L}(X_u))$-generating for $\tilde{H}_u \cap \mathcal{L}(X_u)$, proving (4.1c). We return $\boldsymbol{\tau}_{i_u}$ as the translate for X'_u.

By (7.1b), $\operatorname{supp} X_u \subseteq \operatorname{supp} Y_{i_u} \subseteq (\mathbb{Z} \cap [-\Delta_{i_u}, \Delta_{i_u}])^{d_{i_u}}$, where $\Delta_{i_u} \le C_{\mathrm{sc},d}^{d+1} \Delta = \mathcal{O}_d(\Delta) \le \Delta^{\mathcal{O}(1)}$ for Δ sufficiently large in terms of d, as required.

The time complexity of the post-processing stage is dominated by the time complexity for the main stage.

7.3 Proof of Lemma 7.2

We prove Lemma 7.2 by partitioning a linear image of Y into $\mathcal{O}_d(1)$ boxes and selecting a submultiset from each box.

Set $d' = \dim \operatorname{span} Y$ and apply Lemma 6.4 to $\operatorname{conv}(\operatorname{supp} Y)$, using a scan of $\operatorname{supp} Y$ for each optimization query. The d' chosen input vectors define an isomorphism $M\colon \mathbb{R}^{d'} \to \operatorname{span} Y$, computed in $|\operatorname{supp} Y| b_0^{\mathcal{O}_d(1)}$ time. Compute $\hat{Y} = M^{-1}Y$, and set $H = \operatorname{conv}(\pm\hat{Y})$. Since $\{\boldsymbol{e}_1, \dots, \boldsymbol{e}_{d'}\} \subseteq \operatorname{supp} \hat{Y} \subseteq [-2,2]^{d'}$, we have

$$(d')^{-1}[-1,1]^{d'} \subseteq H \subseteq [-2,2]^{d'}.$$

Grouping nearby vectors. Choose $\varepsilon_d > 0$ sufficiently small in terms of d. By multilinearity of the determinant, partition $[-2,2]^{d'}$ into $\mathcal{O}_d(1)$ boxes so finely that

$$|\det(\boldsymbol{x}_1, \dots, \boldsymbol{x}_{d'}) - \det(\boldsymbol{y}_1, \dots, \boldsymbol{y}_{d'})| \le \varepsilon_d \tag{7.18}$$

whenever $\boldsymbol{x}_j, \boldsymbol{y}_j$ belong to the same box for each j. Write $\mathcal{Q}$ for this partition and $\boldsymbol{m}(Q)$ for the center of a box Q. Define

$$\mathcal{T} = \left\{(Q_1, \dots, Q_{d'}) \in \mathcal{Q}^{d'} : |\det(\boldsymbol{m}(Q_1), \dots, \boldsymbol{m}(Q_{d'}))| \ge 3\varepsilon_d\right\}.$$

For $(Q_1, \dots, Q_{d'}) \in \mathcal{T}$, (7.18) gives $|\det(\boldsymbol{x}_1, \dots, \boldsymbol{x}_{d'})| \ge 2\varepsilon_d$ whenever $\boldsymbol{x}_j \in Q_j$ for every $j \in [d']$. Conversely, if $\boldsymbol{x}_j \in Q_j$ and $|\det(\boldsymbol{x}_1, \dots, \boldsymbol{x}_{d'})| \ge 4\varepsilon_d$, then $(Q_1, \dots, Q_{d'}) \in \mathcal{T}$.

The common multiplicity threshold. Let a_Q count the occurrences of $\hat{Y}$ in Q, and set

$$\ell = \max_{(Q_1,\dots,Q_{d'}) \in \mathcal{T}} \min_{j \in [d']} a_{Q_j}.$$

From each box Q, select $\min\{a_Q, \ell\}$ occurrences, obtaining $\hat{Z}$; let $\hat{W}$ contain the remaining occurrences. Then $|\hat{Z}| \le |\mathcal{Q}|\ell = \mathcal{O}_d(\ell)$, and $Q \cap \operatorname{supp} \hat{W} \ne \varnothing$ exactly when $a_Q > \ell$. By the definition of ℓ, every $(Q_1, \dots, Q_{d'}) \in \mathcal{T}$ has some $j \in [d']$ with $Q_j \cap \operatorname{supp} \hat{W} = \varnothing$.

Zonotope balance. Choose $(Q_1, \dots, Q_{d'}) \in \mathcal{T}$ with $\min_j a_{Q_j} = \ell$. For each Q_j, let $\boldsymbol{w}_j$ be the average of its ℓ selected vectors. Convexity gives $\boldsymbol{w}_j \in Q_j$, so $|\det(\boldsymbol{w}_1, \dots, \boldsymbol{w}_{d'})| \ge 2\varepsilon_d$. Since H and all $\boldsymbol{w}_j$ lie in $[-2,2]^{d'}$, Cramer's rule gives $H \subseteq \{\sum_{j=1}^{d'} t_j \boldsymbol{w}_j : |t_j| \le C_d\}$ for a constant C_d depending only on d. Moreover, the selected vectors in Q_j sum to $\ell \boldsymbol{w}_j$, so their centered zonotope contains $[-\ell\boldsymbol{w}_j/2, \ell\boldsymbol{w}_j/2]$. Consequently,

$$(c_{\mathrm{peel},d}|\hat{Z}|)H \subseteq \sum_{j=1}^{d'} [-\ell\boldsymbol{w}_j/2, \ell\boldsymbol{w}_j/2] \subseteq \mathcal{Z}_0(\hat{Z}).$$

Here $c_{\mathrm{peel},d} > 0$ is sufficiently small in terms of d.

Volume of the remainder. If $\dim \operatorname{span} \hat{W} < d'$, then $\operatorname{vol}_{d'}(\operatorname{conv}(\pm\hat{W})) = 0$. Otherwise, apply Lemma 6.4 to $\operatorname{conv}(\operatorname{supp} \hat{W})$. Its selected vertices $\boldsymbol{v}_1, \dots, \boldsymbol{v}_{d'}$ belong to $\operatorname{supp} \hat{W}$. Thus,

$$\operatorname{conv}(\pm\hat{W}) \subseteq \left\{\sum_{j=1}^{d'} t_j \boldsymbol{v}_j : |t_j| \le 2\right\}.$$

Their boxes form a tuple outside $\mathcal{T}$, so $\delta_{\text{rem}} = |\det(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_{d'})| < 4\varepsilon_d$. The parallelotope on the right has volume $4^{d'}\delta_{\text{rem}} = \mathcal{O}_d(\varepsilon_d)$. Since $\operatorname{vol}_{d'}(H) \geq (2/d')^{d'}$, choosing ε_d sufficiently small gives

$$\operatorname{vol}_{d'}(\operatorname{conv}(\pm\hat{W})) \leq \tfrac{1}{2}\operatorname{vol}_{d'}(H).$$

Set $Z = M\hat{Z}$ and $W = M\hat{W}$. Applying M gives $(c_{\text{peel},d}|Z|)\operatorname{conv}(\pm Y) \subseteq \mathcal{Z}_0(Z)$ and preserves the volume ratio.

Implementation. Compute a_Q for every $Q \in \mathcal{Q}$, enumerate $\mathcal{T}$ to find ℓ, and select the prescribed multiplicity from each box in support order. We have $\mathsf{maxbitlen}(\hat{Y}) = \mathcal{O}_d(b_0)$. Together with the basis computation, this takes $|\operatorname{supp} Y| b_0^{\mathcal{O}_d(1)}$ time.

8 Macbeath Regions

Recall the definition of Macbeath regions [Mac52].

Definition 8.1 (Macbeath regions)**.** For a convex body $K \subseteq \mathbb{R}^d$, $\boldsymbol{x} \in K$, and $\lambda \geq 0$, the *λ-scaled Macbeath region* at $\boldsymbol{x}$ is defined as

$$M_K^\lambda(\boldsymbol{x}) = \boldsymbol{x} + \lambda((K - \boldsymbol{x}) \cap (\boldsymbol{x} - K)).$$

Macbeath regions are convex and centrally symmetric.

8.1 Expansion Containment

Ewald, Larman, and Rogers [ELR70] proved the expansion-containment property of Macbeath regions. In the parameterized form of Abdelkader and Mount [AM18, Lemma 1], for $\boldsymbol{x}, \boldsymbol{y} \in K$ and $0 < \lambda < 1$, if $M_K^\lambda(\boldsymbol{x}) \cap M_K^\lambda(\boldsymbol{y}) \neq \varnothing$, then for $\alpha \geq 0$ and $\beta = \frac{2+\alpha(1+\lambda)}{1-\lambda}$, $M_K^{\alpha\lambda}(\boldsymbol{x}) \subseteq M_K^{\beta\lambda}(\boldsymbol{y})$.

We state and prove two variants of the expansion-containment property. The proofs are similar to the original proof in [ELR70].

The first variant relaxes the $\lambda < 1$ restriction, but additionally requires the center of one Macbeath region to be contained in the other.

Lemma 8.2. *Let $K \subseteq \mathbb{R}^d$ be a convex body, $\boldsymbol{x}, \boldsymbol{y} \in K$, and $\lambda \geq 0$. If $\boldsymbol{x} \in M_K^\lambda(\boldsymbol{y})$, then, for $\alpha \geq 1$, $M_K^\alpha(\boldsymbol{x}) \subseteq M_K^{(2\lambda+1)\alpha}(\boldsymbol{y})$.*

Proof. Since $\boldsymbol{x} \in M_K^\lambda(\boldsymbol{y})$, we have $\boldsymbol{x} = \boldsymbol{y} + \lambda\boldsymbol{u}$ for some $\boldsymbol{u}$ with $\boldsymbol{y} \pm \boldsymbol{u} \in K$. For every $\boldsymbol{v}$ with $\boldsymbol{x} \pm \boldsymbol{v} \in K$, convexity of K gives

$$\boldsymbol{y} \pm \frac{\boldsymbol{v}}{\lambda+1} = \frac{\lambda(\boldsymbol{y} - \boldsymbol{u}) + (\boldsymbol{x} \pm \boldsymbol{v})}{\lambda+1} \in K.$$

Thus both $\boldsymbol{y} + \boldsymbol{u}$ and $\boldsymbol{y} + \boldsymbol{v}/(\lambda+1)$ lie in $M_K^1(\boldsymbol{y})$. Then, convexity of $M_K^1(\boldsymbol{y})$ gives

$$\boldsymbol{y} + \frac{\boldsymbol{x} + \alpha\boldsymbol{v} - \boldsymbol{y}}{\lambda + \alpha(\lambda+1)} = \frac{\lambda}{\lambda + \alpha(\lambda+1)}(\boldsymbol{y} + \boldsymbol{u}) + \frac{\alpha(\lambda+1)}{\lambda + \alpha(\lambda+1)}\left(\boldsymbol{y} + \frac{\boldsymbol{v}}{\lambda+1}\right) \in M_K^1(\boldsymbol{y}).$$

Since every point of $M_K^\alpha(\boldsymbol{x})$ has the form $\boldsymbol{x} + \alpha\boldsymbol{v}$ with $\boldsymbol{x} \pm \boldsymbol{v} \in K$, this gives

$$M_K^\alpha(\boldsymbol{x}) \subseteq M_K^{\lambda+\alpha(\lambda+1)}(\boldsymbol{y}) \subseteq M_K^{(2\lambda+1)\alpha}(\boldsymbol{y}). \qquad \square$$

The second variant involves two shrunken Macbeath regions whose images modulo a lattice Λ have non-empty intersection. In general, this condition is not sufficient for the expansion-containment property, but we can recover it by imposing a balance condition on one of the centers, as defined below.

Definition 8.3 (Balancedness)**.** Let $K \subseteq \mathbb{R}^d$ be a convex body, $\Lambda \subseteq \mathbb{R}^d$ be a lattice, and $\gamma \geq 1$. A point $\boldsymbol{y} \in K$ is *γ-balanced in K modulo Λ* if $K \cap (\boldsymbol{y} + \Lambda) \subseteq M_K^{\gamma}(\boldsymbol{y})$.

Lemma 8.4. *Let $K \subseteq \mathbb{R}^d$ be a convex body, $\Lambda \subseteq \mathbb{R}^d$ be a lattice, $\boldsymbol{x}, \boldsymbol{y} \in K$, and $\gamma \geq 1$. If $\boldsymbol{y}$ is γ-balanced modulo $\Lambda/2$ and $\pi_\Lambda(M_K^{1/2}(\boldsymbol{x})) \cap \pi_\Lambda(M_K^{1/2}(\boldsymbol{y})) \neq \varnothing$, then, for $\alpha \geq 1$, $M_K^{\alpha}(\boldsymbol{x}) \subseteq M_K^{13\gamma\alpha}(\boldsymbol{y})$.*

Proof. The intersection hypothesis gives $\boldsymbol{y} + \boldsymbol{h}/2 = \boldsymbol{x} + \boldsymbol{h}'/2 + \boldsymbol{\ell}$ with $\boldsymbol{y} \pm \boldsymbol{h}, \boldsymbol{x} \pm \boldsymbol{h}' \in K$ and $\boldsymbol{\ell} \in \Lambda$. Then, by convexity of K,

$$\boldsymbol{c} \coloneqq \boldsymbol{y} - \boldsymbol{\ell}/2 = \frac{1}{2}\left(\boldsymbol{x} + \frac{\boldsymbol{h}'}{2}\right) + \frac{1}{2}\left(\boldsymbol{y} - \frac{\boldsymbol{h}}{2}\right) \in K.$$

Since $\boldsymbol{c} \in K \cap (\boldsymbol{y} + \Lambda/2)$ and $\boldsymbol{y}$ is γ-balanced modulo $\Lambda/2$, we have $\boldsymbol{c} \in M_K^{\gamma}(\boldsymbol{y})$, that is, $\boldsymbol{y} \pm (\boldsymbol{c} - \boldsymbol{y})/\gamma \in K$. Hence, by convexity of K,

$$\boldsymbol{y} - \frac{\boldsymbol{x} - \boldsymbol{y}}{4\gamma + 2} = \frac{4\gamma(\boldsymbol{y} - (\boldsymbol{c} - \boldsymbol{y})/\gamma) + (\boldsymbol{y} - \boldsymbol{h}) + (\boldsymbol{x} + \boldsymbol{h}')}{4\gamma + 2} \in K.$$

Since $\boldsymbol{y}, \boldsymbol{x} \in K$, this gives $\boldsymbol{x} \in M_K^{4\gamma+2}(\boldsymbol{y})$. Then, by Lemma 8.2,

$$M_K^{\alpha}(\boldsymbol{x}) \subseteq M_K^{(8\gamma+5)\alpha}(\boldsymbol{y}) \subseteq M_K^{13\gamma\alpha}(\boldsymbol{y}). \qquad \square$$

8.2 Finding a balanced center

We use the following lemma to find a balanced center.

Lemma 8.5. *For any $d \geq 1$, there are $q_d \in \mathbb{Z}_{\geq 1}$ and $\gamma_d \geq 1$ such that the following holds.*

Let $K \subseteq \mathbb{R}^d$ be a convex body, $\Lambda \subseteq \mathbb{R}^d$ be a lattice, and $\boldsymbol{x}_0 \in K$. There exists $\boldsymbol{y} \in K \cap (\boldsymbol{x}_0 + q_d^{-1}\Lambda)$ that is γ_d-balanced in K modulo Λ, and satisfies $K \cap (\boldsymbol{x}_0 + \Lambda) \subseteq M_K^{\gamma_d}(\boldsymbol{y})$.

If K is a rational polytope given by a linear optimization oracle and $\Lambda, \boldsymbol{x}_0$ are rational, then one can compute such $\boldsymbol{y}$ in $b_0^{\mathcal{O}_d(1)} T_{\mathrm{query}}(K)$ time, where $b_0 = \mathsf{maxbitlen}(K, \Lambda, \boldsymbol{x}_0)$.

Proof. By translation, we may assume $\boldsymbol{x}_0 = \boldsymbol{0}$ without loss of generality. Choose an integer β at least as large as $\beta_1, \ldots, \beta_d$ from Lemma 6.5, and set $q = q_d = 2d\beta$ and $\gamma_d = 2\beta + 1$. Set $V = \operatorname{span}(K \cap \Lambda)$, choose a basis $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_{d'}$ of V from $K \cap \Lambda$, and set $F = K \cap V$. We enlarge this section as needed, preserving the invariant

$$K \cap \Lambda \subseteq F, \qquad \boldsymbol{v}_i \in \Lambda \cap (F - F) \quad (1 \leq i \leq d'). \tag{8.1}$$

Apply Lemma 6.5 to the convex body F in V, obtaining $\boldsymbol{c} \in F$ and an origin-symmetric parallelotope B with $\boldsymbol{c} + B \subseteq F \subseteq \boldsymbol{c} + \beta B$. Write $\boldsymbol{c} = \sum_{i=1}^{d'} t_i \boldsymbol{v}_i$ and round each qt_i to a nearest integer n_i. Define $\boldsymbol{y} = q^{-1} \sum_{i=1}^{d'} n_i \boldsymbol{v}_i$. By (8.1), $\boldsymbol{v}_i \in 2\beta B$, so

$$\boldsymbol{y} - \boldsymbol{c} \in \frac{d'\beta}{q} B \subseteq \frac{1}{2} B.$$

Hence $\boldsymbol{y} + B/2 \subseteq F \subseteq \boldsymbol{y} + (\beta + 1/2)B$, giving

$$\boldsymbol{y} \in F \cap q^{-1}\Lambda, \qquad F \subseteq M_K^{\gamma_d}(\boldsymbol{y}). \tag{8.2}$$

Search for $\boldsymbol{x} \in K \cap (\boldsymbol{y} + \Lambda)$ outside V. If such a point exists, append $\boldsymbol{v}_{d'+1} = \boldsymbol{x} - \boldsymbol{y}$ and repeat the rounding in the enlarged section. Since $\boldsymbol{v}_{d'+1} \in \Lambda \setminus V$, appending it preserves independence. The enlarged section contains the previous section and both $\boldsymbol{x}, \boldsymbol{y}$, so $\boldsymbol{v}_{d'+1} = \boldsymbol{x} - \boldsymbol{y}$ lies in its difference body and (8.1) remains valid. There are at most $\operatorname{rank}\Lambda$ enlargements. When no such point exists, $K \cap (\boldsymbol{y} + \Lambda) \subseteq F$, so (8.2) proves balance, and (8.1) gives the other stated containment.

For rational inputs, carry out the same construction algorithmically. For the initial spanning set, start with $V = \{\boldsymbol{0}\}$ and optimize both signs of $\langle \boldsymbol{u}, \boldsymbol{x} \rangle$, for a basis of vectors $\boldsymbol{u} \in V^{\perp}$, over $K \cap \Lambda$. A positive optimum gives the next independent vector; if all optima are zero, the intersection lies in V. Use the same tests over $K \cap (\boldsymbol{y} + \Lambda)$ for each subsequent search. Lemmas 6.3 and 6.9 implement these queries and the optimization oracles for F. All points selected from these intersections and all rounded centers lie in $K \cap q^{-1}\Lambda$, and each selected $\boldsymbol{v}_i$ lies in $\Lambda \cap (K - K)$. By (6.1), these points and vectors have $\mathsf{maxbitlen} = \mathcal{O}_d(b_0)$. The same bound holds for the section equations, their vertices, and the parallelotope approximations by fixed-dimensional rational linear algebra and Lemmas 6.3 and 6.5. There are at most $\operatorname{rank}\Lambda + 1$ roundings; the stated time bound follows from Lemma 6.5 and the same oracle bounds. □

8.3 Constructing a cap

For a convex body $K \subseteq \mathbb{R}^d$, a *cap* of K is the intersection of K with a halfspace, $Q = K \cap \{\boldsymbol{x} \in \mathbb{R}^d : \langle \boldsymbol{u}, \boldsymbol{x} \rangle \geq a\}$, and is represented by its defining halfspace $(\boldsymbol{u}, a)$.

Classical cap–Macbeath relations compare a sufficiently shallow cap with a Macbeath region centered at the centroid of its base [AdFM17, Lemma 2.3]. Here the center is prescribed; we construct a cap comparable to its Macbeath region.

Lemma 8.6. *Let $K \subseteq \mathbb{R}^d$ be a convex body and let $\boldsymbol{y}$ be an interior point of K. There exists $\boldsymbol{u} \neq \boldsymbol{0}$ such that, for every $t \geq 1$, the cap $Q_t = \{\boldsymbol{x} \in K : \langle \boldsymbol{u}, \boldsymbol{x} - \boldsymbol{y} \rangle \geq -t\}$ satisfies*

$$K \cap M_K^t(\boldsymbol{y}) \subseteq Q_t \subseteq M_K^{(4d+3)t}(\boldsymbol{y}). \tag{8.3}$$

If K is a rational polytope given by a linear optimization oracle and $\boldsymbol{y}$ is rational, one can compute $\boldsymbol{u}$ with $\mathsf{maxbitlen}(\boldsymbol{u}) = \mathcal{O}_d(b_0)$ in $b_0^{\mathcal{O}_d(1)} T_{\mathrm{query}}(K)$ time, where $b_0 = \mathsf{maxbitlen}(K, \boldsymbol{y})$.

Proof. Set $A = K - \boldsymbol{y}$ and $M = M_K^1(\boldsymbol{y}) - \boldsymbol{y} = A \cap (-A)$. The polar $P = \{\boldsymbol{a} \in \mathbb{R}^d : \langle \boldsymbol{a}, \boldsymbol{x} \rangle \leq 1 \text{ for every } \boldsymbol{x} \in A\}$ is a convex body containing the origin in its interior. Choose a maximum-volume simplex in P, with vertices $\boldsymbol{p}_0, \ldots, \boldsymbol{p}_d$. Replacing one vertex shows that every $\boldsymbol{a} \in P$ has barycentric coordinates of absolute value at most 1. For rational inputs, apply Lemma 6.4 to $P \times \{1\}$ instead. The selected points give a simplex with barycentric coordinates of absolute value at most 2.

Set $\boldsymbol{c} = (d+1)^{-1} \sum_{i=0}^{d} \boldsymbol{p}_i$. For $\boldsymbol{a} = \sum_{i=0}^{d} \alpha_i \boldsymbol{p}_i \in P$, where $\sum_i \alpha_i = 1$ and $\alpha_i \leq 2$, we have

$$\boldsymbol{q} \coloneqq \frac{2\sum_i \boldsymbol{p}_i - \boldsymbol{a}}{2d+1} = \sum_i \frac{2 - \alpha_i}{2d+1} \boldsymbol{p}_i \in P, \qquad \boldsymbol{a} = (2d+2)\boldsymbol{c} - (2d+1)\boldsymbol{q}.$$

If $\boldsymbol{c} \neq \boldsymbol{0}$, set $\boldsymbol{u} = \boldsymbol{c}$. For $\boldsymbol{x} \in A$ with $\langle \boldsymbol{u}, \boldsymbol{x} \rangle \geq -1$, the polar inequalities and this identity give

$$-(4d+3) \leq \langle \boldsymbol{a}, \boldsymbol{x} \rangle \leq 1 \qquad \text{for every } \boldsymbol{a} \in P.$$

By polarity, $\boldsymbol{x} \in (4d+3)M$. If $\boldsymbol{c} = \mathbf{0}$, the same identity gives $P \subseteq -(2d+1)P$, so $A \subseteq (2d+1)M$; choose any nonzero $\boldsymbol{p}_i$ as $\boldsymbol{u}$. In both cases $\boldsymbol{u} \neq \mathbf{0}$ and $\boldsymbol{u} \in P$, so M lies in the cap. Thus

$$M \subseteq \{\boldsymbol{x} \in A : \langle \boldsymbol{u}, \boldsymbol{x} \rangle \geq -1\} \subseteq (4d+3)M.$$

Translation by $\boldsymbol{y}$ proves the claim for $t = 1$. For $t \geq 1$, apply these containments to $\boldsymbol{y} + (\boldsymbol{x} - \boldsymbol{y})/t$, which belongs to K whenever $\boldsymbol{x} \in K$, by convexity.

For the algorithmic claim, K and P are rational polytopes. Vertices of P solve fixed-size systems determined by vertices of A, so $\mathsf{maxbitlen}(P) = \mathcal{O}_d(b_0)$. Optimizing over A gives strong separation for P; optimization–separation equivalence therefore gives $T_{\text{query}}(P) = b_0^{\mathcal{O}_d(1)} T_{\text{query}}(K)$. An oracle for P also optimizes over $P \times \{1\}$. Lemma 6.4 and rational linear algebra give the stated running time. Since the selected $\boldsymbol{p}_i$ are vertices of P, we have $\mathsf{maxbitlen}(\boldsymbol{p}_i) = \mathcal{O}_d(b_0)$ and hence $\mathsf{maxbitlen}(\boldsymbol{u}) = \mathcal{O}_d(b_0)$. □

9 Quotient Cap Covering

Our quotient variant of economic cap covering is stated as follows:

Theorem 9.1. *Given a lattice $\Lambda \subseteq \mathbb{R}^d$ of rank less than d, a convex body $K \subseteq \mathbb{R}^d$, and $0 < \rho < 1$, write $D = \rho(K - K)/2$ and $r = \rho \operatorname{vol}_d(\pi_\Lambda K)/\operatorname{vol}_d(\pi_\Lambda D)$.*

Then, there exist a collection of $\mathcal{O}_d(r)$ caps $Q_i = K \cap \{\boldsymbol{x} : \langle \boldsymbol{u}_i, \boldsymbol{x} \rangle \geq a_i\}$ with enclosing parallelotopes $P_i \supseteq Q_i$, and a point set $R \subseteq \mathbb{R}^d$ of size $|R| = \mathcal{O}_d(r)$, such that the following properties hold:

(i) $\sum_i \operatorname{vol}_d(\pi_\Lambda P_i) \leq \mathcal{O}_d(\rho) \operatorname{vol}_d(\pi_\Lambda K)$,
(ii) *each Q_i contains a translate of $c_d D$, for some $0 < c_d \leq 1$ depending only on d,*
(iii) $K \setminus \bigcup_i Q_i \subseteq (K \ominus D) + \Lambda$,
(iv) $R + 3dD \subseteq K$,
(v) *and,* $(K \setminus \bigcup_i Q_i) \cap (\bigcup_i P_i + D + \Lambda) \subseteq R + D + \Lambda$.

If K is a rational polytope given by a linear optimization oracle and Λ, ρ are rational, then, writing $b_0 = \mathsf{maxbitlen}(K, \Lambda, \rho)$, one can compute R and all $\boldsymbol{u}_i, a_i, P_i$ in $r \cdot b_0^{\mathcal{O}_d(1)} T_{\text{query}}(K)$ time, with $\mathsf{maxbitlen}(\boldsymbol{r}) = \mathcal{O}_d(b_0)$ for every $\boldsymbol{r} \in R$ and $\mathsf{maxbitlen}(\boldsymbol{u}_i, a_i, P_i) = \mathcal{O}_d(b_0)$ for every i.

The goal of this section is to prove Theorem 9.1.

9.1 Preliminaries

Lemma 9.2. *Let $\Lambda \subseteq \mathbb{R}^d$ be a lattice and $K \subseteq \mathbb{R}^d$ be a convex body. For $0 < a \leq b$,*

$$\frac{\operatorname{vol}_d(\pi_\Lambda(aK))}{a^d} \geq \frac{\operatorname{vol}_d(\pi_\Lambda(bK))}{b^d}.$$

Proof. For a compact convex set A, let $n_A(\boldsymbol{x}) = |(\boldsymbol{x} + \Lambda) \cap A|$. Partitioning A by lattice translates of a fundamental domain gives $\operatorname{vol}_d(\pi_\Lambda(A)) = \int_A n_A(\boldsymbol{x})^{-1} \, d\boldsymbol{x}$. For $\boldsymbol{x} \in K$, convexity gives $a(K - \boldsymbol{x}) \subseteq b(K - \boldsymbol{x})$, so $n_{aK}(a\boldsymbol{x}) \leq n_{bK}(b\boldsymbol{x})$. Hence

$$\frac{\operatorname{vol}_d(\pi_\Lambda(aK))}{a^d} = \int_K \frac{d\boldsymbol{x}}{n_{aK}(a\boldsymbol{x})} \geq \int_K \frac{d\boldsymbol{x}}{n_{bK}(b\boldsymbol{x})} = \frac{\operatorname{vol}_d(\pi_\Lambda(bK))}{b^d}.$$ □

Lemma 9.3. *Let $\Lambda \subseteq \mathbb{R}^d$ be a lattice, and $K, D \subseteq \mathbb{R}^d$ be convex bodies, where D is origin-symmetric, and K contains a translate of $\rho^{-1}D$ for some $0 < \rho < 1$. Then*

$$\operatorname{vol}_d(\pi_\Lambda(K + D) \setminus \pi_\Lambda(K \ominus D)) \leq \mathcal{O}_d(\rho) \operatorname{vol}_d(\pi_\Lambda K).$$

Proof. Choose $\boldsymbol{c}$ with $\boldsymbol{c}+\rho^{-1}D \subseteq K$. Set $V = \mathrm{vol}_d(\pi_\Lambda K)$. Convexity gives $K+D \subseteq \boldsymbol{c}+(1+\rho)(K-\boldsymbol{c})$ and $\boldsymbol{c}+(1-\rho)(K-\boldsymbol{c}) \subseteq K \ominus D$. Thus Lemma 9.2 and translation invariance give

$$\mathrm{vol}_d(\pi_\Lambda(K+D)) \le (1+\rho)^d V, \qquad \mathrm{vol}_d(\pi_\Lambda(K \ominus D)) \ge (1-\rho)^d V.$$

Since $K \ominus D \subseteq K + D$,

$$\mathrm{vol}_d(\pi_\Lambda(K+D) \setminus \pi_\Lambda(K \ominus D)) \le ((1+\rho)^d - (1-\rho)^d)V = \mathcal{O}_d(\rho)V. \qquad \square$$

The following coordinates cover the quotient by equal-volume boxes with disjoint interiors. Integer subdivision numbers make their indices periodic in the lattice directions.

Lemma 9.4. *Let $\Lambda \subseteq \mathbb{R}^d$ be a lattice of rank d', and $P \subseteq \mathbb{R}^d$ be a d-dimensional origin-symmetric parallelotope. There are positive integers $k_1, \dots, k_{d'}$ and a basis $\boldsymbol{v}_1, \dots, \boldsymbol{v}_d$ of $\mathbb{R}^d$ such that $k_1\boldsymbol{v}_1, \dots, k_{d'}\boldsymbol{v}_{d'}$ form a basis of Λ and $F = \mathcal{Z}((\boldsymbol{v}_i)_{i=1}^d)$ satisfies $F - F \subseteq P$ and $\mathrm{vol}_d(F) = \Theta_d(\mathrm{vol}_d(\pi_\Lambda(P)))$.*

If Λ, P are rational, one can compute all $k_i, \boldsymbol{v}_i$ with $\mathsf{maxbitlen}(k_1, \dots, k_{d'}, \boldsymbol{v}_1, \dots, \boldsymbol{v}_d) = \mathcal{O}_d(b_0)$ in $b_0^{\mathcal{O}_d(1)}$ time, where $b_0 = \mathsf{maxbitlen}(\Lambda, P)$.

Proof. Choose a linear map T with $TP = [-1,1]^d$, and apply Lemma 6.8 to $T\Lambda$, obtaining a basis $\boldsymbol{\beta}_1, \dots, \boldsymbol{\beta}_{d'}$. Set $a_i = \|\boldsymbol{\beta}_i\|_\infty$ and $\boldsymbol{u}_i = \boldsymbol{\beta}_i/a_i$. Writing $U = [\boldsymbol{u}_1 \ \cdots \ \boldsymbol{u}_{d'}]$, the product bound in Lemma 6.8 gives

$$\sqrt{\det(U^\mathsf{T}U)} = \frac{\det(T\Lambda)}{\prod_i a_i} \ge 2^{-(d')^2}.$$

Complete U to a matrix A using standard basis vectors complementary to a largest $d' \times d'$ minor. Cauchy–Binet gives $|\det A| = \Omega_d(1)$, while every entry of A has absolute value at most one. The entry bound and Cramer's rule give $A[-1/d, 1/d]^d \subseteq [-1,1]^d \subseteq A[-C_d, C_d]^d$ for a constant C_d depending only on d.

In the coordinates given by A, the lattice $T\Lambda$ is generated by $a_i\boldsymbol{e}_i$, $i \le d'$. For $s > 0$, direct integration gives

$$\mathrm{vol}_d(\pi_{T\Lambda}(A[-s,s]^d)) = |\det A|(2s)^{d-d'} \prod_{i=1}^{d'} \min\{2s, a_i\}.$$

The two containments therefore imply $\mathrm{vol}_d(\pi_{T\Lambda}([-1,1]^d)) = \Theta_d(|\det A| \prod_i \min\{1, a_i\})$. Set $\boldsymbol{\ell}_i = T^{-1}\boldsymbol{\beta}_i$, $k_i = \lceil 2da_i \rceil$, $\boldsymbol{v}_i = \boldsymbol{\ell}_i/k_i$ for $i \le d'$, and $\boldsymbol{v}_i = T^{-1}A\boldsymbol{e}_i/(2d)$ otherwise. Every $T\boldsymbol{v}_i$ has infinity norm at most $1/(2d)$, so $F - F \subseteq P$. Also,

$$\mathrm{vol}_d(F) = |\det T|^{-1}|\det A|(2d)^{d'-d} \prod_{i=1}^{d'} \frac{a_i}{k_i} = \Theta_d(\mathrm{vol}_d(\pi_\Lambda(P))).$$

For rational inputs, choose T rational and compute the reduced basis by Lemma 6.8. The remaining fixed-dimensional rational linear algebra gives the stated time and $\mathsf{maxbitlen}$ bounds. $\square$

9.2 Enumerating shallow boxes

We enumerate the shallow boxes by skipping long intervals of deep boxes.

Lemma 9.5. *In the setup of Lemma 9.4, suppose $d' < d$, and let K be a convex body containing a translate of $\rho^{-1}P$ for some (unknown) $\rho > 0$. Define*

$$\mathcal{G} = \sum_{i=1}^{d'} \{0, \dots, k_i - 1\}\boldsymbol{v}_i + \sum_{i=d'+1}^{d} \mathbb{Z}\boldsymbol{v}_i,$$

$$\mathcal{S} = \left\{ \boldsymbol{z} \in \mathcal{G} : \begin{array}{l} (\boldsymbol{z}+F) \cap (K+\Lambda) \neq \varnothing, \text{ and} \\ \boldsymbol{z}+F+\boldsymbol{\lambda} \not\subseteq K \ominus 2P, \forall \boldsymbol{\lambda} \in \Lambda \end{array} \right\}.$$

If K is a rational polytope with a linear optimization oracle, and Λ, P are rational, then one can compute $\mathcal{S}$ in

$$\left(|\mathcal{S}| + \rho \frac{\operatorname{vol}_d(\pi_\Lambda K)}{\operatorname{vol}_d(F)} + 1 \right) b_0^{\mathcal{O}_d(1)} T_{\text{query}}(K)$$

time, where $b_0 = \mathsf{maxbitlen}(K, \Lambda, P)$.

Proof. Write z_i for the coordinates of $\boldsymbol{z}$ in the basis $(\boldsymbol{v}_i)_i$. Set $V = \operatorname{vol}_d(\pi_\Lambda K)$. The projected boxes $\pi_\Lambda(\boldsymbol{z}+F)$, $\boldsymbol{z} \in \mathcal{G}$, cover $\mathbb{R}^d/\Lambda$, have disjoint interiors, and each has volume $\operatorname{vol}_d(F)$. In the setup of Lemma 9.5, a *column* is a set of the form $\boldsymbol{z} + \mathbb{Z}\boldsymbol{v}_d$ with $\boldsymbol{z} \in \mathcal{G}$.

For $\boldsymbol{z} \in \mathcal{G}$, call $\boldsymbol{z}$ *occupied* if $(\boldsymbol{z}+F) \cap (K+\Lambda) \neq \varnothing$, and *shallow* if $\boldsymbol{z} \in \mathcal{S}$. There are finitely many occupied points $\boldsymbol{z} \in \mathcal{G}$. If $\boldsymbol{z}$ is occupied and not shallow, then a translate of $\boldsymbol{z}+F+2P$ lies in K, so its predecessor $\boldsymbol{z} - \boldsymbol{v}_d$ is occupied because $\boldsymbol{v}_d \in P$. Thus the first occupied $\boldsymbol{z}$ in each occupied column is shallow, so there are at most $|\mathcal{S}|$ occupied columns. For $s\rho < 1$, apply Lemma 9.3 with margin sP and parameter $s\rho$; for $s\rho \geq 1$, bound the volume by $V \leq s\rho V$. This gives

$$\operatorname{vol}_d(\pi_\Lambda K \setminus \pi_\Lambda(K \ominus sP)) \leq \mathcal{O}_d(s\rho)V \quad (s \geq 1).$$

Optimization queries. An occupancy query finds the next occupied column in lexicographic order, or the least occupied coordinate in a column above a given threshold. For occupied $\boldsymbol{z}$, a depth query maximizes the integer $t \geq 1$ subject to $\boldsymbol{z}+F+\boldsymbol{\lambda}+tP \subseteq K$ for some $\boldsymbol{\lambda} \in \Lambda$. $\boldsymbol{z}$ is shallow exactly when the query is infeasible or its maximum is one. Normalizing P to $[-1,1]^d$ bounds t by the diameter of the image of K, which is $2^{\mathcal{O}_d(b_0)}$.

Enumerating the boxes. In each occupied column, start at its least occupied coordinate $z_d = n$ and run the depth query. If the query is infeasible or $t = 1$, record $\boldsymbol{z}$ and continue at the least occupied coordinate at least $n+1$. For $t \geq 2$, skip the interval $[n, n+t-2]$ and continue at the least occupied coordinate at least $n+t-1$. Indeed, for $0 \leq j \leq t-2$, the maximizing lift satisfies

$$\boldsymbol{z} + j\boldsymbol{v}_d + F + \boldsymbol{\lambda} + 2P \subseteq \boldsymbol{z} + F + \boldsymbol{\lambda} + tP \subseteq K,$$

because $\boldsymbol{v}_d \in P$. Thus every skipped $\boldsymbol{z} + j\boldsymbol{v}_d$ is occupied and not shallow. The algorithm visits or skips every occupied $\boldsymbol{z}$ and records exactly the points $\boldsymbol{z} \in \mathcal{S}$.

For a skipped interval of length $t-1$, let $J = \bigcup_{j=0}^{t-2} (\boldsymbol{z} + j\boldsymbol{v}_d + F + \boldsymbol{\lambda})$. Then $\pi_\Lambda J \subseteq \pi_\Lambda K$ has volume $(t-1)\operatorname{vol}_d(F)$. For $\boldsymbol{u} \in J$, we have $\boldsymbol{z}+F+\boldsymbol{\lambda}-\boldsymbol{u} \subseteq (t-1)P$. If $\boldsymbol{u}+\boldsymbol{\mu} \in K \ominus 2tP$ for some $\boldsymbol{\mu} \in \Lambda$, then

$$\boldsymbol{z} + F + \boldsymbol{\lambda} + \boldsymbol{\mu} + (t+1)P \subseteq \boldsymbol{u} + \boldsymbol{\mu} + 2tP \subseteq K,$$

contrary to maximality of t. For $s = 2 \cdot 2^j$, the projected sets $\pi_\Lambda J$ with $s \leq t < 2s$ have disjoint interiors and lie in $\pi_\Lambda K \setminus \pi_\Lambda(K \ominus 4sP)$. Each has volume at least $(s/2)\operatorname{vol}_d(F)$, so the shell bound gives at most $\mathcal{O}_d(\rho V / \operatorname{vol}_d(F))$ intervals in each range. There are $\mathcal{O}_d(b_0)$ ranges. Together with $\mathcal{S}$ and the occupied columns, this bounds the number of queries by $\mathcal{O}_d(b_0)(1 + |\mathcal{S}| + \rho V / \operatorname{vol}_d(F))$.

Implementing the queries. Occupancy is expressed by $\boldsymbol{x} \in K$, $\boldsymbol{\lambda} \in \Lambda$, and $\boldsymbol{x} - \boldsymbol{z} - \boldsymbol{\lambda} \in F$. Impose $0 \leq z_i \leq k_i - 1$ for $i \leq d'$ and project the feasible rational polytope onto $(z_1, \ldots, z_d)$ and the coordinates of $\boldsymbol{\lambda}$ in the given basis of Λ, using Lemma 6.3; then apply Lemma 6.9 to that projection. Minimizing coordinates successively and fixing prefixes therefore enumerates occupied columns in

lexicographic order and finds the least occupied coordinate in a column above any given threshold. Each search uses $\mathcal{O}_d(1)$ applications of Lemmas 6.3 and 6.9.

For the depth query, it suffices to impose membership in K of the sums of every vertex of $\boldsymbol{z}+F$ and t times every vertex of P. These $\mathcal{O}_d(1)$ points depend affinely on the query variables, so Lemmas 6.3 and 6.9 implement this query.

All variables can be restricted to coordinate boxes of radius $2^{\mathcal{O}_d(b_0)}$. If $(\boldsymbol{z}+F+\boldsymbol{\lambda})\cap K\neq\varnothing$, then $\boldsymbol{z}+\boldsymbol{\lambda}\in K+P$; for $i>d'$, z_i is bounded by the range of the ith coordinate of K in the basis $(\boldsymbol{v}_i)_i$ minus $[0,1]$. For $i\le d'$, we have $0\le z_i<k_i$. These bounds also bound the coordinates of $\boldsymbol{\lambda}$. Together with the bound on t, they give the claimed coordinate boxes. Each query consequently takes $b_0^{\mathcal{O}_d(1)}T_{\text{query}}(K)$ time and returns coordinates a with $\mathsf{maxbitlen}(a)=\mathcal{O}_d(b_0)$.

The algorithm does not use ρ. The query bound gives the stated time. □

9.3 Constructing the covering

Proof of Theorem 9.1. Write $V=\mathrm{vol}_d(\pi_\Lambda K)$ and $v_D=\mathrm{vol}_d(\pi_\Lambda D)$. Let $\beta=\lceil\beta_d\rceil$, where β_d is from Lemma 6.5.

Shallow boxes. Use Lemma 6.5 to choose $\boldsymbol{c},\tilde{B}$ with $\boldsymbol{c}+\tilde{B}\subseteq K\subseteq\boldsymbol{c}+\beta\tilde{B}$, and set $B=\beta\rho\tilde{B}$. Taking differences gives $D\subseteq B\subseteq\beta D$. Set $B_0=2dB$ and $\theta=2d\beta\rho$. If $\theta\ge 1/(2\beta+8)$, return $R=\varnothing$, the single cap K, and its enclosing parallelotope $\boldsymbol{c}+\beta\tilde{B}$. Any halfspace containing this parallelotope represents the cap K. The cap contains $\boldsymbol{c}+\beta^{-1}D$, and Lemma 9.2 bounds the enclosure volume by $\beta^dV=\mathcal{O}_d(\rho V)$. Lemma 9.2 gives $v_D\le\beta^dV$, so $r=\Omega_d(1)$ in this case. Both covering assertions hold because the cap is K.

Assume henceforth that $\theta<1/(2\beta+8)$. Apply Lemma 9.4 to $\Lambda,B/\beta$, and let $\mathcal{G}$ be as in Lemma 9.5. Then

$$F-F\subseteq D,\qquad \mathrm{vol}_d(F)=\Theta_d(v_D),\qquad B_0=\theta\tilde{B}. \tag{9.1}$$

The volume estimate follows from $D\subseteq B\subseteq\beta D$ and Lemma 9.2. The projected boxes $\pi_\Lambda(\boldsymbol{z}+F)$, $\boldsymbol{z}\in\mathcal{G}$, have disjoint interiors and volume $\mathrm{vol}_d(F)$.

Since $B_0=2d\beta(B/\beta)$, the same $k_i,\boldsymbol{v}_i,F$ satisfy Lemma 9.4 with $P=B_0$, with the volume comparison following from Lemma 9.2. Use Lemma 9.5, with margin B_0 and parameter θ, to enumerate $\mathcal{S}$. For $\boldsymbol{z}\in\mathcal{G}\setminus\mathcal{S}$ with $(\boldsymbol{z}+F)\cap(K+\Lambda)\neq\varnothing$, $\boldsymbol{z}+F$ has a Λ-translate contained in $K\ominus 2B_0$. For each $\boldsymbol{z}\in\mathcal{S}$, set $C=\boldsymbol{z}+F$. Then

$$\pi_\Lambda C\subseteq\pi_\Lambda(K+B_0)\setminus\pi_\Lambda(K\ominus 4B_0). \tag{9.2}$$

Indeed, a lift of one point into $K\ominus 4B_0$ would place the whole box, with margin $2B_0$, inside K, since $C-C\subseteq B_0$. Since $K+B_0\subseteq K+4B_0$, disjoint projected box interiors and Lemma 9.3, with margin $4B_0$ and parameter $4\theta<1$, give $|\mathcal{S}|=\mathcal{O}_d(\rho V/v_D)$. To see that $\mathcal{S}\neq\varnothing$, choose $\boldsymbol{z}\in\mathcal{G}$ with $(\boldsymbol{z}+F)\cap(K+\Lambda)\neq\varnothing$ and with minimal last coordinate in its column $\boldsymbol{z}+\mathbb{Z}\boldsymbol{v}_d$; such a minimum exists because K is bounded and $d'<d$. If $\boldsymbol{z}\notin\mathcal{S}$, some translate of $\boldsymbol{z}+F+2B_0$ lies in K, so $(\boldsymbol{z}-\boldsymbol{v}_d+F)\cap(K+\Lambda)\neq\varnothing$, since $\boldsymbol{v}_d\in F-F\subseteq B_0$, a contradiction. Thus $1\le|\mathcal{S}|=\mathcal{O}_d(r)$, so $r=\Omega_d(1)$.

Balanced centers. Let $q=q_d$ and $\gamma=\gamma_d$ be the constants from Lemma 8.5. Enlarge K homothetically about $\boldsymbol{c}$:

$$K^+=\boldsymbol{c}+(1+2\theta)(K-\boldsymbol{c}),\qquad K+2B_0\subseteq K^+\subseteq K+2\beta B_0.$$

Set $\Lambda' = 2q\Lambda$. For each $\boldsymbol{z} \in \mathcal{S}$, let $C = \boldsymbol{z} + F$ and let $\boldsymbol{a}_C$ be its center. For each class $\boldsymbol{\lambda}_0 + q\Lambda$ in $\Lambda/q\Lambda$, choose a point $\boldsymbol{x}_0 \in (K + B_0) \cap (\boldsymbol{a}_C + \boldsymbol{\lambda}_0 + q\Lambda)$ when possible. Apply Lemma 8.5 to $K^+, q\Lambda, \boldsymbol{x}_0$. The returned center $\boldsymbol{y} \in K^+ \cap (\boldsymbol{a}_C + \Lambda)$ is γ-balanced modulo $\Lambda'/2$ and satisfies $K^+ \cap (\boldsymbol{x}_0 + q\Lambda) \subseteq M^{\gamma}_{K^+}(\boldsymbol{y})$. Associate these $\mathcal{O}_d(1)$ candidates with C.

Set $\alpha = 2\gamma + 1$ and $M_{\boldsymbol{y}} = (K^+ - \boldsymbol{y}) \cap (\boldsymbol{y} - K^+)$. For $\boldsymbol{x} \in K \cap (C + \Lambda)$, choose $\boldsymbol{p} \in \boldsymbol{a}_C + \Lambda$ with $\boldsymbol{x} - \boldsymbol{p} \in B_0$. Then $\boldsymbol{p} \in K + B_0$ and $\boldsymbol{x} \in \boldsymbol{p} + B_0 \subseteq M^1_{K^+}(\boldsymbol{p})$. The seed in $\boldsymbol{p}$'s class gives $\boldsymbol{p} \in M^{\gamma}_{K^+}(\boldsymbol{y})$, so Lemma 8.2 gives $\boldsymbol{x} \in M^{\alpha}_{K^+}(\boldsymbol{y})$. Also $\boldsymbol{x}_0 + B_0 \subseteq M^1_{K^+}(\boldsymbol{x}_0) \subseteq M^{\alpha}_{K^+}(\boldsymbol{y})$; taking differences and dividing by two gives $B_0 \subseteq \alpha M_{\boldsymbol{y}}$. Thus

$$K \cap (C + \Lambda) \subseteq \bigcup_{\boldsymbol{y} \text{ stored at } C} M^{\alpha}_{K^+}(\boldsymbol{y}), \qquad B_0 \subseteq \alpha M_{\boldsymbol{y}}. \tag{9.3}$$

Apply Lemma 6.5 to $M_{\boldsymbol{y}}$. Taking differences of the resulting containments gives an origin-symmetric parallelotope $B_{\boldsymbol{y}}$ with $B_{\boldsymbol{y}} \subseteq M_{\boldsymbol{y}} \subseteq \beta B_{\boldsymbol{y}}$. Set

$$\eta = 13\gamma\alpha, \qquad V_{\boldsymbol{y}} = \boldsymbol{y} + B_{\boldsymbol{y}}/2, \qquad S_{\boldsymbol{y}} = \eta\beta B_{\boldsymbol{y}}.$$

Then $V_{\boldsymbol{y}} \subseteq M^{1/2}_{K^+}(\boldsymbol{y})$ and $B_0 \subseteq S_{\boldsymbol{y}}$.

Greedy selection. Choose an active candidate $\boldsymbol{y}$, enumerate the projected boxes intersecting $\pi_\Lambda(\boldsymbol{y} + S_{\boldsymbol{y}})$, and delete every active candidate $\boldsymbol{y}'$ in their lists for which $\pi_{\Lambda'}(V_{\boldsymbol{y}}) \cap \pi_{\Lambda'}(V_{\boldsymbol{y}'}) \neq \varnothing$. The intersection test is lattice feasibility in the explicit polytope $V_{\boldsymbol{y}} - V_{\boldsymbol{y}'}$. By Lemma 8.4, every intersecting candidate satisfies

$$M^{\alpha}_{K^+}(\boldsymbol{y}') \subseteq M^{13\gamma\alpha}_{K^+}(\boldsymbol{y}) \subseteq \boldsymbol{y} + S_{\boldsymbol{y}}.$$

Its indexed box is visited because $\boldsymbol{y}' \in \boldsymbol{a}_C + \Lambda$. Thus every intersecting candidate is deleted. Repeat until no active candidate remains. The selected $V_{\boldsymbol{y}}$ have disjoint images modulo Λ'. By Eq. (9.3), their $M^{13\gamma\alpha}_{K^+}(\boldsymbol{y})$ cover $K \cap (C + \Lambda)$ for every $C = \boldsymbol{z} + F$ with $\boldsymbol{z} \in \mathcal{S}$.

Volume bound. By Eq. (9.2), every candidate satisfies $(\boldsymbol{y} + \Lambda) \cap (K \ominus 4B_0) = \varnothing$. Suppose $\boldsymbol{x} \in V_{\boldsymbol{y}}$ and $\boldsymbol{x} + \boldsymbol{\ell} \in K \ominus (2\beta + 8)B_0$ for $\boldsymbol{\ell} \in \Lambda'$. Since $2\boldsymbol{y} - \boldsymbol{x} \in K^+ \subseteq K + 2\beta B_0$, taking their midpoint gives $\boldsymbol{y} + \boldsymbol{\ell}/2 \in K \ominus 4B_0$, a contradiction. The selected projected parallelotopes therefore lie in

$$\pi_{\Lambda'}(K + 2\beta B_0) \setminus \pi_{\Lambda'}(K \ominus (2\beta + 8)B_0).$$

Enlarging the outer padding to $(2\beta + 8)B_0$ and applying Lemma 9.3 with parameter $(2\beta + 8)\theta < 1$ bounds this volume by $\mathcal{O}_d(\rho V)$, since $\boldsymbol{c} + \theta^{-1}B_0 \subseteq K$, $\theta = \mathcal{O}_d(\rho)$, and $\mathrm{vol}_d(\pi_{\Lambda'}K) \leq [\Lambda : \Lambda']V = \mathcal{O}_d(V)$. Since $S_{\boldsymbol{y}} = 2\eta\beta(V_{\boldsymbol{y}} - \boldsymbol{y})$, Lemma 9.2 and disjointness give

$$\sum_{\boldsymbol{y} \text{ selected}} \mathrm{vol}_d(\pi_\Lambda S_{\boldsymbol{y}}) \leq (2\eta\beta)^d \sum_{\boldsymbol{y} \text{ selected}} \mathrm{vol}_d(\pi_{\Lambda'} V_{\boldsymbol{y}}) = \mathcal{O}_d(\rho V). \tag{9.4}$$

Returning caps in K. Set $a = (1 + 2\theta)^{-1}$ and replace each selected center $\boldsymbol{y}$ by $\boldsymbol{z}_{\boldsymbol{y}} = \boldsymbol{c} + a(\boldsymbol{y} - \boldsymbol{c})$. Apply Lemma 8.6 to $K^+, \boldsymbol{y}$, obtaining a normal $\boldsymbol{u}_{\boldsymbol{y}}$ whose cap contains $\boldsymbol{y} + M_{\boldsymbol{y}}$ and lies in $\boldsymbol{y} + hM_{\boldsymbol{y}}$, where $h = 4d + 3$. Choose a constant $m \geq \alpha(13\gamma + 2\beta)$ depending only on d and return the halfspace defining

$$Q_{\boldsymbol{y}} = K \cap \{\boldsymbol{x} : \langle \boldsymbol{u}_{\boldsymbol{y}}, \boldsymbol{x} - \boldsymbol{z}_{\boldsymbol{y}} \rangle \geq -m\}.$$

The homothety maps K^+ onto K and gives $M_K^1(\boldsymbol{z_y}) - \boldsymbol{z_y} = aM_{\boldsymbol{y}}$. The scaled cap containment in Lemma 8.6, with scaling factor m/a, therefore gives

$$K \cap (\boldsymbol{z_y} + mM_{\boldsymbol{y}}) \subseteq Q_{\boldsymbol{y}} \subseteq \boldsymbol{z_y} + hmM_{\boldsymbol{y}}, \qquad \boldsymbol{z_y} + aM_{\boldsymbol{y}} \subseteq Q_{\boldsymbol{y}}.$$

Moreover, $\boldsymbol{y} - \boldsymbol{z_y} \in 2\theta(K - \boldsymbol{c}) \subseteq 2\beta B_0 \subseteq 2\beta\alpha M_{\boldsymbol{y}}$, and hence

$$M_{K^+}^{13\gamma\alpha}(\boldsymbol{y}) \subseteq \boldsymbol{z_y} + \alpha(13\gamma + 2\beta)M_{\boldsymbol{y}} \subseteq \boldsymbol{z_y} + mM_{\boldsymbol{y}}.$$

Thus the returned caps cover every point of K outside $((K \ominus 2B_0) + \Lambda)$: a point outside this set belongs to $\mathcal{S} + F + \Lambda$. Since $D \subseteq 2B_0$, this proves the first covering assertion. Since $D \subseteq B \subseteq \alpha M_{\boldsymbol{y}}$, each $Q_{\boldsymbol{y}}$ contains $\boldsymbol{z_y} + (a/\alpha)D$. Since $a > 1/3$, a sufficiently small $c_d > 0$ gives the stated cap thickness. Return the explicit parallelotope

$$P_{\boldsymbol{y}} = \boldsymbol{z_y} + hm\beta B_{\boldsymbol{y}} \supseteq Q_{\boldsymbol{y}}.$$

Each $P_{\boldsymbol{y}}$ is a translate of $(hm/\eta)S_{\boldsymbol{y}}$, so Lemma 9.2 and Eq. (9.4) gives $\sum_{\boldsymbol{y}} \mathrm{vol}_d(\pi_\Lambda P_{\boldsymbol{y}}) = \mathcal{O}_d(\rho V)$.

Interior translates near the caps. For each returned $P_{\boldsymbol{y}}$, enumerate all $\boldsymbol{z} \in \mathcal{G}$ for which $\boldsymbol{z} + F$ intersects $P_{\boldsymbol{y}} + B + \Lambda$. Every such box is contained in $P_{\boldsymbol{y}} + 2B + \Lambda$, since $F - F \subseteq B$. Write $P_{\boldsymbol{y}} = \boldsymbol{z_y} + hm\beta B_{\boldsymbol{y}}$ as above. The containments $B \subseteq \alpha M_{\boldsymbol{y}} \subseteq \alpha\beta B_{\boldsymbol{y}}$ show that $P_{\boldsymbol{y}} + 2B$ is contained in a constant dilation of $P_{\boldsymbol{y}}$ about $\boldsymbol{z_y}$. Thus Lemma 9.2 and the disjoint projected box interiors bound the total number of enumerated points $\boldsymbol{z}$, including repetitions, by $\mathcal{O}_d(\sum_{\boldsymbol{y}} \mathrm{vol}_d(\pi_\Lambda P_{\boldsymbol{y}})/v_D) = \mathcal{O}_d(r)$.

Start with $R = \varnothing$. At the center $\boldsymbol{g}$ of each enumerated box, test whether $\boldsymbol{g} + \boldsymbol{\lambda} + 3dB \subseteq K$ for some $\boldsymbol{\lambda} \in \Lambda$. If so, add $\boldsymbol{r} = \boldsymbol{g} + \boldsymbol{\lambda}$ to R. Since $D \subseteq B$, every $\boldsymbol{r} \in R$ satisfies $\boldsymbol{r} + 3dD \subseteq K$. To prove the second covering assertion, let $\boldsymbol{x} \in K \cap (\bigcup_{\boldsymbol{y}} P_{\boldsymbol{y}} + D + \Lambda)$ lie outside the caps. The cap construction gives $\boldsymbol{x} + \boldsymbol{\mu} \in K \ominus 2B_0$ for some $\boldsymbol{\mu} \in \Lambda$. Choose an enumerated box and $\boldsymbol{\nu} \in \Lambda$ with $\boldsymbol{x} \in \boldsymbol{z} + F + \boldsymbol{\nu}$, and let $\boldsymbol{g}$ be its center. Since $\boldsymbol{g} + \boldsymbol{\nu} - \boldsymbol{x} \in F - F \subseteq D \subseteq B$ and $(3d+1)B \subseteq 2B_0$,

$$\boldsymbol{g} + \boldsymbol{\nu} + \boldsymbol{\mu} + 3dB \subseteq \boldsymbol{x} + \boldsymbol{\mu} + 2B_0 \subseteq K.$$

The test at $\boldsymbol{g}$ succeeds, and its returned center $\boldsymbol{r} \in \boldsymbol{g} + \Lambda$ satisfies $\boldsymbol{x} \in \boldsymbol{r} + D + \Lambda$.

Running time. Computing $\mathcal{S}$ takes $b_0^{\mathcal{O}_d(1)} r\, T_{\mathrm{query}}(K)$ time by Lemma 9.5 and the bound on $|\mathcal{S}|$. There are $\mathcal{O}_d(\rho V/v_D)$ seeds and candidates. Each seed query, center construction, and cap approximation takes $b_0^{\mathcal{O}_d(1)} T_{\mathrm{query}}(K)$ time by Lemmas 6.3, 8.5, 8.6 and 6.9. The oracle for $M_{\boldsymbol{y}}$ follows from Lemma 6.3, and Lemma 6.5 constructs $B_{\boldsymbol{y}}$ within the same bound. At a selected center, every visited box projects into $\pi_\Lambda(\boldsymbol{y} + S_{\boldsymbol{y}} + B) \subseteq \pi_\Lambda(\boldsymbol{y} + 2S_{\boldsymbol{y}})$. Disjoint projected box interiors and Eq. (9.1) bound their number by $\mathcal{O}_d(\mathrm{vol}_d(\pi_\Lambda S_{\boldsymbol{y}})/v_D)$. Enumerate the corresponding points $\boldsymbol{z} \in \mathcal{G}$ in lexicographic order of their coordinates in $(\boldsymbol{v}_i)_i$, using Lemmas 6.3 and 6.9, with lattice-translation coordinates as auxiliary integer variables. Each $\boldsymbol{z}$ has $\mathcal{O}_d(1)$ candidates, and each intersection test takes $b_0^{\mathcal{O}_d(1)}$ time. Summing Eq. (9.4) gives total time $b_0^{\mathcal{O}_d(1)} r\, T_{\mathrm{query}}(K)$ for the cap construction.

The boxes intersecting $P_{\boldsymbol{y}} + B + \Lambda$ are enumerated by the same successive integer minimizations, now over explicit parallelotopes. Enumerating each $\boldsymbol{z}$ takes $b_0^{\mathcal{O}_d(1)}$ time. For the center $\boldsymbol{g}$ of $\boldsymbol{z} + F$, the safety test imposes $\boldsymbol{g} + \boldsymbol{\lambda} + \boldsymbol{v} \in K$ for the 2^d vertices $\boldsymbol{v}$ of $3dB$. Lemmas 6.3 and 6.9 find a feasible $\boldsymbol{\lambda} \in \Lambda$, or report none, in $b_0^{\mathcal{O}_d(1)} T_{\mathrm{query}}(K)$ time. The $\mathcal{O}_d(r)$ enumerated points $\boldsymbol{z}$ therefore add $b_0^{\mathcal{O}_d(1)} r\, T_{\mathrm{query}}(K)$ time. All centers, parallelotopes, and query polytopes have $\mathsf{maxbitlen} = \mathcal{O}_d(b_0)$ by (6.1), Lemmas 6.3, 6.5 and 6.9, Lemmas 8.5 and 8.6, and Lemmas 9.4 and 9.5, together with fixed-dimensional rational linear algebra. Thus the points $\boldsymbol{r} \in R$, cap coefficients $\boldsymbol{u_y}, \langle \boldsymbol{u_y}, \boldsymbol{z_y} \rangle - m$, and parallelotopes $P_{\boldsymbol{y}}$ have the required $\mathsf{maxbitlen}$ bounds. □

10 Covering for Subset Sums

A generating submultiset makes deep lattice points of a zonotope attainable by Lemma 3.5. We construct arithmetic progressions of attained values and cover every lattice point whose value is omitted by caps with explicit enclosing parallelotopes.

Theorem 10.1. *Let two multisets $X' \subseteq X$ with $\operatorname{supp}(X) \subseteq \mathbb{Z}^d$ and $\operatorname{span}(X) = \mathbb{R}^d$, a nonzero linear map $f\colon \mathbb{R}^d \to \mathbb{R}$ with integer coefficients, and integers $\ell \le u$ be given. Write $\Gamma = \mathcal{L}(X)$, $H = \operatorname{conv}(\pm \operatorname{supp} X)$, $D = |X'|H$, $\Lambda = \Gamma \cap \ker f$, and $K = \mathcal{Z}(X) \cap f^{-1}([\ell, u])$.*

Suppose X' is d-generating for $H \cap \Gamma$, and K contains translates of H and $\rho^{-1}D$ for some (unknown) $\rho > 0$. Then, there is an algorithm that constructs:

(i) *An integer set G, represented as an AP decomposition of size $\mathcal{O}_d(\rho(u-\ell))$, such that*

$$G \subseteq f((K \ominus D) \cap \Gamma) \subseteq f(\Sigma(X)) \cap [\ell, u], \tag{10.1}$$

where the second containment follows from Lemma 3.5.

(ii) *Caps $Q_i = K \cap \{\boldsymbol{x} : \langle \boldsymbol{u}_i, \boldsymbol{x}\rangle \ge a_i\}$ with enclosing parallelotopes $P_i \supseteq Q_i$ such that*

$$\{\boldsymbol{x} \in K \cap \Gamma : f(\boldsymbol{x}) \notin G\} \subseteq \bigcup_i Q_i, \tag{10.2}$$

$$\sum_i \frac{\operatorname{vol}_d(\pi_\Lambda P_i)}{\det \Gamma} \le \mathcal{O}_d(\rho(u-\ell)), \tag{10.3}$$

and every Q_i contains a translate of $c_d H$, where $0 < c_d \le 1$ depends only on d.

Writing $b_0 = \mathsf{maxbitlen}(X, X', f, \ell, u)$, the construction algorithm takes time

$$b_0^{\mathcal{O}_d(1)} \rho(u-\ell) \left(1 + \frac{|\operatorname{supp} X|}{|f(D \cap \Gamma)|}\right), \tag{10.4}$$

and $\mathsf{maxbitlen}(\boldsymbol{u}_i, a_i, P_i) = \mathcal{O}_d(b_0)$ for all i.

We use the following bound to count lattice values from quotient volumes.

Lemma 10.2. *Let $\Lambda \subseteq \Gamma \subseteq \mathbb{R}^d$ be two lattices and $K \subseteq \mathbb{R}^d$ be an origin-symmetric convex body with $\operatorname{span}(K \cap \Gamma) = \mathbb{R}^d$. Then $|\pi_\Lambda(K \cap \Gamma)| \det \Gamma \le \mathcal{O}_d(\operatorname{vol}_d(\pi_\Lambda K))$.*

Proof. By Lemma 6.11, Γ has a basis $\boldsymbol{u}_1, \dots, \boldsymbol{u}_d$ and positive integer widths m_i such that $P = \sum_i [-m_i, m_i]\boldsymbol{u}_i$ satisfies $K \subseteq P \subseteq \kappa_d K$. Thus $F = \sum_i [0, \boldsymbol{u}_i) \subseteq P$ is a half-open fundamental parallelepiped of Γ. Choose one $\boldsymbol{x} \in K \cap \Gamma$ from each class modulo Λ. The sets $\pi_\Lambda(\boldsymbol{x} + F)$ are pairwise disjoint, each of volume $\det \Gamma$, and lie in $\pi_\Lambda(2P)$. Cover $2P$ by $\mathcal{O}_d(1)$ translates of $\kappa_d^{-1} P \subseteq K$. Subadditivity and translation invariance of quotient volume give the bound. □

Proof of Theorem 10.1. Write $N = |\operatorname{supp} X|$. Compute Γ and $\Lambda = \Gamma \cap \ker f$ using Lemmas 6.6 and 6.7, and set $W = \operatorname{width}_f(K) \le u - \ell$. By Lemmas 6.2 and 6.3, $\mathsf{maxbitlen}(K) = \mathcal{O}_d(b_0)$ and K has an optimization oracle with $T_{\text{query}}(K) = N b_0^{\mathcal{O}_d(1)}$. Write $V = \operatorname{vol}_d(\pi_\Lambda K)$. Since $f(\Gamma)$ is a subgroup of $\mathbb{Z}$, a slab of f-width W has quotient volume at most $W \det \Gamma$, so $V \le W \det \Gamma$. Since $f(\Gamma) \subseteq \mathbb{Z}$ and $2 \le \operatorname{width}_f(D) \le \rho W$, we have $|f(D \cap \Gamma)| \le \rho W + 1 = \mathcal{O}(\rho W)$. Hence $N = \mathcal{O}(\rho W N / |f(D \cap \Gamma)|)$.

Choosing the scale. By Lemmas 6.3 and 6.4, compute an origin-symmetric parallelotope B_0 with

$$(K-K)/2 \subseteq B_0 \subseteq d(K-K).$$

Choose the smallest σ with $2dD \subseteq \sigma B_0$ by scanning the vectors $|X'|\boldsymbol{x}$, $\boldsymbol{x} \in \operatorname{supp} X$, in the coordinates of B_0. Since $D \subseteq \rho(K-K)/2$, we have $\sigma \le 2d\rho$. Set $D_0 = \sigma(K-K)/2$, $B = \sigma B_0$, and $v_{D_0} = \operatorname{vol}_d(\pi_\Lambda D_0)$. Then

$$D \subseteq (2d)^{-1}B \subseteq D_0 \subseteq B \subseteq 2dD_0.$$

These computations take $b_0^{\mathcal{O}_d(1)}N$ time. If $\sigma \ge 1$, use Lemma 6.5 to compute $\boldsymbol{c}, B_1$ with $\boldsymbol{c}+B_1 \subseteq K \subseteq \boldsymbol{c}+\beta_d B_1$. Return $G = \varnothing$, the single cap $Q_1 = K$ represented by $f(\boldsymbol{x}) \ge \min f(K)$, and $P_1 = \boldsymbol{c} + \lceil \beta_d \rceil B_1$. This cap contains a translate of H by hypothesis. Covering P_1 by $\mathcal{O}_d(1)$ translates of $\boldsymbol{c}+B_1 \subseteq K$ gives $\operatorname{vol}_d(\pi_\Lambda P_1) = \mathcal{O}_d(V)$, so $\operatorname{vol}_d(\pi_\Lambda P_1)/\det\Gamma = \mathcal{O}_d(W) = \mathcal{O}_d(\rho(u-\ell))$. The additional computation takes $b_0^{\mathcal{O}_d(1)}N$ time.

Assume henceforth that $\sigma < 1$. Apply Theorem 9.1 to Λ, K, σ. It returns a set R of centers and caps Q_i enclosed in parallelotopes P_i, with

$$\boldsymbol{a} + 3dD_0 \subseteq K \quad (\boldsymbol{a} \in R), \qquad \sum_i \operatorname{vol}_d(\pi_\Lambda P_i) = \mathcal{O}_d(\sigma V).$$

The number of centers and caps is $\mathcal{O}_d(\sigma V / v_{D_0})$. Each cap contains a translate of $c_d D_0$, hence of $c_d H$. The volume bound and $V \le W \det\Gamma$ give (10.3).

Values in the enclosing and interior boxes. Let $T = f(K \cap \Gamma)$. We use the values E from the enclosures P_i and the interior seed values S to determine which gaps in E belong to T. Enumerate

$$E = \bigcup_i f(P_i \cap \Gamma), \qquad S = \bigcup_{\boldsymbol{a} \in R} f((\boldsymbol{a}+B) \cap \Gamma)$$

by Lemma 6.10. Each box has $\mathcal{O}_d(1)$ explicit inequalities. Since $\boldsymbol{a} + B + D_0 \subseteq \boldsymbol{a} + 3dD_0 \subseteq K$,

$$S \subseteq f((K \ominus D_0) \cap \Gamma).$$

Cap thickness and $D_0 \subseteq B$ imply that every enumerated box P satisfies $D_0 \subseteq c_d^{-1}(P-P)$, whose lattice points therefore span $\mathbb{R}^d$. Translate a nonempty $P \cap \Gamma$ by one of its points, apply Lemma 10.2, and cover $c_d^{-1}(P-P)$ by $\mathcal{O}_d(1)$ translates of P to obtain

$$|f(P \cap \Gamma)| \le \mathcal{O}_d\left(\frac{\operatorname{vol}_d(\pi_\Lambda P)}{\det\Gamma}\right).$$

Covering B by $\mathcal{O}_d(1)$ translates of $(2d)^{-1}B \subseteq D_0$ gives $\operatorname{vol}_d(\pi_\Lambda B) = \mathcal{O}_d(v_{D_0})$. Summing over the caps and centers, the number of values, including repetitions, is $\mathcal{O}_d(\sigma V/\det\Gamma) = \mathcal{O}_d(\sigma W)$. Enumeration and sorting take $b_0^{\mathcal{O}_d(1)}(\sigma V/v_{D_0} + \sigma W)$ time.

Filling gaps in the value set. Choose $\boldsymbol{w} \in \pm\operatorname{supp} X$ with $p = f(\boldsymbol{w}) > 0$. Then $p \in \mathbb{Z}$ and $\boldsymbol{w} \in H \subseteq D \subseteq D_0$. If $t \in T \setminus E$, none of its lattice lifts in K belongs to a cap. The deep-covering guarantee of Theorem 9.1 therefore gives a lift in $(K \ominus D_0) \cap \Gamma$. Adding or subtracting $\boldsymbol{w}$ shows that $t-p, t+p \in T$. It follows that every maximal gap in E within a residue class modulo p is wholly contained in T or disjoint from T. Every unbounded gap is disjoint from T, since T is bounded.

Sort E by residue modulo p, then by value. For consecutive values $e < e'$ in one residue class, consider the nonempty gap

$$e+p, e+2p, \dots, e'-p.$$

This gap belongs to T exactly when $e+p \in S$. To prove the forward implication, take $\boldsymbol{x} \in K \cap \Gamma$ with $f(\boldsymbol{x}) = e+p$ and $\boldsymbol{z} \in P_i \cap \Gamma$ with $f(\boldsymbol{z}) = e$. Then $\boldsymbol{x}-\boldsymbol{z}-\boldsymbol{w} \in \Lambda$, so $\boldsymbol{x} \in P_i + D_0 + \Lambda$. Since $e+p \notin E$, the point $\boldsymbol{x}$ is in no cap. The local covering in Theorem 9.1 gives $\boldsymbol{x} \in R + D_0 + \Lambda \subseteq R + B + \Lambda$, and hence $e + p \in S$. The reverse implication follows from $S \subseteq T$.

Include every such gap whose first value lies in S, and include each value of S as a singleton progression. Their union is $G = (T \setminus E) \cup S$. Both parts have D_0-deep lattice lifts, so

$$G \subseteq f((K \ominus D) \cap \Gamma).$$

Now suppose $\boldsymbol{x} \in K \cap \Gamma$ and $f(\boldsymbol{x}) \notin G$. Every occupied gap was included, so $f(\boldsymbol{x}) \in E \setminus S$. Choose $\boldsymbol{z} \in P_i \cap \Gamma$ of the same value. Then $\boldsymbol{x} - \boldsymbol{z} \in \Lambda$, and the local covering puts $\boldsymbol{x}$ in $R + D_0 + \Lambda$ or a cap. The first possibility would give $f(\boldsymbol{x}) \in S$, so $\boldsymbol{x}$ belongs to a cap.

There are $\mathcal{O}(|E| + |S|) = \mathcal{O}_d(\sigma W)$ progressions, all with common difference p. Apply Lemma 3.7 to obtain an AP decomposition of G. The deep-value inclusion and Lemma 3.5 give (10.1); the omitted-value argument gives (10.2). Sorting, membership tests in S, and the AP operations take $b_0^{\mathcal{O}_d(1)} \sigma W$ time.

Running time. The geometric construction takes $b_0^{\mathcal{O}_d(1)}(\sigma V / v_{D_0})N$ time. Together with preprocessing and the value enumerations, the total is

$$b_0^{\mathcal{O}_d(1)} \left[N + \frac{\sigma V}{v_{D_0}} N + \sigma W \right].$$

By Lemma 10.2 applied to D, and $D \subseteq D_0$,

$$|f(D \cap \Gamma)| \det \Gamma \leq \mathcal{O}_d(\mathrm{vol}_d(\pi_\Lambda D)) \leq \mathcal{O}_d(v_{D_0}).$$

Using $V \leq W \det \Gamma$, $W \leq u - \ell$, $\sigma \leq 2d\rho$, and the preceding bound on N gives (10.4). The invoked algorithms and fixed-dimensional rational arithmetic give the stated $\mathcal{O}_d(b_0)$ bounds on the cap coefficients $\boldsymbol{u}_i, a_i$, and the parallelotopes P_i. □

10.1 Subset-sum witness recovery

Lemma 10.3. *In the setup of Theorem 10.1, given $t \in G$ and $\boldsymbol{\tau} \in \Gamma$ with $\boldsymbol{\tau} + (dH \cap \Gamma) \subseteq \Sigma(X')$, one can recover a submultiset $Y \subseteq X$ with $\sum_{\boldsymbol{y} \in Y} f(\boldsymbol{y}) = t$ in $b_0^{\mathcal{O}_d(1)}(|\operatorname{supp} X| + \sum_{\boldsymbol{v} \in X'} |f(\boldsymbol{v})|)$ time.*

The proof of Lemma 10.3 uses the following standard lemma:

Lemma 10.4. *Given $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n \in \mathbb{Z}^d$ and $\boldsymbol{y} \in \mathcal{Z}((\boldsymbol{v}_i)_{i=1}^n) \cap \mathbb{Q}^d$, one can find rational $\boldsymbol{\alpha} \in [0,1]^n$ such that $\boldsymbol{y} = \sum_{i=1}^n \alpha_i \boldsymbol{v}_i$ and $|\{i \in [n] : 0 < \alpha_i < 1\}| \leq d$ in $\widetilde{\mathcal{O}}_d(n \cdot b_0^{\mathcal{O}_d(1)})$ time, with $\mathsf{maxbitlen}(\alpha_i) = \mathcal{O}_d(b_0)$ for each $i \in [n]$, where $b_0 = \mathsf{maxbitlen}(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n, \boldsymbol{y})$.*

Proof. We split the generators into two parts, choose their separate contributions to $\boldsymbol{y}$, and solve the two smaller representation problems recursively. Choosing a vertex at each split makes the final coefficient vector a vertex.

For $I \subseteq [n]$, let $X_I = (\boldsymbol{v}_i)_{i \in I}$ and write $Z_I = \mathcal{Z}(X_I)$. To represent $\boldsymbol{y}_I \in Z_I$ when $|I| > 1$, split I into I_1, I_2 of sizes differing by at most one. A possible contribution $\boldsymbol{z}$ from I_1 must belong to Z_{I_1} and leave $\boldsymbol{y}_I - \boldsymbol{z} \in Z_{I_2}$. Thus the possible choices form the nonempty polytope

$$Z_{I_1} \cap (\boldsymbol{y}_I - Z_{I_2}).$$

Optimizing over Z_{I_1} and Z_{I_2} separately gives an oracle for $Z_{I_1} \times Z_{I_2}$ by Lemma 6.2. Apply Lemma 6.3 with the equality $\boldsymbol{z}_1 + \boldsymbol{z}_2 = \boldsymbol{y}_I$ and the projection $(\boldsymbol{z}_1, \boldsymbol{z}_2) \mapsto \boldsymbol{z}_1$; a zero-objective query finds a vertex $\boldsymbol{z}$ of the displayed intersection. Recursively represent $\boldsymbol{z}$ using I_1 and $\boldsymbol{y}_I - \boldsymbol{z}$ using I_2. At a singleton $I = \{i\}$, solve $\alpha_i \boldsymbol{v}_i = \boldsymbol{y}_I$, choosing $\alpha_i = 0$ when $\boldsymbol{v}_i = \boldsymbol{0}$.

We prove by induction that the returned coefficient vector is a vertex of $\{\boldsymbol{\alpha} : \sum_{i \in I} \alpha_i \boldsymbol{v}_i = \boldsymbol{y}_I,\ 0 \le \alpha_i \le 1\}$. The singleton case follows from the choice just described. For the induction step, suppose the returned vector is the midpoint of two feasible coefficient vectors. Their contributions from I_1 are both feasible choices for the split and average to $\boldsymbol{z}$. Since $\boldsymbol{z}$ is a vertex, these contributions must both equal $\boldsymbol{z}$; their contributions from I_2 must therefore both equal $\boldsymbol{y}_I - \boldsymbol{z}$. The two coefficient vectors consequently solve the same two recursive subproblems. In each part their midpoint is the returned vertex, so induction makes their coefficients equal. This proves the claim.

At a vertex, the vectors $\boldsymbol{v}_i$ with $0 < \alpha_i < 1$ are linearly independent, so at most d coefficients are partial.

All other coefficients are integers, equal to 0 or 1. Subtract their contributions from $\boldsymbol{y}$ and solve for the at most d remaining coefficients in a nonsingular square subsystem. Cramer's rule writes them over a common denominator q obtained from a denominator of $\boldsymbol{y}$ and a determinant of at most d input vectors. Since $0 \le \alpha_i \le 1$, this gives $\mathsf{maxbitlen}(q, q\alpha_1, \dots, q\alpha_n) = \mathcal{O}_d(b_0)$. Every recursive target is a partial sum of this same final representation, so it can also be written over the denominator q and satisfies $\mathsf{maxbitlen}(\boldsymbol{y}_I) = \mathcal{O}_d(b_0 + \log n)$. Thus a node on I takes $\widetilde{\mathcal{O}}_d(|I|)(b_0 + \log n)^{\mathcal{O}_d(1)}$ time by Lemmas 6.2 and 6.3. The total size of the index sets at each level is n, and the balanced recursion has $\mathcal{O}(\log n)$ levels, giving the stated bound. □

Proof of Lemma 10.3. By Theorem 10.1, there exists $\boldsymbol{q} \in (K \ominus D) \cap \Gamma$ with $f(\boldsymbol{q}) = t$. The generating condition gives $\boldsymbol{\tau} \in \Sigma(X')$, so $\mathcal{Z}(X') - \boldsymbol{\tau} \subseteq D$. Since $K \subseteq \mathcal{Z}(X)$, we have

$$\boldsymbol{q} + \mathcal{Z}(X') - \boldsymbol{\tau} \subseteq \mathcal{Z}(X) = \mathcal{Z}(X \setminus X') + \mathcal{Z}(X').$$

Canceling $\mathcal{Z}(X')$ by support functions gives $\boldsymbol{q} - \boldsymbol{\tau} \in \mathcal{Z}(X \setminus X')$. Thus there is a point $\boldsymbol{x} \in \mathcal{Z}(X \setminus X') \cap \Gamma$ with $f(\boldsymbol{x}) = t - f(\boldsymbol{\tau})$.

Write $F = \Sigma(f(X'))$ and $R = \sum_{\boldsymbol{v} \in X'} |f(\boldsymbol{v})|$. Split each multiplicity in X' into $\mathcal{O}(b_0)$ binary chunks. Complement the negative chunks and discard the zero chunks; then F is a translate of the subset sums of positive integer chunk values whose total is R. Compute the subset sums by complete sparse convolutions (Lemma 3.1) along a balanced binary tree, and store the subset-sum set at each node. At each level the sum of the support sizes is $\mathcal{O}(R)$, since the total span is R and there are at most R positive chunks. To recover the two summands at a node, scan one child's set and test the complementary value in the other. Thus construction and backtracing take $b_0^{\mathcal{O}_d(1)}(|\operatorname{supp} X| + R)$ time. Undo the complements when recovering a witness.

Compute Γ by Lemma 6.6 and use Lemmas 6.2, 6.3 and 6.9 to find

$$\boldsymbol{x} \in \mathcal{Z}(X \setminus X') \cap \Gamma, \qquad f(\boldsymbol{x}) = t - f(\boldsymbol{\tau}).$$

These computations take $b_0^{\mathcal{O}_d(1)} |\operatorname{supp} X|$ time. By (6.1), $\mathsf{maxbitlen}(\boldsymbol{x}) = \mathcal{O}_d(b_0)$. Apply Lemma 10.4 to $\boldsymbol{w}_{\boldsymbol{v}} = \mu_{X \setminus X'}(\boldsymbol{v})\boldsymbol{v}$, $\boldsymbol{v} \in \operatorname{supp}(X \setminus X')$, obtaining coefficients $\alpha_{\boldsymbol{v}}$. Set $\beta_{\boldsymbol{v}} = \mu_{X \setminus X'}(\boldsymbol{v})\alpha_{\boldsymbol{v}}$. Then $0 \le \beta_{\boldsymbol{v}} \le \mu_{X \setminus X'}(\boldsymbol{v})$ and

$$\sum_{\boldsymbol{v}} \beta_{\boldsymbol{v}} \boldsymbol{v} = \boldsymbol{x},$$

$$|\{\boldsymbol{v} : 0 < \beta_{\boldsymbol{v}} < \mu_{X \setminus X'}(\boldsymbol{v})\}| \le d.$$

Set $a_{\boldsymbol{v}} = \lfloor \beta_{\boldsymbol{v}} \rfloor$. Then

$$\boldsymbol{\delta} = \boldsymbol{x} - \sum_{\boldsymbol{v}} a_{\boldsymbol{v}} \boldsymbol{v} = \sum_{\boldsymbol{v}} (\beta_{\boldsymbol{v}} - a_{\boldsymbol{v}}) \boldsymbol{v} \in dH \cap \Gamma.$$

Thus $\boldsymbol{\tau} + \boldsymbol{\delta} \in \Sigma(X')$. Recover a witness in X' for $f(\boldsymbol{\tau} + \boldsymbol{\delta})$ by backtracing in F. Combine it with $(a_{\boldsymbol{v}})$ on $X \setminus X'$. The resulting value is

$$\sum_{\boldsymbol{v}} a_{\boldsymbol{v}} f(\boldsymbol{v}) + f(\boldsymbol{\tau} + \boldsymbol{\delta}) = t.$$

Coefficient recovery takes $b_0^{\mathcal{O}_d(1)} |\operatorname{supp} X|$ time. Together with the construction of F and backtracing, this proves the stated time bound. □

11 Solving One Group

In this section, we prove Lemma 4.3.

We need the following lemma, which follows from a direct generalization of Bringmann's algorithm [Bri17] with its derandomization by Chan [Cha26]. We include a proof sketch in Appendix C.

Lemma 11.1. *Given $\boldsymbol{t} \in \mathbb{Z}^d_{\geq 0}$, write $B = \prod_{j=1}^d [0, t_j]$. For a multiset X with $\operatorname{supp} X \subseteq B \cap \mathbb{Z}^d$, one can compute $\Sigma(X) \cap B$ in $\widetilde{\mathcal{O}}_d(|\operatorname{supp} X| + |B \cap \mathbb{Z}^d|)$ time.*

Moreover, if $\boldsymbol{t} \in \Sigma(X)$, one can recover a witness $Y \subseteq X$ with $\sum_{\boldsymbol{y} \in Y} \boldsymbol{y} = \boldsymbol{t}$ in the same time complexity.

11.1 Computing boundary subset sums

We use the following lemma to compute the set of subset sums in a cap of the zonotope.

Lemma 11.2. *Given a multiset X of vectors in $\mathbb{Z}^d$, and a linear map $f \colon \mathbb{R}^d \to \mathbb{R}$ with integer coefficients, write $\Gamma = \mathcal{L}(X)$ and $K = \mathcal{Z}(X)$. Let $\boldsymbol{u} \in \mathbb{Q}^d \setminus \{\boldsymbol{0}\}$, $a \in \mathbb{Q}$, and a rational parallelotope P such that $P \supseteq Q \coloneqq K \cap \{\boldsymbol{x} : \langle \boldsymbol{u}, \boldsymbol{x} \rangle \geq a\}$ be given.*

Then, one can compute an integer set A satisfying

$$f(\Sigma(X) \cap Q) \subseteq A \subseteq f(\Sigma(X))$$

in $(|\operatorname{supp} X| + |f((P - P) \cap \Gamma)|)\, b_0^{\mathcal{O}_d(1)}$ time, where $b_0 = \mathsf{maxbitlen}(X, f, \boldsymbol{u}, a, P)$.

Moreover, given $t \in A$, one can recover a submultiset $Y \subseteq X$ with $\sum_{\boldsymbol{y} \in Y} f(\boldsymbol{y}) = t$ in the same time complexity.

Proof. Set $B = (P - P)/2$. Index $\operatorname{supp} X$ by $\boldsymbol{x}_i$, with multiplicities μ_i, and set $N = |\operatorname{supp} X|$. Let $\epsilon_i = 1$ if $\langle \boldsymbol{u}, \boldsymbol{x}_i \rangle > 0$, and $\epsilon_i = 0$ otherwise. Set

$$\boldsymbol{r} = \sum_i \epsilon_i \mu_i \boldsymbol{x}_i, \qquad \boldsymbol{v}_i = (1 - 2\epsilon_i)\boldsymbol{x}_i, \qquad c_i = |\langle \boldsymbol{u}, \boldsymbol{x}_i \rangle|.$$

Then $\boldsymbol{r}$ maximizes $\langle \boldsymbol{u}, \cdot \rangle$ on K. Set $T = \langle \boldsymbol{u}, \boldsymbol{r} \rangle - a$. If $T < 0$, return $A = \varnothing$. Otherwise, $\boldsymbol{r} \in Q \subseteq P$. For every selection $0 \leq k_i \leq \mu_i$, its complemented point $\boldsymbol{q} = \boldsymbol{r} + \sum_i k_i \boldsymbol{v}_i$ belongs to $\Sigma(X)$ and satisfies

$$\langle \boldsymbol{u}, \boldsymbol{r} - \boldsymbol{q} \rangle = \sum_i k_i c_i.$$

Thus $\boldsymbol{q} \in Q$ precisely when its cost is at most T. Every subcollection of such a selection also has cost at most T. Its point lies in $Q \subseteq P$, so its displacement from $\boldsymbol{r}$ lies in $P - P = 2B$.

Discard types with $c_i > T$. Each remaining vector $\boldsymbol{v}_i$ belongs to $2B$, and every subset sum in Q has a complemented witness using only remaining types. Let E be their real span and Γ_0 their generated lattice. If $f|_E = 0$, return $A = \{f(\boldsymbol{r})\}$ with selected counts $(\epsilon_i \mu_i)_i$. Otherwise, use a

rational basis of E as coordinates, and apply Lemma 6.12 to $2B \cap E$, Γ_0, and $f|_E$. Write its output as $\Phi\colon E \to \mathbb{R}^{d'}$, $\bar{f}$, and the box $\prod_{j=1}^{d'}[-U_j, U_j]$. Thus

$$\Phi\Gamma_0 = \mathbb{Z}^{d'}, \qquad f|_E = \bar{f} \circ \Phi, \qquad \Phi(2B \cap E) \subseteq \prod_{j=1}^{d'}[-U_j, U_j].$$

The grid size $V = \prod_j (2U_j + 1)$ satisfies

$$V \le \mathcal{O}_d(|f(2B \cap \Gamma_0)|) \le \mathcal{O}_d(|f(2B \cap \Gamma)|).$$

We now drop the total cost constraint $\sum_i k_i c_i \le T$ and impose only the coordinate bounds U_j. Partition the projected vectors among their $2^{d'}$ coordinate orthants. In each orthant, reverse the negative coordinate signs and apply Lemma 11.1 with coordinate bounds U_j and their multiplicities. Undo the signs, apply $\bar{f}$, and remove repeated scalar values, storing one contributing vector for each value. Let the resulting sets be $S_1, \dots, S_\ell$, where $\ell \le 2^{d'}$, and return

$$A = f(\boldsymbol{r}) + S_1 + \cdots + S_\ell.$$

For any subset sum in Q, each orthant part of its complemented witness has displacement in $2B \cap E$. Its projected coordinates therefore satisfy the bounds U_j, so its value belongs to the corresponding S_j. This proves completeness. Every returned value combines selections from disjoint groups of types, so it is a value of a legal complemented selection. Undoing complementation proves soundness.

Each intermediate scalar sumset is the image under $\bar{f}$ of a subset of $\prod_j[-\ell U_j, \ell U_j] \cap \mathbb{Z}^{d'}$, which has $\mathcal{O}_d(V)$ points. Shift the scalar supports to nonnegative integers and compute these sumsets by complete sparse convolutions (Lemma 3.1) in $b_0^{\mathcal{O}_d(1)}(N+V)$ time, including the orthant computations. For a requested value, scan the stored child supports to find summands at each merge. In each orthant, reverse the negative coordinate signs of the stored projected vector to obtain a target $\boldsymbol{s}$, apply the same sign changes to the orthant input vectors, discard those outside $\prod_j[0, s_j]$, and apply Lemma 11.1 with target $\boldsymbol{s}$. Nonnegativity preserves attainment, and the target box has at most V lattice points. Distribute the selected multiplicities among their original types and undo complementation. The scans and the $\mathcal{O}_d(1)$ box queries have the same time bound. Every numerical datum a in the lattice computations and transformations satisfies $\mathsf{maxbitlen}(a) = \mathcal{O}_d(b_0)$, and these computations fit the same time bound. □

11.2 Proof of Lemma 4.3

We now prove Lemma 4.3 (except for the witness-recovery part, which is deferred to the next subsection).

Fix X, X', f as in Lemma 4.3. Write $\Gamma = \mathcal{L}(X)$, $n = |X|$, $s = |X'|$, and $N = |\operatorname{supp} X|$. Compute an HNF basis matrix $B\colon \mathbb{R}^{d'} \to \operatorname{span}\Gamma$ by Lemma 6.6, where $d' = \operatorname{rank}\Gamma$. Replace X, X', f by $B^{-1}X, B^{-1}X', f \circ B$, respectively. Renaming d' as d and using the same notation for the transformed data, we have $\Gamma = \mathbb{Z}^d$. Set $H = \operatorname{conv}(\pm \operatorname{supp} X)$, $D = sH$, and $K = \mathcal{Z}(X)$. Write $w_X = \sum_{\boldsymbol{x} \in X} f(\boldsymbol{x}) = \operatorname{width}_f(K)$.

Set $\rho = s\lambda/n$. Equation (4.1b) gives $D \subseteq \rho\mathcal{Z}_0(X)$. Thus K contains a translate of $\rho^{-1}D$. Taking convex hulls in the generating condition shows that $\mathcal{Z}(X')$ contains a translate of dH, since the vertices of H lie in Γ. Hence K contains a translate of H. Apply Theorem 10.1 to $(X, X', f, 0, w_X)$. Write G for the returned set, p for the common difference of its AP decomposition, and $(Q_i, P_i)_{i \in \mathcal{I}}$ for the returned caps and enclosing parallelotopes. Write $Q_i = K \cap \{\boldsymbol{x} : \langle \boldsymbol{u}_i, \boldsymbol{x} \rangle \ge a_i\}$ and set

$B_i = (P_i - P_i)/2$. If all progressions in the decomposition of G are singletons, set $p = 1$ and coalesce adjacent values; otherwise $p \le w_X$.

For each $i \in \mathcal{I}$, apply Lemma 11.2 to $(X, f, \boldsymbol{u}_i, a_i, P_i)$, obtaining A_i. Set

$$A_{\mathrm{loc}} = \Big(\bigcup_{i\in\mathcal{I}} A_i\Big) \setminus G,$$

using membership queries in G (Lemma 3.7).

By Lemma 11.2 and Theorem 10.1, $A_{\mathrm{loc}} \cup G \subseteq f(\Sigma(X))$.

Conversely, let $t \in f(\Sigma(X)) \setminus G$ and choose $\boldsymbol{q} \in \Sigma(X) \subseteq K \cap \Gamma$ with $f(\boldsymbol{q}) = t$. By Theorem 10.1, some $i \in \mathcal{I}$ satisfies $\boldsymbol{q} \in Q_i$. Then Lemma 11.2 gives $t \in A_i$, so $t \in A_{\mathrm{loc}}$.

Thus $A_{\mathrm{loc}} \cup G = f(\Sigma(X))$. Apply Lemma 3.7 to the AP decomposition of G and the singleton progressions for A_{loc}, all with common difference p.

Time and progression count. Properness gives $N \le \Delta$ and

$$N \le |f(D \cap \Gamma)| = \mathcal{O}_d(\rho w_X),$$

since $\operatorname{supp} X \subseteq D \cap \Gamma$ and $2 \le \operatorname{width}_f(D) \le \rho w_X$. The input bounds give $\mathsf{maxbitlen}(X, X', f, 0, w_X) = \mathcal{O}_d(\log \Delta)$. The cover takes $\rho w_X (\log \Delta)^{\mathcal{O}_d(1)}$ time by Theorem 10.1.

Set $\Lambda = \Gamma \cap \ker f$. The cap thickness guarantee gives $H \subseteq c_d^{-1} B_i$, since $2c_d H \subseteq Q_i - Q_i \subseteq 2B_i$. Thus $c_d^{-1} B_i \cap \Gamma$ spans $\mathbb{R}^d$. Properness gives $N \le |f(H \cap \Gamma)|$. Apply Lemma 10.2 to $(c_d^{-1} + 2)B_i$. Since P_i is a translate of B_i, Lemma 9.2 and the volume bound for the enclosures in Theorem 10.1 give

$$\sum_{i\in\mathcal{I}} (N + |f(2B_i \cap \Gamma)|) = \mathcal{O}_d(\rho w_X). \tag{11.1}$$

The local computations and membership queries take $\rho w_X (\log \Delta)^{\mathcal{O}_d(1)}$ total time by Lemmas 11.2 and 3.7 and (11.1).

Translate each nonempty $Q_i \cap \Gamma$ by one of its points into $(Q_i - Q_i) \cap \Gamma \subseteq 2B_i \cap \Gamma$. Since $A_{\mathrm{loc}} \subseteq f(K \cap \Gamma) \setminus G$, Theorem 10.1 and (11.1) give

$$|A_{\mathrm{loc}}| \le \sum_{i\in\mathcal{I}} |f(Q_i \cap \Gamma)| \le \sum_{i\in\mathcal{I}} |f(2B_i \cap \Gamma)| = \mathcal{O}_d(\rho w_X).$$

Together with Theorem 10.1 (i), this bounds the output decomposition size by $\mathcal{O}_d(\rho w_X)$. Constructing it takes $\rho w_X (\log \Delta)^{\mathcal{O}_d(1)}$ time by Lemma 3.7. Since $w_X \le n\Delta$,

$$\rho w_X = \mathcal{O}_d(s\lambda\Delta). \tag{11.2}$$

Hence construction takes $\widetilde{\mathcal{O}}_d(s\lambda\Delta)$ time and the decomposition has size $\mathcal{O}_d(s\lambda\Delta)$.

11.3 Witness recovery

We now prove the witness-recovery part of Lemma 4.3.

Given $X, X', f, \boldsymbol{\tau}$ and $t \in f(\Sigma(X))$, repeat the geometric preparation and covering construction in Section 11.2. Under the HNF change of coordinates, replace $\boldsymbol{\tau}$ by $B^{-1}\boldsymbol{\tau}$ as well. If $t \in G$, apply Lemma 10.3 to $(X, X', f, \boldsymbol{\tau}, t)$ to recover a witness of value t.

If $t \notin G$, then Theorem 10.1 gives an index $i \in \mathcal{I}$ with $t \in f(\Sigma(X) \cap Q_i)$. Apply Lemma 11.2 to $(X, f, \boldsymbol{u}_i, a_i, P_i)$ successively for $i \in \mathcal{I}$ until $t \in A_i$, and use the witness query of Lemma 11.2.

Rebuilding the geometric data and computing the local sets are part of $\mathsf{OneGroup}(X, X', f)$, whose construction bound in Lemma 4.3 is $\widetilde{\mathcal{O}}_d(s\lambda\Delta)$. By Lemma 11.2, the witness query for A_i takes the same time as its construction, so it fits within this bound. When $t \in G$, the bound $\sum_{\boldsymbol{v}\in X'} |f(\boldsymbol{v})| \le s\Delta$ gives $\widetilde{\mathcal{O}}_d(s\Delta + |\operatorname{supp} X|)$ time by Lemma 10.3. Since $\lambda \ge 1$ and $|\operatorname{supp} X| \le \Delta$, each recovery takes $\widetilde{\mathcal{O}}_d(s\lambda\Delta)$ time, as claimed.

12 Merging Groups

We prove Lemma 4.4 by covering an interval of the combined zonotope, converting its caps into candidate pairs, and computing the resulting sumsets by sparse convolution (Lemma 3.1).

12.1 Geometric pair covers

Use the inputs and value map f of Lemma 4.4. If $I = \varnothing$, return the empty set. Set $V = X \sqcup Y$, $\Gamma_X = \mathcal{L}(X)$, $s = |X'|$, and $N = |\operatorname{supp} X| + |\operatorname{supp} Y|$. Since $\operatorname{supp} Y \subseteq \Gamma_X$, $\mathcal{L}(V) = \Gamma_X$. Compute an HNF basis matrix $B\colon \mathbb{R}^{d'} \to \operatorname{span}\Gamma_X$ by Lemma 6.6, where $d' = \operatorname{rank}\Gamma_X$. Replace X, X', Y, V, f by $B^{-1}X, B^{-1}X', B^{-1}Y, B^{-1}V, f \circ B$, respectively. The transformed lattice is $\mathbb{Z}^{d'}$. Retain the same notation for the transformed data and write d for d', so $\Gamma_X = \mathbb{Z}^d$. Set $H = \operatorname{conv}(\pm\operatorname{supp} V)$ and $D = sH$.

Write $w_X = \sum_{\boldsymbol{v}\in X} f(\boldsymbol{v})$ and set

$$J^+ = J + [0, w_X], \qquad K = \mathcal{Z}(V) \cap f^{-1}(J^+).$$

Since $f(\operatorname{supp} X) \subseteq [1, \Delta]$, we have $f(\mathcal{Z}(X)) = [0, w_X]$. Thus

$$\mathcal{Z}(X) + (\mathcal{Z}(Y) \cap f^{-1}(J)) \subseteq K \subseteq \mathcal{Z}(V).$$

The first body is nonempty because $J \subseteq f(\mathcal{Z}(Y))$. In particular, K contains a translate of $\mathcal{Z}(X)$. Moreover, $J^+ \subseteq f(\mathcal{Z}(V))$, so $\operatorname{width}_f(K) = |J^+| = w_X + |J|$. Set $\rho = s\lambda/|X|$. The balance condition (4.1b) gives $D \subseteq \rho\mathcal{Z}_0(X)$, so K contains a translate of $\rho^{-1}D$. Properness of f on $H \cap \Gamma_X$ and $\operatorname{supp} X, \operatorname{supp} Y \subseteq H \cap \Gamma_X \subseteq D \cap \Gamma_X$ give

$$N \le 2|f(D \cap \Gamma_X)| \le 2(\rho|J^+| + 1) = \mathcal{O}_d(\rho|J^+|),$$

where $2 \le \operatorname{width}_f(D) \le \rho|J^+|$.

Taking convex hulls in the generating condition shows that $\mathcal{Z}(X')$ contains a translate of dH, since the vertices of H lie in Γ_X. Hence K contains a translate of H. Apply Theorem 10.1 to V, X', f and the endpoints of J^+. Write G for the returned set and $Q_i = K \cap \{\boldsymbol{x} : \langle \boldsymbol{u}_i, \boldsymbol{x}\rangle \ge a_i\}$, $i \in \mathcal{I}$, for the returned caps, with enclosing parallelotopes P_i. Set $B_i = (P_i - P_i)/2$, an origin-symmetric translate of P_i. Choose a common difference p for the AP decomposition of G. If all progressions in the decomposition of G are singletons, set $p = 1$ and coalesce adjacent values; otherwise $p \le |J^+|$. Use Lemmas 6.3 and 6.9 to find $\boldsymbol{q}_i \in Q_i \cap \Gamma_X$ for each nonempty lattice intersection, and let $\mathcal{I}_0$ index these caps.

For each $i \in \mathcal{I}_0$, choose the integral maximizer

$$\boldsymbol{x}_i = \sum_{\substack{\boldsymbol{v}\in X \\ \langle \boldsymbol{u}_i, \boldsymbol{v}\rangle > 0}} \boldsymbol{v}$$

of $\langle \boldsymbol{u}_i, \boldsymbol{x}\rangle$ over $\mathcal{Z}(X)$. Compute $\boldsymbol{x}_i$ by scanning $\operatorname{supp} X$. Enumerate $U_i = f(2B_i \cap \Gamma_X)$ by Lemma 6.10, and define the candidate lists

$$C_{X,i} = f(\boldsymbol{x}_i) + U_i, \qquad C_{Y,i} = f(\boldsymbol{q}_i - \boldsymbol{x}_i) + U_i.$$

To prove pair coverage, take $\boldsymbol{z}_X \in \mathcal{Z}(X) \cap \Gamma_X$ and $\boldsymbol{z}_Y \in \mathcal{Z}(Y) \cap \Gamma_X$ with $f(\boldsymbol{z}_Y) \in J$, and write $\boldsymbol{q} = \boldsymbol{z}_X + \boldsymbol{z}_Y$. If $f(\boldsymbol{q}) \notin G$, then $\boldsymbol{q} \in K$ and Theorem 10.1 gives an $i \in \mathcal{I}_0$ with $\boldsymbol{q} \in Q_i \subseteq P_i$. The point $\boldsymbol{p} = \boldsymbol{x}_i + \boldsymbol{z}_Y$ also belongs to K and satisfies $\langle \boldsymbol{u}_i, \boldsymbol{p} \rangle \geq \langle \boldsymbol{u}_i, \boldsymbol{q} \rangle$. Thus $\boldsymbol{p} \in Q_i \subseteq P_i$. Since $\boldsymbol{q}_i, \boldsymbol{q} \in Q_i \subseteq P_i$ and $P_i - P_i = 2B_i$, we obtain

$$\begin{aligned} \boldsymbol{z}_Y - (\boldsymbol{q}_i - \boldsymbol{x}_i) = \boldsymbol{p} - \boldsymbol{q}_i \in 2B_i, \\ \boldsymbol{z}_X - \boldsymbol{x}_i = \boldsymbol{q} - \boldsymbol{p} \in 2B_i. \end{aligned}$$

Both differences lie in Γ_X, so $(f(\boldsymbol{z}_X), f(\boldsymbol{z}_Y)) \in C_{X,i} \times C_{Y,i}$.

12.2 The merge algorithm

Use the given AP decompositions to test candidate membership in S_X and S_Y by Lemma 3.7, and compute $S_{X,i} \coloneqq C_{X,i} \cap S_X$ and $S_{Y,i} \coloneqq C_{Y,i} \cap S_Y$ in increasing order. For each $i \in \mathcal{I}_0$, compute $T_i = S_{X,i} + S_{Y,i}$ by one-dimensional sparse convolution (Lemma 3.1), after translating the inputs to nonnegative integers.

Set $S^+ \coloneqq G \cup \bigcup_{i \in \mathcal{I}_0} T_i$. Represent each $t \in T_i$ by the singleton progression $\{t\}$ with common difference p. Combine these with the progressions representing G, intersect them with I, and apply Lemma 3.7 to construct an AP decomposition of $S^+ \cap I$.

Correctness and running time. By construction, $T_i \subseteq S_X + S_Y \subseteq f(\Sigma(V))$. Together with Theorem 10.1 (i), this gives $S^+ \subseteq f(\Sigma(V))$. Conversely, let $t \in f(\Sigma(V)) \cap I$ and split a witnessing submultiset between X and Y, with sums $\boldsymbol{z}_X, \boldsymbol{z}_Y$. The interval condition (4.2) gives $f(\boldsymbol{z}_Y) \in J$, so $(f(\boldsymbol{z}_X), f(\boldsymbol{z}_Y)) \in S_X \times S_Y$. If $t \notin G$, the pair coverage proved above gives $i \in \mathcal{I}_0$ with $(f(\boldsymbol{z}_X), f(\boldsymbol{z}_Y)) \in C_{X,i} \times C_{Y,i}$. Hence $(f(\boldsymbol{z}_X), f(\boldsymbol{z}_Y)) \in S_{X,i} \times S_{Y,i}$, so $t \in T_i$. Thus $S^+ \cap I = (S_X + S_Y) \cap I = f(\Sigma(V)) \cap I$.

The local sumsets satisfy

$$C_{X,i} + C_{Y,i} \subseteq f(\boldsymbol{q}_i) + f(4B_i \cap \Gamma_X).$$

Set $\Lambda = \Gamma_X \cap \ker f$. Each Q_i contains a translate of $c_d H$, so $H \subseteq c_d^{-1} B_i$ by taking differences. Thus $4c_d^{-1} B_i \cap \Gamma_X$ spans $\mathbb{R}^d$. Since $N \leq 2|f(H \cap \Gamma_X)|$ and B_i is a translate of P_i, Lemmas 10.2 and 9.2 and the volume bound for the P_i in Theorem 10.1 give

$$\sum_{i \in \mathcal{I}_0} |f(4B_i \cap \Gamma_X)| + N|\mathcal{I}| = \mathcal{O}_d(\rho |J^+|).$$

Consequently,

$$\sum_{i \in \mathcal{I}_0} |C_{X,i} + C_{Y,i}| = \mathcal{O}_d(\rho |J^+|). \tag{12.1}$$

Since $|C_{X,i}| + |C_{Y,i}| \leq 2|C_{X,i} + C_{Y,i}|$, the total number of candidate values and the number of list pairs are also $\mathcal{O}_d(\rho |J^+|)$.

The inequality $N \leq 2|f(D \cap \Gamma_X)|$ and Theorem 10.1, with $b_0 = \mathcal{O}_d(\log \Delta)$, bound the covering time by $\rho |J^+| (\log \Delta)^{\mathcal{O}_d(1)}$. We optimize over K and its caps using Lemmas 6.2 and 6.3, in $N(\log \Delta)^{\mathcal{O}_d(1)}$ time per query. The bound on $N|\mathcal{I}|$ implies that all lattice-feasibility queries and scans for $\boldsymbol{x}_i$ take $\rho |J^+| (\log \Delta)^{\mathcal{O}_d(1)}$ time. The lattice-value enumerations and list construction take the same time, by Lemma 6.10 and (12.1).

The bound $N = \mathcal{O}_d(\rho |J^+|)$ also absorbs the $N(\log \Delta)^{\mathcal{O}_d(1)}$ time for preparing V, J^+ and computing the lattice basis.

Since $w_X \le |X|\Delta$ and $|J^+| = w_X + |J|$,

$$\rho|J^+| = \rho(w_X + |J|) = \mathcal{O}_d\Big(s\lambda\left(\Delta + \frac{|J|}{|X|}\right)\Big). \tag{12.2}$$

Equation (12.1) bounds filtering and sorting time by $\rho|J^+|(\log\Delta)^{\mathcal{O}_d(1)}$. Since $T_i \subseteq C_{X,i} + C_{Y,i}$,

$$\sum_i |T_i| \le \sum_i |C_{X,i} + C_{Y,i}| \le \mathcal{O}_d(\rho|J^+|). \tag{12.3}$$

Thus sparse convolution (Lemma 3.1) gives the same total time bound for computing these sumsets.

By the progression bound in Theorem 10.1 and (12.3), the output AP decomposition has size $\mathcal{O}_d(\rho|J^+|)$ and can be constructed in $\rho|J^+|(\log\Delta)^{\mathcal{O}_d(1)}$ time by Lemma 3.7. Equation (12.2) gives the stated construction time and progression count.

Recovering two summands. Given the inputs of MergeWitness and $v \in (S_X + S_Y) \cap I$, repeat the covering and candidate-list constructions. In the lattice-coordinate change above, also replace $\boldsymbol{\tau}$ by $B^{-1}\boldsymbol{\tau}$. If $v \in G$, apply Lemma 10.3 to $(V, X', f, \boldsymbol{\tau}, v)$. It returns a submultiset $T \subseteq V$ with $\sum_{\boldsymbol{w}\in T} f(\boldsymbol{w}) = v$. Split T between X and Y, obtaining $T_X \subseteq X$ and $T_Y \subseteq Y$. Return $v_X = f(\sum_{\boldsymbol{w}\in T_X} \boldsymbol{w})$ and $v_Y = f(\sum_{\boldsymbol{w}\in T_Y} \boldsymbol{w})$. Then $v_X + v_Y = v \in I$, and (4.2) gives $v_Y \in J$. Hence $v_X \in S_X$ and $v_Y \in S_Y$. Since $\sum_{\boldsymbol{v}\in X'} |f(\boldsymbol{v})| \le s\Delta$, this takes $\widetilde{\mathcal{O}}_d(s\Delta + N)$ time.

If $v \notin G$, for each $i \in \mathcal{I}_0$ and $a \in C_{X,i}$, test $a \in S_X$ and $v - a \in S_Y$ by Lemma 3.7, and return $(a, v-a)$ when both tests succeed. Pair coverage guarantees a successful test. Equation (12.1) gives $\sum_i |C_{X,i}| = \mathcal{O}_d(\rho|J^+|)$, so the construction and these tests take $\rho|J^+|(\log\Delta)^{\mathcal{O}_d(1)}$ time.

By (12.2) and $\lambda \ge 1$, recovery takes time

$$\widetilde{\mathcal{O}}_d\Big(s\lambda\left(\Delta + \frac{|J|}{|X|}\right)\Big).$$

A Proof of Lemma 5.3

We prove Lemma 5.3 by derandomizing Randolph and Węgrzycki's algorithmic Freiman theorem [RW24, Theorem 5 and full-version Appendix A]. We replace the random modeling step of [RW24, Lemma A.2] by conditional expectations and give a certified finite-precision implementation of the Fourier step in [RW24, Lemma A.3].

For a GAP P, let $\mathsf{maxbitlen}(P) = \mathsf{maxbitlen}(x_P, f_P, B_P)$.

Lemma A.1. *Given a rank-d GAP $P \subseteq \mathbb{Z}$ with $b_0 = \mathsf{maxbitlen}(P)$, one can in $\mathrm{vol}(P)b_0^{\mathcal{O}_d(1)}$ time either certify 2-properness or compute a GAP $Q \supseteq P$ of rank $d' < d$ with $\mathrm{vol}(Q) = \mathcal{O}_d(\mathrm{vol}(P))$ and $\mathsf{maxbitlen}(Q) = \mathcal{O}_d(b_0)$.*

Lemma A.2. *Let $P \subseteq \mathbb{Z}$ be a rank-d GAP, $0 \in A \subseteq P$, and $b_0 = \mathsf{maxbitlen}(P)$. One can compute in $\mathrm{vol}(P)b_0^{\mathcal{O}_d(1)}$ time an origin-symmetric 4-proper GAP $Q \supseteq A$ of rank $d' \le d$ with $\mathrm{vol}(Q) = \mathcal{O}_d(\mathrm{vol}(P))$, $\mathsf{maxbitlen}(Q) = \mathcal{O}_d(b_0)$, and $\mathcal{L}(f_Q^{-1}(A)) = \mathbb{Z}^{d'}$.*

Proof of Lemma 5.3. The singleton case is immediate, so assume $|A| \ge 2$. The Plünnecke–Ruzsa inequality [TV06, Corollaries 6.28–6.29] gives $|rA - sA| \le K^{r+s}|A|$ for nonnegative integers r, s with $r + s > 0$. Thus every signed sumset of A with a fixed number of summands has $\mathcal{O}_K(|A|)$ elements. Compute the sumsets used below by sparse nonnegative convolution (Lemma 3.1) in $\mathcal{O}_K(|A|)\operatorname{poly}\log\Delta$ time.

A deterministic Freiman model. The Fourier computation must take time nearly linear in $|A|$, even when the elements of A have much larger magnitudes. We therefore pass to a modulus $p = \mathcal{O}_K(|A|)$, choosing a subset $A' \subseteq A$ with $|A'| \geq |A|/16$. A map $\psi\colon A' \to \mathbb{Z}_p$ is a Freiman 8-isomorphism if it preserves equalities between eight-term sums in both directions. We use such a map to lift a progression from $2\psi(A') - 2\psi(A')$ back to the integers. Following [RW24, Lemma A.2], let $A_{\mathrm{diff}} = 8A - 8A$, choose by a deterministic sieve a prime p with $8|A_{\mathrm{diff}}| < p < 16|A_{\mathrm{diff}}|$, and let $q = 2^L$ be the least power of two exceeding $\max_{x \in A_{\mathrm{diff}}} |x|$. For each positive integer m, write $[z]_m$ for the unique representative of z mod m in $[-m/2, m/2)$. We choose an odd λ mod q such that, for every $x \in A_{\mathrm{diff}} \setminus \{0\}$,

$$p \nmid [\lambda x]_q. \tag{A.1}$$

To show that a suitable λ exists, write $x = 2^v u$, with u odd, and $Q = 2^{L-v}$. Since $[\lambda x]_q = 2^v [\lambda u]_Q$ and p is odd, $p \mid [\lambda x]_q$ if and only if $p \mid [\lambda u]_Q$. For a uniformly chosen odd λ mod q, the residue $[\lambda u]_Q$ is uniform among the odd residues modulo Q, so $\Pr[p \mid [\lambda x]_q] < 2/p$. Define $N(\lambda) = |\{x \in A_{\mathrm{diff}} \setminus \{0\} : p \mid [\lambda x]_q\}|$. Then

$$\mathbb{E}_\lambda N(\lambda) < \frac{2|A_{\mathrm{diff}}|}{p} < \frac{1}{4}.$$

We find such a λ deterministically by fixing its bits from low to high, beginning with the oddness bit. At each step, choose the next bit that does not increase the conditional expectation of $N(\lambda)$. After the lowest $t \geq 1$ bits of λ have been fixed to λ_0, set $h = \min\{t, L - v\}$ for each $x = 2^v u$. Then

$$\Pr[p \mid [\lambda x]_q \mid \lambda \equiv \lambda_0 \pmod{2^t}] = \frac{|\{z \in \mathbb{Z} \cap [-Q/2, Q/2) : p \mid z,\ z \equiv \lambda_0 u \pmod{2^h}\}|}{Q/2^h}.$$

To count the numerator, write $z = pj$. Since p is odd, the congruence fixes one residue class of j modulo 2^h; counting its representatives with $-Q/2 \leq pj < Q/2$ uses integer division. Thus both conditional expectations are computed exactly in $\mathcal{O}(|A_{\mathrm{diff}}|)\,\mathrm{poly}\log\Delta$ time per bit. Once all bits are fixed, $N(\lambda) = 0$, proving (A.1).

Partition $[-q/2, q/2)$ into sixteen equal half-open intervals, and let $A' \subseteq A$ be the largest inverse image under $x \mapsto [\lambda x]_q$. Then $|A'| \geq |A|/16$, and the map $\psi(x) = [\lambda x]_q \bmod p$ for $x \in A'$ is a Freiman 8-isomorphism. For two eight-term sums from A', let $x \in A_{\mathrm{diff}}$ be their difference. All sixteen canonical residues lie in one interval of length $q/16$, so the difference of their residue sums has absolute value below $q/2$ and equals $[\lambda x]_q$. Thus the ψ-image sums agree if and only if $p \mid [\lambda x]_q$, which by (A.1) is equivalent to $x = 0$.

The Fourier step. Let $U = \psi(A')$ and $\alpha = |U|/p = \Omega_K(1)$. For $r \in \mathbb{Z}_p$, define

$$\widehat{1_U}(r) = \frac{1}{p} \sum_{x \in U} e^{-2\pi i r x/p}.$$

Compute dyadic complex approximations $\tilde{a}_r$ with certified error $|\tilde{a}_r - \widehat{1_U}(r)| \leq \alpha^{3/2}/16$. Set

$$R = \{r \neq 0 : |\tilde{a}_r| \geq 3\alpha^{3/2}/16\}.$$

Thus $|\widehat{1_U}(r)| \geq \alpha^{3/2}/8$ for $r \in R$ and $|\widehat{1_U}(r)| < \alpha^{3/2}/4$ for $r \notin R \cup \{0\}$. Parseval gives $|R| \leq 64\alpha^{-2}$ and

$$\sum_{r \notin R \cup \{0\}} |\widehat{1_U}(r)|^4 \leq \alpha^4/16.$$

For $R \subseteq \mathbb{Z}_p$ and $\delta > 0$, write

$$\mathrm{Bohr}_p(R, \delta) = \{x \in \mathbb{Z}_p : \min_{k \in \mathbb{Z}} |rx/p - k| \le \delta \text{ for every } r \in R\}.$$

The inverse Fourier transform of $|\widehat{1_U}|^4$ is a positive multiple of the number of representations $x = u_1 + u_2 - u_3 - u_4$ with $u_i \in U$. For $x \in \mathrm{Bohr}_p(R, 1/4)$ and $r \in R$, $\mathrm{Re}\, e^{2\pi i r x/p} \ge 0$. Since $\widehat{1_U}(0) = \alpha$,

$$\mathrm{Re} \sum_{r \in \mathbb{Z}_p} |\widehat{1_U}(r)|^4 e^{2\pi i r x/p} \ge \alpha^4 - \sum_{r \notin R \cup \{0\}} |\widehat{1_U}(r)|^4 \ge \frac{15}{16}\alpha^4 > 0.$$

Hence

$$\mathrm{Bohr}_p(R, 1/4) \subseteq 2U - 2U.$$

Compute the length-p transform by a power-of-two convolution of length $\mathcal{O}(p)$ using Bluestein's reduction. With b-bit floating-point arithmetic and roots of unity approximated within 2^{-b}, the FFT error bound [BZ10, Theorem 3.6] gives absolute error $p^{\mathcal{O}(1)}2^{-b}$. Approximating the input chirp factors contributes $\mathcal{O}(p2^{-b})$ additional error. The roots can be computed in $\mathrm{poly}(b + \log p)$ time each [BZ10, Section 4.4]. Choosing $b = \mathcal{O}(\log p + \log \alpha^{-1}) = \mathcal{O}_K(\log p)$ with a sufficiently large constant therefore gives the required coefficient error $\alpha^{3/2}/16$ in $\mathcal{O}_K(p \,\mathrm{poly}\log p)$ time.

Lattice enumeration and covering. Compute in $\widetilde{\mathcal{O}}_K(p)$ time a proper GAP in $\mathbb{Z}_p$ of rank $\mathcal{O}_K(1)$ and size $\Omega_K(p)$ inside $\mathrm{Bohr}_p(R, 1/4)$, using [RW24, Lemma A.4] when $R \neq \varnothing$ and a centered interval when $R = \varnothing$.

Let $E = 2A' - 2A'$ and define $\tau\colon E \to 2U - 2U$ by $\tau(x) = [\lambda x]_q \bmod p$. For $a_i \in A'$, the residues $[\lambda a_i]_q$ lie in one interval of length $q/16$, so

$$\tau(a_1 + a_2 - a_3 - a_4) = \psi(a_1) + \psi(a_2) - \psi(a_3) - \psi(a_4).$$

Comparing two representations from E uses four terms of A' on each side, so the Freiman property gives $\tau(x) = \tau(y)$ if and only if $x = y$. Thus τ is bijective. An equality between two sums of two points of E uses eight terms on each side, so τ and τ^{-1} preserve these equalities. Lift the progression by τ^{-1}. These additive relations give a GAP representation on the same coefficient box, hence with the same rank. Injectivity preserves its size and makes this representation proper. It lies in E.

For computation, enumerate E by sparse convolution (Lemma 3.1) and tabulate each $x \in E$ under the key $\tau(x)$. This evaluates τ^{-1} on the progression in $\mathcal{O}_K(|A|)\,\mathrm{poly}\log\Delta$ total time.

To obtain a progression containing A, use the fixed-order greedy implementation of the covering lemma [RW24, Lemma A.5]. Its hypothesis holds because, for the lifted progression $Q \subseteq 2A' - 2A'$,

$$|Q + A| \le |3A - 2A| = \mathcal{O}_K(|A|) = \mathcal{O}_K(|Q|).$$

It gives a cover $A \subseteq T + (Q - Q)$ with $T \subseteq A$ and $|T| = \mathcal{O}_K(1)$. Explicitly, compute $Q - Q$ by sparse convolution (Lemma 3.1), repeatedly choose the least uncovered element $t \in A$, and mark $t + (Q - Q)$ as covered. The chosen translates $t + Q$ are pairwise disjoint subsets of $A + Q$, so there are at most $|A + Q|/|Q| = \mathcal{O}_K(1)$ choices. Since $Q - Q \subseteq 4A - 4A$ has $\mathcal{O}_K(|A|)$ elements, use balanced search trees to mark these elements in $\mathcal{O}_K(|A|)\,\mathrm{poly}\log\Delta$ time. Adjoining the elements of T to $Q - Q$ as directions with coefficients in $\{0, 1\}$ gives a GAP $P_0 \supseteq A$ of rank $\mathcal{O}_K(1)$, volume $\mathcal{O}_K(|A|)$, and $\mathsf{maxbitlen}(P_0) = \mathcal{O}_K(\log\Delta)$, in time $\mathcal{O}_K(|A|)\,\mathrm{poly}\log\Delta$. Indeed, $\mathrm{vol}(Q) = |Q| = \mathcal{O}_K(|A|)$, and doubling its coefficient widths and adjoining $|T| = \mathcal{O}_K(1)$ binary directions multiply the volume by $\mathcal{O}_K(1)$.

Properness and generation. Apply Lemma A.2 to (P_0, A), obtaining an origin-symmetric 4-proper GAP $Q \supseteq A$ of rank $d = \mathcal{O}_K(1)$, and return $P = 2Q$. Then P is origin-symmetric and 2-proper and contains $A + A$. Since $Q \subseteq P$ and $f_P = f_Q$, properness of P gives $f_P^{-1}(a) = f_Q^{-1}(a)$ for every $a \in A$. Thus $\mathcal{L}(f_P^{-1}(A)) = \mathbb{Z}^d$. By Lemmas 5.2 and A.2, $\operatorname{vol}(P) \le 2^d \operatorname{vol}(Q) = \mathcal{O}_K(|A|)$. The reduction takes $\operatorname{vol}(P_0)\,\mathsf{maxbitlen}(P_0)^{\mathcal{O}_K(1)} = \mathcal{O}_K(|A|)\operatorname{poly}\log\Delta$ time, so the total running time has the same bound. □

Proofs of the GAP reductions

In both proofs, normalize the input coefficient box to $\prod_i [0, w_i]$.

Proof of Lemma A.1. Write the normalized input as $P = x_P + f_P(\mathbb{Z}^d \cap \prod_i [0, w_i])$ and set $B = \prod_i [-w_i, w_i]$. The zero-map case is immediate.

Otherwise, apply Lemma 6.12 to B, $\mathbb{Z}^d$, and f_P, obtaining Φ, $\bar{f}$, and an integer box $C \subseteq \mathbb{R}^{d'}$. If $d' = d$, then Φ is invertible, so the injectivity of $\bar{f}$ on $C \cap \mathbb{Z}^d$ implies the injectivity of f_P on $B \cap \mathbb{Z}^d$. This certifies 2-properness: B is an integer translate of $\prod_i [0, 2w_i]$.

If $d' < d$, return

$$Q = x_P + \bar{f}(C \cap \mathbb{Z}^{d'}).$$

It contains P because $\prod_i [0, w_i] \subseteq B$ and $\Phi B \subseteq C$. The box-size guarantee and $\prod_i (2w_i + 1) \le 2^d \operatorname{vol}(P)$ give

$$\operatorname{vol}(Q) = |C \cap \mathbb{Z}^{d'}| \le \mathcal{O}_d(|f_P(B \cap \mathbb{Z}^d)|) \le \mathcal{O}_d(\operatorname{vol}(P)).$$

The running time and $\mathsf{maxbitlen}$ bounds in Lemma 6.12 imply the claimed bounds. □

Proof of Lemma A.2. If $A = \{0\}$, return the rank-zero GAP $\{0\}$. Otherwise, write $P = x_P + f_P(\mathbb{Z}^d \cap \prod_i [0, w_i])$. For each $a \in A$, choose a coefficient lift $\boldsymbol{u}_a$ with $x_P + f_P(\boldsymbol{u}_a) = a$, and set

$$\tilde{A} = \{\boldsymbol{u}_a - \boldsymbol{u}_0 : a \in A\}, \qquad \Gamma = \mathcal{L}(\tilde{A}), \qquad B = \prod_i [-w_i, w_i] \cap \operatorname{span}(\Gamma).$$

Then $f_P(\tilde{A}) = A$, and $B \cap \Gamma$ spans $\operatorname{span}(\Gamma)$. The restriction of f_P to this space is nonzero. Use lattice coordinates for Γ and apply Lemma 6.12, obtaining Φ, $\bar{f}$, and an integer box $C \subseteq \mathbb{R}^{d'}$. Define

$$Q = \bar{f}(C \cap \mathbb{Z}^{d'}).$$

The containments $\Phi\tilde{A} \subseteq \Phi B \subseteq C$ and $f_P|_{\operatorname{span}(\Gamma)} = \bar{f} \circ \Phi$ give $A \subseteq Q$. This GAP is origin-symmetric. Injectivity of $\bar{f}$ on $4C \cap \mathbb{Z}^{d'}$ gives 4-properness. Its coefficient lifts satisfy

$$\mathcal{L}(f_Q^{-1}(A)) = \mathcal{L}(\Phi\tilde{A}) = \Phi\Gamma = \mathbb{Z}^{d'}.$$

Since $\Gamma \subseteq \mathbb{Z}^d$ and $\prod_i (2w_i + 1) \le 2^d \operatorname{vol}(P)$, the box-size guarantee gives

$$\operatorname{vol}(Q) = |C \cap \mathbb{Z}^{d'}| \le \mathcal{O}_d(|f_P(B \cap \Gamma)|) \le \mathcal{O}_d(\operatorname{vol}(P)).$$

Enumerate the original coefficient box and store one lift under each value $x_P + f_P(\boldsymbol{u})$ in a balanced search tree. Retrieving the lifts of A and forming $\tilde{A}$ takes $\operatorname{vol}(P) b_0^{\mathcal{O}_d(1)}$ time. An HNF basis of Γ and Lemma 6.12 give $\mathsf{maxbitlen}(Q) = \mathcal{O}_d(b_0)$ within the same time bound. □

B Proof of Theorem 5.13

We deduce Theorem 5.13 from Theorem 5.1 by encoding vectors as integers and using s-generation to lift the GAP generators without carries.

Proof of Theorem 5.13. We follow the reduction in [PZ25, Appendix]. Use Theorem 5.1 with exponent $\beta_0 \coloneqq 2\beta k$, and let C_0, d be its constants.

Set $H = 2\lceil \Delta^2 \rceil + 1$ and define

$$\psi(a_1, \dots, a_k) = \sum_{j=1}^{k} H^{j-1} a_j.$$

This linear map is injective on $[-\Delta^2, \Delta^2]^k \cap \mathbb{Z}^k$. Thus $\psi(A)$ is a multiset of non-zero integers in $[-\Delta_0, \Delta_0]$, where

$$\Delta_0 \coloneqq kH^{k-1}\Delta \ll_k \Delta^{2k-1}.$$

For $z \in \mathbb{Z}$, let $\mathrm{rec}_H(z) = (a_1, \dots, a_k)$ be its balanced base-H expansion: $z = \sum_{j=1}^{k} H^{j-1} a_j$, with $a_1, \dots, a_{k-1} \in [-(H-1)/2, (H-1)/2] \cap \mathbb{Z}$ and $a_k \in \mathbb{Z}$. For $\boldsymbol{a} \in [-\Delta^2, \Delta^2]^k \cap \mathbb{Z}^k$, this computes $\mathrm{rec}_H(\psi(\boldsymbol{a})) = \boldsymbol{a}$ in $\mathcal{O}_{\beta,k}(\mathrm{poly}\log \Delta)$ time.

For $C = C(\beta, k)$ sufficiently large, we have $\Delta_0 \le \Delta^{2k}$ and $2C_0 s\Delta \le \Delta^2$. Hence $\Delta_0^{1/\beta_0} \le \Delta^{1/\beta} \le s$, and Theorem 5.1 applies to $\psi(A)$ with (β_0, s, Δ_0).

Theorem 5.1 returns $A_0' \subseteq \hat{A}_0 \subseteq \psi(A)$ with $|A_0'| \le C_0 s$, and an s-proper GAP $P_0 \subseteq \mathbb{Z}$ of rank $m \le d$. Write $P_0 = f_0(B \cap \mathbb{Z}^m) \subseteq \mathbb{Z}$, where $f_0(\boldsymbol{u}) = \langle \boldsymbol{u}, \boldsymbol{q} \rangle$ and $B = \prod_{i=1}^{m} [-w_i, w_i]$ with $w_i \in \mathbb{Z}_{>0}$. Moreover, $\mathrm{supp}(\hat{A}_0) \subseteq P_0$ and A_0' is s-generating for P_0.

By injectivity,

$$\hat{A} \coloneqq \mathrm{rec}_H(\hat{A}_0) \subseteq A, \quad A' \coloneqq \mathrm{rec}_H(A_0') \subseteq \hat{A}.$$

Lifting the generators. Compute $\boldsymbol{g}_i \coloneqq \mathrm{rec}_H(q_i)$ for $i \in [m]$ and define $f \colon \mathbb{R}^m \to \mathbb{R}^k$ by $f(\boldsymbol{z}) = \sum_i z_i \boldsymbol{g}_i$.

By s-generation (Definition 3.4), there is $\boldsymbol{\tau} \in \mathbb{Z}^m$ such that, with $Q = (\boldsymbol{\tau} + sB) \cap \mathbb{Z}^m$, $f_0(Q) \subseteq \Sigma(A_0') = \psi(\Sigma(A'))$. Since $\Sigma(A') \subseteq [-C_0 s\Delta, C_0 s\Delta]^k$, for each $\boldsymbol{x} \in Q$ we have $F(\boldsymbol{x}) \coloneqq \mathrm{rec}_H(f_0(\boldsymbol{x})) \in \Sigma(A')$. If $\boldsymbol{x}, \boldsymbol{x} + \boldsymbol{e}_i \in Q$, then

$$\psi\big(F(\boldsymbol{x} + \boldsymbol{e}_i) - F(\boldsymbol{x})\big) = q_i.$$

Since $\|F(\boldsymbol{x} + \boldsymbol{e}_i) - F(\boldsymbol{x})\|_\infty \le 2C_0 s\Delta \le \Delta^2$, injectivity of ψ gives $F(\boldsymbol{x} + \boldsymbol{e}_i) - F(\boldsymbol{x}) = \mathrm{rec}_H(q_i) = \boldsymbol{g}_i$. Summing these identities along coordinate directions in Q gives

$$F(\boldsymbol{\tau} + \boldsymbol{x}) = F(\boldsymbol{\tau}) + f(\boldsymbol{x}) \quad \text{for } \boldsymbol{x} \in (sB) \cap \mathbb{Z}^m.$$

Verifying the vector progression. We have $\psi(f(\boldsymbol{z})) = f_0(\boldsymbol{z})$ for every $\boldsymbol{z} \in \mathbb{Z}^m$. Output the origin-symmetric GAP $P = f(B \cap \mathbb{Z}^m)$.

Fix $a_0 \in \hat{A}_0$ and let $\boldsymbol{a} = \mathrm{rec}_H(a_0)$ and $\boldsymbol{z} = f_0^{-1}(a_0) \in B \cap \mathbb{Z}^m$. Then $\psi(f(\boldsymbol{z})) = f_0(\boldsymbol{z}) = a_0 = \psi(\boldsymbol{a})$, while $f(\boldsymbol{z}) = F(\boldsymbol{\tau} + \boldsymbol{z}) - F(\boldsymbol{\tau})$ has ℓ_∞-norm at most $2C_0 s\Delta \le \Delta^2$. Since $\|\boldsymbol{a}\|_\infty \le \Delta$, injectivity gives $f(\boldsymbol{z}) = \boldsymbol{a}$. Moreover, $f_0 = \psi \circ f$ is injective on $(sB) \cap \mathbb{Z}^m$, so f is injective there as well. Thus P is s-proper and $\mathrm{supp}(\hat{A}) \subseteq P$.

By Definition 3.4, $f_0^{-1}(A_0')$ is s-generating for $B \cap \mathbb{Z}^m$. Applying f therefore shows that $f(f_0^{-1}(A_0')) = \mathrm{rec}_H(A_0') = A'$ is s-generating for P.

Coordinate bounds. For every $j \in [m]$, we have $2sw_j\boldsymbol{g}_j = F(\boldsymbol{\tau} + sw_j\boldsymbol{e}_j) - F(\boldsymbol{\tau} - sw_j\boldsymbol{e}_j)$. Thus $w_j\|\boldsymbol{g}_j\|_\infty \le C_0\Delta$. Since P is proper, $\boldsymbol{g}_j \in \mathbb{Z}^k \setminus \{\mathbf{0}\}$, so $w_j \le C_0\Delta$.

Set $U_0 = \operatorname{supp}(\hat{A}_0) \cup \{0\}$ and $\ell = \lceil C_0 \rceil$. The s-properness of P and P_0, the s-generation by A_0', and $|A_0'| \le C_0 s$ give

$$|sP| = |sB \cap \mathbb{Z}^m| = |sP_0| \le |\Sigma(A_0')| \le |\ell s U_0|.$$

Since $0 \in U_0$, Corollary 5.7 with exponent β_0 gives

$$|\ell s U_0| \le |\ell s U_0 - \ell s U_0| \ll_{\beta,k} |sU_0|.$$

Also, $sU_0 = \psi(s(\operatorname{supp}(\hat{A}) \cup \{\mathbf{0}\}))$, so $|sU_0| \le |s(\operatorname{supp}(\hat{A}) \cup \{\mathbf{0}\})|$, proving the bound on $|sP|$ after enlarging C.

The cardinality guarantees follow from $|A'| = |A_0'|$, $|\hat{A}| = |\hat{A}_0|$, and Theorem 5.1, enlarging $C = C(\beta, k)$ since $\log \Delta_0 \ll_{\beta,k} \log \Delta$. For the running time, the scalar call takes $\mathcal{O}_{\beta,k}(s|s(\operatorname{supp}(\psi(A)) \cup \{0\})| \operatorname{poly}\log \Delta)$ time. Evaluating ψ and recovering the vectors take time within the same bound. Finally,

$$\begin{aligned} |s(\operatorname{supp}(\psi(A)) \cup \{0\})| &= |\psi(s(\operatorname{supp}(A) \cup \{\mathbf{0}\}))| \\ &\le |s(\operatorname{supp}(A) \cup \{\mathbf{0}\})|. \end{aligned}$$

Thus the running time is $\widetilde{\mathcal{O}}_{\beta,k}(s|s(\operatorname{supp}(A) \cup \{\mathbf{0}\})|)$. □

C Proof of Lemma 11.1

Proof of Lemma 11.1. Write $\operatorname{supp} X = \{\boldsymbol{v}_1, \dots, \boldsymbol{v}_n\}$, with multiplicities μ_i, and set $b_0 = \log(2 + \prod_j (t_j + 1))$ and $S = \Sigma(X) \cap \prod_{j=1}^d [0, t_j]$. Discard zero vectors, for which we will return coefficient zero. For every remaining type, replace its multiplicity by

$$\min\left\{\mu_i, \min_{j : v_{i,j} > 0} \left\lfloor \frac{t_j}{v_{i,j}} \right\rfloor\right\}.$$

Nonnegativity makes this replacement valid for all sums in the box. Split each remaining multiplicity into powers of two and a remainder, obtaining $n' = \mathcal{O}(nb_0)$ indexed vectors whose subset sums represent exactly the allowed selections. Store the original type and chunk size of each vector. If no chunks remain, return $\{\mathbf{0}\}$ with the zero witness.

We encode the coordinates additively so that every encoded sum below the target determines its vector sum uniquely. Let $p_1, \dots, p_d$ be distinct fixed primes, and let q_j be the smallest power of p_j at least $2d(t_j + 1)$. Set

$$P = \prod_{j=1}^d q_j, \qquad E(\boldsymbol{a}) = \sum_{j=1}^d a_j \frac{P}{q_j}, \qquad H = E(\boldsymbol{t}).$$

Then $P = \mathcal{O}_d(\prod_j (t_j + 1))$ and $H < P/2$. If $\boldsymbol{a} \in \mathbb{Z}^d_{\ge 0}$ and $E(\boldsymbol{a}) \le H$, then $a_j P/q_j \le H < P$ gives $0 \le a_j < q_j$ for every j. Since the q_j are pairwise coprime, an encoded sum $h \le H$ therefore determines its vector sum uniquely: its jth coordinate is

$$a_j = \left(h(P/q_j)^{-1} \bmod q_j\right) \in [0, q_j - 1].$$

Apply Chan's deterministic all-targets subset-sum algorithm [Cha26, Theorem 2.7 and Section 4] to the encoded indexed vectors, with target H. Its running time is $\widetilde{\mathcal{O}}(n' + H)$. Recover the coordinates

of all returned sums and keep those satisfying $a_j \le t_j$ for every j. Every sum in the required box has encoding at most H, so the resulting vectors are exactly S.

For witness recovery at $\boldsymbol{t}$, form a balanced binary tree on the indexed chunks. At an internal node with requested sum h, use the same all-targets algorithm to compute the attainable sums at most h in each child. Scan the two sorted supports to find $h = h_0 + h_1$, and recurse on the children with these targets. At each depth, the targets sum to at most the original target, and the node input lengths sum to at most n'. There are $\mathcal{O}(\log n')$ depths, so one query takes $\widetilde{\mathcal{O}}(n' + H)$ time. Combine the selected chunk sizes by original type to recover Y.

Every integer a used after truncating the multiplicities satisfies $\mathsf{maxbitlen}(a) = \mathcal{O}_d(b_0)$. □

D AP Decompositions of All Subset Sums

Recall that the AP decomposition (Definition 3.6) of an integer set S with common difference s has size $|S \setminus (S + s)|$. We prove the following lemma. Here $\ln x = \log_e x$.

Lemma D.1. *Let A be a nonempty multiset of non-zero integers. For some $a \in \operatorname{supp} A$, the AP decomposition of $\Sigma(A)$ with common difference $|a|$ has size at most $\frac{|\Sigma(A)| \ln |\Sigma(A)|}{|A|}$.*

In particular, if $|A| = n$ and A is supported on $\{1, \dots, \Delta\}$, then $\Sigma(A)$ has an AP decomposition of size $\mathcal{O}(\Delta \log(n\Delta))$.

It remains open whether such an AP decomposition can be computed in $\mathcal{O}((n + \Delta)^{1+o(1)})$ time. In our Subset Sum algorithm (Theorem 1.1), we compute an AP decomposition for each group (Lemma 4.3), but when we combine the answers from all the groups in Algorithm 1, we restrict attention to shorter intervals, and consequently lose information about the full $\Sigma(A)$.

To prove Lemma D.1, we use the submultiplicativity theorem of Gyarmati, Matolcsi, and Ruzsa [GMR10]. The same theorem was also used by Bringmann, Fischer, and Nakos [BFN25] to design an improved output-sensitive algorithm for computing $\Sigma(A) \cap [0, t]$ given a multiset A of positive integers.

Theorem D.2 ([GMR10, Theorem 1.2])**.** *Let $n \ge 2$ and let $A_1, \dots, A_n \subseteq \mathbb{Z}$ be finite and nonempty. Define $S = \sum_{j=1}^{n} A_j$ and $S_i = \sum_{j \in [n] \setminus \{i\}} A_j$. Then*

$$|S| \le \left(\prod_{i=1}^{n} |S_i| \right)^{1/(n-1)}.$$

Proof of Lemma D.1. Let $n = |A|$. The case $n = 1$ is immediate. For $n \ge 2$, write $A = \{a_1, \dots, a_n\}$, $S = \Sigma(A)$, and $S_i = \Sigma(A \setminus \{a_i\})$. Applying Theorem D.2 to $A_i = \{0, a_i\}$ gives $\prod_{i=1}^{n} |S_i| \ge |S|^{n-1}$. Taking logarithms and using $\ln \frac{1}{x} \ge 1 - x$ for $0 < x \le 1$ gives

$$\ln |S| \ge \sum_{i=1}^{n} \ln \frac{|S|}{|S_i|} \ge \sum_{i=1}^{n} \left(1 - \frac{|S_i|}{|S|} \right).$$

By averaging, there exists i such that $\frac{1}{n} \ln |S| \ge 1 - |S_i|/|S|$, that is, $|S| - |S_i| \le \frac{1}{n} |S| \ln |S|$. Since $S = S_i \cup (S_i + a_i)$, we have $|S \cap (S + a_i)| \ge |S_i + a_i| = |S_i|$, and therefore

$$|S \setminus (S + a_i)| = |S| - |S \cap (S + a_i)| \le |S| - |S_i| \le \frac{1}{n} |S| \ln |S|,$$

as desired. □

We now show that an efficient algorithm for computing an AP decomposition of $\Sigma(A)$ would imply an efficient algorithm for the following problem posed by Węgrzycki:

Problem 3. Bounded Support Subset Sum
Input: Positive integers $a_1, \ldots, a_n$ and a target integer t.
Task: Compute $\max\{|I| : I \subseteq [n],\ \sum_{i \in I} a_i = t\}$, or report that no such I exists.

Węgrzycki [Węg25, Problem 2] asked whether this problem can be solved in $\mathcal{O}(n^{1-\delta}t + n^{\mathcal{O}(1)})$ time for some absolute constant $\delta > 0$.

Proposition D.3. *Suppose there is a $T(n, \Delta)$-time algorithm for computing an AP decomposition of $\Sigma(A)$ given any multiset A of at most n integers in $[1, \Delta]$. Then,* Bounded Support Subset Sum *can be solved in $\widetilde{\mathcal{O}}(T(n, 3t\sqrt{n}) + n + t\sqrt{n})$ time.*

In particular, if $T(n, \Delta) \leq \widetilde{\mathcal{O}}(n + \Delta)$, then Bounded Support Subset Sum can be solved in $\widetilde{\mathcal{O}}(n + t\sqrt{n})$ time.

Proof. Let $K = \lceil \sqrt{n} \rceil$. Call $a_i \in [1, t/K]$ the small items, and $a_i \in (t/K, t]$ the large items. Let $L[s]$ denote the cardinality of the largest subset of large items with sum exactly s (where $L[s] = -\infty$ if no such subset exists). Similarly define $S[s]$ for small items. We will compute these two arrays. Then, the answer is clearly $\max_{0 \leq s \leq t}\{L[s] + S[t - s]\}$ which can be computed in $\mathcal{O}(t)$ additional time.

Large items. By definition, a subset of sum at most t must contain fewer than K large items. Define the multiset $B = \{Ka_i + 1 : i \in [n] \text{ is a large item}\}$ and $t' = tK + K$. Note that any $B' \subseteq B$ of size $|B'| \geq K$ has sum $\sum_{b \in B'} b > |B'|(K \cdot (t/K) + 1) \geq t'$. Use Bringmann's Subset Sum algorithm [Bri17, Cha26] to compute $\Sigma(B) \cap [0, t']$ in $\widetilde{\mathcal{O}}(n + t') = \widetilde{\mathcal{O}}(n + tK)$ time. Then, for every $s \in \mathbb{Z} \cap [0, t]$ and $k \in \mathbb{Z} \cap [0, K)$, we have $Ks + k \in \Sigma(B) \cap [0, t']$ if and only if there is a subset of exactly k large items that sums to s. Thus we immediately obtain the array $L[s]$ for all $s \in [0, t]$.

Small items. Define $\Delta = 3t\sqrt{n}$, and the multiset $E = \{2na_i + 1 : i \in [n] \text{ is a small item}\}$ which contains positive integers bounded by $2n \cdot t/K + 1 \leq \Delta$. Then, for every $s \in \mathbb{Z} \cap [0, t]$ and $k \in \mathbb{Z} \cap [0, n]$, $2ns + k \in \Sigma(E)$ if and only if there is a subset of exactly k small items that sums to s. Thus, for each possible sum $s \in \mathbb{Z} \cap [0, t]$, the desired maximum cardinality is $S[s] = \max\{v - 2ns : v \in \Sigma(E) \cap [2ns, 2ns + n]\}$. To compute all $S[s]$, it suffices to answer the predecessor queries $\max(\Sigma(E) \cap [0, u])$ for all $u \in Q \coloneqq \{2ns + n : s \in [0, t] \cap \mathbb{Z}\}$.

Compute an AP decomposition $\mathcal{D}$ of $\Sigma(E)$ in $T(n, \Delta) = T(n, 3t\sqrt{n})$ time, and let $p > 0$ denote the common difference. We answer the predecessor queries offline in time near-linear in $|Q| = t + 1$ and $|\mathcal{D}|$, as follows. Sort all endpoints in $\mathcal{D}$ and all query points in Q together in increasing order, and treat them as time steps. Iterate over the time steps in increasing order, and maintain a dynamic set P_u (using a balanced search tree) such that at time step u,

$$P_u = \{a \bmod p : \{a, a + p, \ldots, b\} \in \mathcal{D} \text{ and } a \leq u < b\}.$$

Also maintain

$$r_u = \max\{b : \{a, a + p, \ldots, b\} \in \mathcal{D} \text{ and } b \leq u\}.$$

At query time $u \in Q$, the predecessor of u in $\Sigma(E)$ is precisely

$$\max\left\{r_u, \max\{u - ((u - a) \bmod p) : a \in P_u\}\right\},$$

which can be computed in logarithmic time by making $\mathcal{O}(1)$ predecessor queries to P_u. The dynamic set P_u can be maintained when the time step u reaches an endpoint of an arithmetic progression in $\mathcal{D}$, by inserting or deleting the residue of that progression modulo p.

The total time is therefore $\widetilde{\mathcal{O}}(T(n,\Delta)+n+tK) = \widetilde{\mathcal{O}}(T(n,3t\sqrt{n})+n+t\sqrt{n})$. □

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